

WEO Methodology Rationale

Warmth Engine Observatory

Companion Document to Methodology Manual

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CANONICAL SOURCE DECLARATION

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Novel terminology, frameworks, and methodological innovations introduced herein originated with this publication as of January 2026.

Version 1.4 (May 2026) introduces rationale documentation for the Sovereign Capability Profiles (SCP) framework including three-gate PAA designation, severance testing, dimension qualification principles, and Actor Designation Framework (§5.4), bloc classification methodology with universal membership categories and bloc assignment gate (§3.12), supply-chain response qualification Option G (§3.6.4), the F3 AI Centrality pathway exemption (§4.6.3), the Scope Comparator Hierarchy for Parallel Policy scope assessment (§9.5.2), CJK language conventions (§3.11.1), the Distinct Actor Requirement for Parallel Policy (§9.1.1), Funding Flow denominator clarification (§9.4.2), and Chinese regulatory temporal accommodation (§9.4.1).

The canonical version is the Zenodo publication. Derivative works, summaries, or adaptations should cite this canonical source.

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About the Author

Warmth Engine is the founder of the Warmth Engine Observatory, a platform tracking AI infrastructure coordination dynamics across geopolitical blocs. Other published work by this author includes *The Warmth Engine Series: Analytical Prophecy and the Convergence of AI Sovereignty*, *Infrastructure Colonialism*, and *the Coordination Imperative* (Zenodo DOI: 10.5281/zenodo.17451629), conducted as separate independent research.

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Section 1: Introduction & Document Purpose

1.1 Purpose

This document provides the empirical derivation, theoretical grounding, and calibration justification for all methodology components in the WEO Methodology Manual. It serves as the authoritative reference for understanding *why* methodological choices were made, complementing the Manual's specification of *what* those choices are.

1.2 Document Relationship

Document	Purpose
Methodology Manual (DOI: 10.5281/zenodo.18427565)	Operational specification — what the methodology is
Methodology Rationale	Justification — why the methodology is designed this way

The Methodology Manual is self-contained for operational use. This Rationale document provides empirical derivations, historical precedent analysis, and calibration justifications for users requiring full methodological transparency.

1.3 Rationale Categories

Each methodology component falls into one of six categories:

Category	Definition	Treatment
EMPIRICALLY GROUNDED	Derived from external research, data, or established frameworks	Full derivation with citations
ADAPTED	Modified from established methodology for WEO context	Adaptation rationale with source
WEO SYNTHESIS	Novel combination of established precedents	Synthesis sources documented
WEO-ORIGINAL	Genuinely novel contribution with no prior art	Novelty verification documented
CATEGORY CREATOR	Novel calibration choice for new measurement domain	Explicit acknowledgement with validation commitment
STANDARD PRACTICE	Industry-standard approach requiring no special justification	Brief note citing convention

Section 2: Methodology Balance Statement

2.1 Overall Balance

WEO methodology components distribute approximately ~32% **empirically grounded**, ~36% **precedent-based**, and ~31% **category creator** — with the empirically grounded and precedent-based components collectively representing ~69% external grounding.

Category	Component Count	Percentage
EMPIRICALLY GROUNDED (VERY HIGH/HIGH confidence)	32	~32%
PRECEDENT-BASED (MEDIUM-HIGH/MEDIUM confidence)	36	~36%
CATEGORY CREATOR	31	~31%

Note: Component count reflects independent methodology design choices across Sections 3–10. A single section may document multiple components where each design decision is independently determinable. External claims supporting these components are separately verified in Appendix B. Industry-specific analytical metrics documented in the companion TCS and TCD papers maintain their own rationale and confidence assessments within those publications.

2.2 Confidence Distribution

Confidence Level	Count	Examples
VERY HIGH (>90%)	3	Keohane coordination; Finnemore & Sikkink threshold; Basel G-SIB
HIGH (75-90%)	51	Event classification; software routes; bloc definitions; Vienna Convention; industry taxonomy; workflow integration; OEP Pathways A/B; OEP MIIT framework; source accommodation alignment; Corporate Strategic Disclosure; Level 5 operational evidence; ENT enrichment layer; ENT binary qualification; ENT guardrails; ENT-1 resource demand; ENT-4 environmental dependency; Chinese cascade model; SCP 7-dimension framework; SCP evidence standard; universal membership categories; bloc assignment gate; register evidence standard; 3× scope threshold; 根据 temporal accommodation; F3 pathway exemption; Distinct Actor Requirement; Funding Flow denominator
MEDIUM-HIGH (60-75%)	30	Tier classification; ADAPTED frameworks; CC evidence requirements; mechanism statements; PIET windows; T2 Impact-Demonstrated; OEP rack density; OEP confidence gradation; Option F defence appropriation; conflict-era currency adjustment; ENT taxonomy structure; ENT-3 consequence activation; T2 scope (principal actor asymmetry); ≥3 PAA dimensional threshold; three-gate PAA designation; severance test design; AIK/ACS/Participant taxonomy; sovereign control principle; dimension sustainability taxonomy; third bloc formation; two-level metric hierarchy; inapplicable/indeterminate distinction; Option G supply-chain response
MEDIUM (50-60%)	5	MIXED components; derivative calibration
CATEGORY CREATOR (variable)	10	WEO-ORIGINAL novel IP; PIET gate architecture; PIET thresholds; OEP 10% threshold; conflict-era review interval; ENT-2 linguistic threshold; ENT-5 medium scope; ENT-2/ENT-5 boundary
LOW (<50%)	0	None — all components meet minimum threshold

2.3 Minimum Confidence Thresholds

WEO maintains two distinct minimum confidence thresholds:

Threshold	Applies To	Minimum	Exceptions
Event Verification	Individual events entering database	70%	None — events below 70% excluded
Methodology Components	Empirical backing for calibration values	65%	None

The 70% event threshold ensures database quality. The 65% methodology threshold ensures empirical grounding whilst acknowledging that category creation requires some novel calibration.

2.4 Honest Framing

Where external validation is unavailable, WEO:

1. Adapts proven methodologies from adjacent domains
2. Explicitly flags category creator choices
3. Commits to recalibration as empirical data accumulates
4. Maintains transparency about confidence levels

2.5 Methodology Evolution Commitment

WEO methodology represents current best practice for AI infrastructure coordination tracking, developed through rigorous research and empirical grounding. This is a living framework subject to refinement as:

1. Empirical data accumulates from ongoing observation
2. New analytical approaches emerge in adjacent fields
3. Validation milestones (e.g., 100-event threshold) trigger systematic review
4. User feedback identifies practical improvements

All methodology components include explicit confidence levels and validation commitments. Category Creator elements are flagged for priority review as the evidence base matures. Version history and material changes are documented in methodology version notes.

Section 3: Event Verification Rationale

Manual Reference: Chapter 3

3.1 Source Quality Hierarchy

Manual Section: 3.1

Category: STANDARD PRACTICE

Rationale: The four-tier source hierarchy (Primary → Corroborating → Supplementary → Excluded) follows established verification practice across intelligence community (ICD 203/206), wire services (Reuters, AFP), fact-checking networks (EFCSN), and defence intelligence (Jane's).

Key Precedents:

- ICD 203/206 establish analytic standards emphasising multi-source analysis and explicit disclosure of single-source dependence
 - Reuters Handbook requires minimum source standards by claim type
 - EFCSN Code of Standards establishes multi-source verification requirements
-

3.1.1 Primary Source Substitution

Manual Section: §3.1 (Primary Source Substitution)

Category: ADAPTED (75% confidence)

Rationale: The substitution rule formalises the principle that source quality can substitute for source quantity when primary sources are Grade 1 canonical instruments. A Federal Register entry, an official gazette publication, or an audited financial filing independently verifies the event it documents — its authority derives from its institutional origin, not from external corroboration. Two independent Grade 1 instruments documenting the same event provide stronger verification than one Grade 1 primary plus two Grade 3 news articles paraphrasing it.

Closest Precedents:

Framework	Relevance	Divergence from WEO
ICD 203 (Analytic Standards)	Quality-over-quantity principle for intelligence assessments; single high-authority source can outweigh multiple low-authority sources	Intelligence context, not event verification
Reuters Handbook of Journalism	Convergent independent verification, not fixed source counts; authority of source weighs against number required	Wire service practice, not infrastructure event verification
§5.3.4 Dimension Evidence Standard	Same quality-over-quantity principle applied to structural capability claims; 1+1 base with Grade hierarchy	Dimension claims, not event claims; the substitution rule extends the same logic to events

Key Design Decision:

1. **Grade 1 restriction.** The substitution applies only to Grade 1 sources because their institutional authority provides self-verification. Two corporate press releases (Grade 2) or two industry analyses (Grade 3) do not carry the same self-verifying quality — they require independent corroboration to confirm accuracy. The restriction prevents the substitution from weakening the evidence standard for events where corroboration is genuinely needed.

Confidence Assessment: 75%. The principle is well-established across the precedent frameworks and is already implemented in §3.4.9 for Tier A claims ("2+ Primary OR 1P + 2C"). The substitution rule formalises and extends this existing pathway. Confidence matches §5.4.12 (Dimension Evidence Standard) because the underlying quality-over-quantity logic is identical.

Validation Commitment: Monitor whether the Grade 1 restriction is sufficient to prevent evidence-standard erosion. If events begin qualifying through the substitution pathway with sources that are technically Grade 1 but lack the self-verifying quality of canonical instruments, tighten the qualifying criteria.

3.2 Claim Verification Tiers (A/B/C)

Manual Section: 3.4

Category: STANDARD PRACTICE

Rationale: Three-tier claim verification follows ICD 203/206 analytic standards and UK PHIA probability/confidence frameworks.

Tier	Requirement	Application
Tier A	Primary mandatory	Existential claims
Tier B	Primary preferred	Supporting details
Tier C	Corroborating acceptable	Contextual information

3.3 Four Event Types

Manual Section: 3.6

Category: EMPIRICALLY GROUNDED (HIGH)

Rationale: The four event types (Coordination, Infrastructure, Policy, Corporate) are classified by **mechanism** (HOW the action occurs) and **actor type** (WHO performs it), not by subject matter, downstream effects, or durability.

Event Type	Primary Mechanism	Actor Dimension
Coordination	Negotiation, mutual accommodation amongst ≥ 2 sovereign entities	Multi-nation
Infrastructure	Deployment of physical/digital systems	Government/sovereign entity
Policy	Unilateral legislative, regulatory, or executive action	Single sovereign entity
Corporate	Private sector action (partnerships, releases, market events)	Private company

Key Distinction: An agreement *about* infrastructure = Coordination (mechanism is negotiation). A policy *requiring* infrastructure = Policy (mechanism is regulation). A company *deploying* infrastructure = Corporate (actor is private). A government *deploying* infrastructure = Infrastructure (actor is sovereign).

Empirical Precedent:

Event Type	Primary Precedent	Confidence
Coordination	Abbott & Snidal legalization framework (2000); Keohane cooperation theory (1984); CAMEO codes 03-05	HIGH
Policy	Lowi's Policy Typology (1964) — classifies by HOW coercion is applied, not subject matter	HIGH
Corporate	IPE literature (Strange 1988, Bartley 2018); CAMEO BUS/MNC actor codes	MEDIUM-HIGH
Infrastructure	Emerging DPI literature; Bakonyi & Darwich (2024) — distinguishes infrastructure materiality (physical construction) from infrastructure governance (rule-setting)	MEDIUM

Derivation: This mechanism-based classification follows established precedent from CAMEO/GDELT event coding (which classifies by action verb, not topic) and Lowi's Policy Typology (which classifies by HOW coercion is applied, not subject matter). Infrastructure as a distinct category is supported by emerging Digital Public Infrastructure (DPI) scholarship, which distinguishes infrastructure materiality from infrastructure governance.

3.4 Event Type × Tier Applicability Matrix

Manual Section: 3.6 (Event Type × Tier Applicability)

Category: LOGICALLY NECESSARY

Rationale: The constraint that Policy events cannot qualify as Tier 1 follows from definitional necessity:

- **T1 requires:** Cross-bloc participation with mutual accommodation
- **Policy is:** Unilateral rule-making by single jurisdiction
- **Unilateral:** Cannot satisfy "mutual" by definition
- **Therefore:** Policy ≠ T1

Policy events with cross-bloc *effects* route to T2 (typically Restriction) or T3. Coordinated multilateral policy alignment involving mutual accommodation qualifies as Coordination Event Type, not Policy.

3.5 Coordination Event Criteria

Manual Section: 3.7

3.5.1 Keohane Definition

Category: EMPIRICALLY GROUNDED (VERY HIGH)

Rationale: WEO adopts Keohane's (1984) coordination definition — observable policy adjustments through negotiation or mutual accommodation, where each entity's actions facilitate achievement of partners' objectives.

This definition has been IR theory consensus for 40+ years and provides clear operationalisation distinguishing:

- Coordination from harmony (automatic alignment)
- Coordination from parallel independent action (similar policies without negotiation)
- Coordination from aspirational statements (declarations without mechanisms)

Citation: Keohane, R. O. (1984). *After Hegemony: Cooperation and Discord in the World Political Economy*. Princeton University Press.

3.5.2 33% Convergence Threshold

Category: EMPIRICALLY GROUNDED (HIGH)

Rationale: The approximately one-third threshold for Type 3 coordination (demonstrated convergence) derives from Finnemore & Sikkink (1998: 901), who establish that "tipping rarely occurs before one-third of the total states in the system adopt the norm." Their analysis across women's suffrage (1930 tipping point), the anti-personnel landmine ban (May 1997), and laws of war norms demonstrates this empirical pattern.

WEO operationalises this at 33% (~50 of ~150 AI-active nations) as a practical threshold, acknowledging this reflects an empirical approximation rather than a precise theoretical requirement.

Citation: Finnemore, M., & Sikkink, K. (1998). "International Norm Dynamics and Political Change." *International Organization* 52(4): 887-917.

3.5.3 Six-Level Implementation Hierarchy

Category: EMPIRICALLY GROUNDED (VERY HIGH)

Rationale: The six-level evidence hierarchy (Announced → Adopted → Ratified → In Force → Implemented → Verified) follows Vienna Convention treaty law, OECD policy tracking standards, and SIPRI arms control verification literature.

Key Precedents:

- Vienna Convention Articles 11-18 codify consent mechanisms (signature, ratification, accession); Article 24 governs entry into force
- Policy monitoring frameworks distinguish announcement, adoption, and implementation phases (cf. OECD policy tracking methodology)
- SIPRI Arms Control Verification hierarchy tracks commitment through operational verification

Key threshold: Binding agreements require Level 3 (ratified) minimum; non-binding frameworks require Level 5 (implemented) minimum.

3.5.4 Type 2 Level 5 — Operational Evidence Requirement

Manual Section: §3.7 (Coordination Criteria — Implementation Verification Standards)

Category: WEO SYNTHESIS (82% confidence)

Rationale: The four-pathway operational evidence framework synthesises formally codified mechanisms from multiple established governance architectures into a novel unified qualification gate. The underlying principle — that publication or catalogue adoption of an international standard does not constitute operational implementation — is formally recognised across multiple institutions: ISO/IEC Guide 21-1:2005 (Clause 1) explicitly excludes rules for the "use" of standards in production, trade, or legislation; the ISO Systematic Review (ISO/IEC Directives Part 1, Subclause 2.9) distinguishes adoption from use as separate assessment questions; and the academic decoupling literature (Boiral 2003; Aravind & Christmann 2011; Wijen 2014) provides extensive empirical documentation of the phenomenon.

The architectural precedent for maintaining independent assessment dimensions for adoption and implementation derives most directly from the FATF Mutual Evaluation Methodology (adopted February 2013, updated October 2022), which separates Technical Compliance (rules adopted into national law) from Effectiveness (rules operating in practice) using independent rating scales and evidence-based assessment across 11 Immediate Outcomes. The FATF Methodology states explicitly that the effectiveness assessment "differs fundamentally" from the assessment of technical compliance. WEO adapts this two-dimensional architectural principle from national AML/CFT compliance to individual standard implementation assessment. Each of the four evidence pathways draws on independently established institutional mechanisms: Pathway (a) on the IAF MLA Level 5 endorsement process (IAF PL 3) and ISO Survey of Certifications; Pathway (b) on the EU New Legislative Framework's OJEU citation gate (Regulation (EU) 1025/2012, confirmed by CJEU in *James Elliott Construction C-613/14* (2016) and *Public.Resource.Org v Commission C-588/21 P* (2024)); Pathway (c) on the ISO/IEC Directives Part 2:2021 mandatory normative reference framework (Clause 2 references "constitute requirements"); and Pathway (d) on mechanisms such as the EU Outdoor Noise Directive's mandatory Declaration of Conformity reporting and the FATF effectiveness assessment's evidence-of-outcomes requirement.

The novel elements are: (i) the combination of all four evidence categories into a single disjunctive qualification gate (no existing framework bundles all four); (ii) the anchoring requirement for Pathway (c), requiring normative dependency chains to terminate in a standard with independent operational evidence (no existing framework formalises this concept); and (iii) the application of this gate to individual international consensus standards as a unit of analysis for geopolitical coordination intelligence (existing analogues operate at the national regulatory system level or within single regulatory domains).

Closest Precedents:

Framework	Institution	Relevance	Divergence from WEO
FATF Mutual Evaluation Methodology (2013/2022)	FATF	Strongest architectural analogue: two independent assessment dimensions (technical compliance vs. effectiveness) with separate evidence requirements and rating scales	Operates at national regulatory system level, not individual standard level; uses graduated four-point scale, not binary gate; domain-specific to AML/CFT
EU New Legislative Framework (Regulation 1025/2012)	European Commission / CJEU	Strongest single-pathway precedent: OJEU citation as formal, judicially confirmed qualification gate between publication and regulatory operational effect	Applies only to harmonised European standards within EU regulatory scope; single-dimension gate (regulatory reference only)
IAF MLA Level 5 / IAF PL 3	IAF (now GACI)	Direct precedent for requiring evidence of certification activity as qualification criterion for scheme endorsement; requires "clear international, regional or national market support" and "evidence of operating"	Gate operates at scheme viability level, not minimum assessment count; limited to management system and product certification standards within IAF scope
ISO/IEC Directives Part 2:2021	ISO/IEC	Direct codification of normative dependency mechanism — normative references "constitute requirements"	Does not formalise anchored vs. unanchored chains; no operational evidence requirement for referenced standards
Basel Committee RCAP	BCBS	Three-tier assessment approach (adoption timeliness → transposition consistency → bank-level outcomes) confirming institutional recognition of the adoption/implementation gap	Explicitly acknowledges it does not assess supervisory effectiveness; banking regulation only

Key Design Decisions:

1. **Disjunctive gate (any one pathway sufficient).** Unlike FATF's 11 Immediate Outcomes (all assessed), WEO requires only one of four pathways. This reflects the binary nature of the implementation question for individual standards: the question is whether ANY operational effect exists, not a comprehensive assessment of implementation depth.
2. **Pathway (c) anchoring.** A normative dependency chain cannot be self-referencing without an anchor. At least one standard in the chain must independently satisfy Pathway (a), (b), or (d). This prevents a cluster of published-but-unimplemented standards from bootstrapping each other to Level 5 through circular normative references. No existing framework formalises this concept.
3. **Binary gate vs. graduated scale.** WEO uses pass/fail rather than FATF's graduated four-point scale. This is a deliberate design choice — WEO needs a classification decision (implemented or not), not an implementation quality assessment. The EU NLF's OJEU citation gate provides direct precedent for binary treatment in standards governance.
4. **Retrospective application.** The operational evidence requirement applies to all events in the database, not only to new candidates. Standards that entered under V1.2 criteria where publication alone satisfied Level 5 are assessed against the four pathways. Events failing all four pathways are removed from the database. This ensures the database reflects verified operational coordination, not catalogued intentions.

Confidence Assessment: 82%. The principle that publication does not equal implementation is formally codified in multiple institutions (ISO/IEC Guide 21-1, FATF Methodology, EU NLF), and three of the four pathways have direct or strong analogous institutional precedent. Confidence is bounded below 90% because: (a) no single framework bundles all four pathways, making this a synthesis rather than an adaptation; (b) the anchoring requirement for Pathway (c) is genuinely novel; and (c) the application to individual standards as a unit of analysis for geopolitical intelligence has no direct precedent. Confidence is above 75% because the architectural principle is well-established, the FATF analogue is strong, and each pathway independently maps onto formally codified mechanisms.

Validation Commitment: Monitor for (1) any ISO or IEC governance reform that creates a formal implementation tracking mechanism for the full standards catalogue; (2) evolution of the merged IAF/ILAC entity (Global Accreditation Cooperation Inc.) and any changes to scheme endorsement criteria; (3) academic literature proposing multi-pathway implementation assessment frameworks; (4) FATF 5th Round methodology updates that may refine effectiveness assessment criteria; and (5) any extension of the EU NLF gate mechanism to non-harmonised standards.

3.6 Two-Stage AI Connection Filter

Manual Section: 3.5

Category: EMPIRICALLY GROUNDED (HIGH)

3.6.1 10% / \$100M Thresholds

Rationale: The 10% allocation threshold and \$100M absolute threshold derive from convergent regulatory precedent:

Source	Threshold	Application
SEC ASC 280-10-50-12	10%	Segment reporting materiality
Basel III	10%	Large exposure definition threshold
IMF BPM6	10%	Direct investment classification (paragraph 6.12: ≥10% voting power)

The \$100M absolute threshold ensures large-scale projects with material AI components (e.g., \$400M AI line in \$5B fab) are captured even when percentage is below 10%.

\$100M Validation Sources:

- Institutional infrastructure fund minimum deal sizes represent institutional-scale investment thresholds
- Market transaction evidence: POET Technologies \$75M AI hardware financing (October 2025), Applied Digital \$112.5M AI data centre drawdown (October 2025)
- EU practice often uses ≥€50M as a threshold for IPCEI notification (GBER 2023) and direct participant status in national calls — WEO's \$100M represents approximately 2× this practical threshold
- Distribution analysis of verified AI infrastructure investments: \$50M-\$100M band contains lowest investment density, confirming natural breakpoint at \$100M

Citations:

- SEC filings: POET Technologies (October 2025), Applied Digital (October 2025)
- EU Important Projects of Common European Interest (IPCEI) framework

3.6.2 Option E — Corporate Strategic Disclosure

Manual Section: §3.5, Stage 2 Option E

Category: ADAPTED (HIGH)

Rationale: Option E adapts established regulatory evidentiary practice to the Stage 2 AI connection question for Corporate events. Corporate financial instruments — PPAs, equity agreements, procurement contracts — specify capacity, price, and delivery terms without designating workload type. This is structurally parallel to the gaps addressed by PIET (export controls list by HS code) and OEP (infrastructure documentation is application-agnostic): the instrument class does not produce AI-specific language by convention.

The evidentiary approach — accepting regulated corporate disclosure as evidence of the strategic purpose of a transaction whose operative instrument is silent on that purpose — has well-established precedent across international regulatory frameworks. Five of seven frameworks examined make the approach mandatory or statutory; no framework rejects it in principle.

Framework	Established Practice
US Antitrust (HSR Act Items 4(c)/4(d); 2023 Merger Guidelines)	Mandatory filing of strategic analyses; ordinary-course documents more probative than retrospective materials
EU Merger Control (Form CO §5.4; <i>Facebook/WhatsApp</i> €110M fine)	Mandatory filing of board presentations; regulated disclosures treated as binding representations
Sanctions Enforcement (OFAC; Section 13(r) Securities Exchange Act)	Statutory pipeline from regulated corporate disclosure to enforcement action
Securities Regulation (SAB 99; Reg S-K Item 303; <i>Leidos</i> (2018))	Entity's own characterisation of transaction significance directly affects regulatory obligations
WTO Dispute Settlement (<i>Australia – Plain Packaging</i> , DS435/441)	Internal corporate documents cited by panel to establish industry awareness; examination beyond measure's text

Option E's requirements implement five convergent safeguards identified across all examined frameworks: regulated filing status, contemporaneity, transaction specificity, senior authorisation, and corroboration. The 1+1 evidence structure (regulated disclosure + corroborating signal) is proportionate to the lower underlying uncertainty — PIET and OEP address cases where the AI connection is genuinely ambiguous, whilst Option E addresses cases where the implementing entity acknowledges AI purpose in regulated disclosure. The calibration principle follows *Philip Morris v. Australia* (ICSID): evidentiary thresholds proportionate to the consequence of the determination.

Documented limitation: *Japan – Alcoholic Beverages II* (WT/DS8/AB/R, 1996) rejected subjective intent inquiry. Option E focuses on operational function (role in AI infrastructure), not subjective corporate intent — consistent with the "design, architecture, and structure" framework that also underpins the T2 Impact-Demonstrated mechanism (§3.7.4).

Confidence Assessment: HIGH. Five of seven frameworks make the approach mandatory or statutory. The adaptation to WEO's Stage 2 classification context is a bounded domain transfer — the evidentiary logic is established; the application to AI connection determination is the contribution.

3.6.3 Option F — Defence Appropriation Evidence

Manual Section: §3.5, Stage 2 Option F

Category: ADAPTED (MEDIUM-HIGH)

Rationale: Policy events with genuine AI connections that are funded through defence appropriation processes face a structural documentation constraint that is universal, not specific to WEO. Defence appropriation documents fund AI-specific programmes through publicly verifiable financial instruments. Once appropriated funds enter implementation, operational detail moves behind classification barriers — a permanent feature of military spending architecture across all major defence economies. The appropriation stage is therefore the sole publicly verifiable evidence window for the AI connection.

Any framework attempting to track military AI activity from public documentation encounters the same structural barrier: the evidence required for post-implementation verification does not exist in publicly accessible form. The constraint is empirically documented across five surveyed economies and is consistent with the broader finding that observable documentation surface contracts during re-militarisation whilst underlying activity expands (§4.11).

Option F accepts binding defence financial documents as AI connection evidence at Stage 2, capturing events at the sole point where the AI connection is publicly documented. The adaptation applies existing primary source evidentiary standards — enacted legislation and executive budget submissions qualify as Tier A evidence per §3.4.4 — to a specific document class not previously recognised in the AI Connection Filter's option structure.

Boundary logic — eligibility by Event Type and document class:

The classification barrier affects all event types within the defence domain. However, existing pathways already serve Coordination and Infrastructure events:

- **Coordination events** in the defence context (e.g., multilateral defence AI partnerships) typically qualify via Keohane auto-qualification because the agreement or communiqué itself is public and contains AI-specific language. Classification constrains downstream implementation detail, not the coordination event's qualification evidence.
- **Infrastructure events** in the defence context (e.g., classified military AI facilities) are served by OEP (Option D), which establishes AI connection through operational physical evidence where official documentation is absent or application-agnostic.
- **Policy events** funded through defence appropriations have no fallback pathway when Options A, B, and C fail. These are the events where the appropriation document is the sole primary source evidence of AI connection.

Option F fills this remaining gap. Eligibility is determined by Event Type (Policy, established at Stage 1) and document class (defence appropriation documents), consistent with how Options D, E, and PIET scope their eligibility. Domain classification is a separate, independent assessment at Stage 4.

Empirical calibration:

The pass criteria were calibrated against a verified survey of defence appropriation documents across five major economies, applying the same multi-agent cross-validation approach documented for the principal actor framework (§5.4):

Economy	Primary Document	AI Granularity	Usable for Option F
United States	DoD RDT&E R-1 Justification Books	EXPLICIT	Yes
United Kingdom	MoD Main Supply Estimates / Policy	IDENTIFIABLE	Yes
European Union	EDF Work Programme 2026	EXPLICIT	Yes
Japan	MOD Defense Programs & Budget FY2026	EXPLICIT	Yes
Australia	Defence Portfolio Budget Statements	EMBEDDED	No

Three-level granularity scale:

- **EXPLICIT:** Dedicated AI budget line with a financially isolated monetary value
- **IDENTIFIABLE:** AI explicitly named as a designated beneficiary of a shared funding pool
- **EMBEDDED:** AI spending subsumed within undifferentiated general categories; not separately identified

The survey confirmed that the classification constraint operates consistently across different legislative traditions. Nations whose defence budget architecture produces only EMBEDDED granularity do not qualify via Option F — this reflects their financial architecture, not a methodology limitation.

Calibration Evidence	Contribution
US DoD R-1 Justification Books (PE 0603040A: AI/ML Advanced Technologies, \$20.487M FY2026)	Confirmed EXPLICIT granularity in executive budget submissions at programme element level
UK Defence Innovation Fund (£400M, ring-fenced with AI as mandated beneficiary)	Confirmed IDENTIFIABLE granularity via binding departmental policy with explicit AI designation
EU EDF Work Programme 2026 (EDF-2026-RA-SIMTRAIN-MSAI-STEP, €16M)	Confirmed EXPLICIT granularity in Commission Implementing Decisions with isolated AI call topic budgets
Japan MOD FY2026 Budget ("Initiatives for the Utilisation of AI" section)	Confirmed EXPLICIT granularity in executive budget submissions with hyper-granular AI breakdown
Australia Defence PBS 2025-26 (ASCA merged into Program 2.8)	Confirmed EMBEDDED granularity — AI not separately identified; correctly excluded
US DoD AI spending structure (\$25.2B total, \$1.8B explicit topline — ~93% embedded in broader programmes)	Confirmed that classification renders post-implementation AI spending verification structurally impossible from public sources

Key design decisions:

1. **IDENTIFIABLE granularity accepted alongside EXPLICIT.** The document must explicitly name AI in the financial commitment; financial isolation of the AI component is not required. Ring-fenced capability pools listing AI as a legally mandated beneficiary satisfy the AI connection test because the designation is documented in the instrument — not inferred.
2. **Three-category document typology.** Enacted appropriation legislation, executive budget submissions, and departmental work programmes all qualify, provided each carries binding appropriation authority. The survey demonstrated that AI-specific granularity resides in programme-level documents, not top-level appropriation act text.
3. **No minimum monetary threshold.** Option F establishes AI connection; significance is assessed separately in Chapter 4. This preserves the clean separation between §3.5 (AI connection) and Chapter 4 (significance assessment).
4. **Eligibility by Event Type and document class, not domain.** Option F restricts to Policy events (Event Type, determined at Stage 1) evidenced by defence appropriation documents (document class). Domain classification is an independent Stage 4 assessment. This is consistent with how Options D, E, and PIET scope their eligibility — none use domain as a restriction.
5. **Economy-agnostic application.** The pathway applies uniformly. Nations with EMBEDDED AI budget granularity do not qualify. This accurately reflects the documentation landscape rather than imposing a jurisdictional restriction. Economies whose defence budget architecture operates at structural opacity levels consistent with the documentation asymmetry documented in §3.15 and Manual §4.11 — currently, no Eastern Bloc defence budget document surveyed produces granularity above EMBEDDED — are not excluded by jurisdictional rule but by the same documentation landscape constraint that the source accommodation framework addresses elsewhere in the methodology.

Confidence Assessment: MEDIUM-HIGH (65–75%). The driving constraint — classification rendering post-implementation verification impossible — is empirically universal and not subject to calibration uncertainty. The pass criteria are calibrated against verified document formats across five economies. The main residual uncertainty is whether the IDENTIFIABLE boundary will produce consistent classification across future edge cases where the AI designation in a shared funding pool is ambiguous.

Validation Commitment: Review pass criteria against the first five events qualifying via Option F, with specific attention to IDENTIFIABLE-granularity edge cases.

3.6.4 Option G — Supply-Chain Response Evidence

Manual Section: §3.5, Option G

Category: ADAPTED (72% confidence)

Source methodology: PIET (§3.5.1) — structural template for qualifying events where official instrument text contains no AI-specific language but the AI infrastructure connection is real and documentable.

Adaptation rationale: PIET addresses restriction events (export controls, materials bans) where the AI connection manifests through post-implementation trade flow disruption. Option G addresses the symmetric case: response events (minerals strategies, subsidy frameworks, stockpile legislation, supply-chain security acts) where the AI connection manifests through the controlled material's documented role in AI infrastructure manufacturing. Both pathways solve the same structural problem — instrument text that addresses the material, not the downstream application — through different evidence mechanisms appropriate to their event class.

PIET requires 12–18 month post-implementation evidence because restriction events need empirical trade flow data to confirm impact. Option G does not require a waiting period because the AI infrastructure relevance of the controlled material is established through pre-existing materials science and supply-chain documentation, not through post-implementation observation. Gallium's role in GaN power semiconductors, germanium's role in fibre optic amplification, and rare earth elements' role in permanent magnets are documented physical facts, not empirical claims requiring post-implementation verification.

Key design decisions:

1. **Two-condition structure (AI infrastructure input + instrument text absence).** Condition 1 establishes that the controlled material is genuinely AI-infrastructure-relevant. Condition 2 confirms that Option G is the appropriate pathway — that the event cannot qualify through the stricter text-based Options A, B, or C. This mirrors PIET's eligible event class structure: event type (Policy), event subtype (response/security rather than restriction), controlled items (AI infrastructure inputs), and prior option failure.
2. **AI-Critical Materials Register as convenience list, not exhaustive boundary.** The register pre-establishes AI infrastructure relevance for common materials, avoiding redundant causal-chain documentation in each processing cycle. Materials not on the register can still qualify through individual documentation under Condition 1(a)+(b). This balances processing efficiency with scope control.
3. **Scope control through instrument specificity.** Option G requires the instrument to be specifically directed at the controlled material — not a general industrial policy that incidentally covers AI-relevant materials. This prevents over-broadening (e.g., a national infrastructure package with a minerals chapter should not qualify the entire package). The specificity standard mirrors Option A's $\geq 10\%$ threshold logic: only the AI-infrastructure-relevant component qualifies.
4. **No post-implementation waiting period.** Unlike PIET, Option G's evidence base is pre-existing scientific and technical documentation. The AI infrastructure application of gallium, germanium, or rare earth elements does not require post-implementation observation to confirm — it is documented in materials science literature, manufacturer specifications, and supply-chain analyses from recognised institutions.
5. **Symmetric treatment of restriction and response.** PIET qualifies restriction events; Option G qualifies response events. Together they provide symmetric coverage of both sides of supply-chain coordination dynamics — which aligns with WEO's stated objective of tracking coordination dynamics across blocs without systematic bias towards one side's instruments.

Coordination Connection implications: Supply-chain response events qualifying through Option G enter the database where they can be assessed for Coordination Connections under standard CC methodology (§9). Candidate connection types include CC Type 5 (Response) where explicit trigger identification exists in official statements, and CC Type 7 (Parallel Policy) where convergent policy action across jurisdictions is documented.

Evidence basis: GAP-2026-05-13-001 documented 11 blocked candidates across three policy domains (critical minerals, energy-AI policy, minerals coordination). The gap was identified when supply-chain response candidates were assessed against the AI Connection Filter and systematically failed Options A, B, and C due to instrument text addressing the controlled material rather than AI. Materials-to-AI-infrastructure causal chains are documented in USGS Mineral Commodity Summaries, BGS Critical Minerals Strategy, EU CRMA legislative history (Regulation 2024/1252 recitals), CSET semiconductor supply chain analysis, and manufacturer technical documentation.

Confidence: 72%. Adapted from PIET (§3.5.1), which is published methodology rated MIXED with components at MEDIUM-HIGH confidence (60–75%) per Rationale §3.7. The adaptation is bounded and structurally parallel: same problem (instrument text does not reference AI), different evidence mechanism (pre-existing documentation vs post-implementation observation), same outcome (AI Connection established at primary source standard). Confidence is bounded below 80% because: (a) the AI-Critical Materials Register is novel and untested — the register's scope may need adjustment after first application; (b) the "instrument specificity" scope control has not been tested against edge cases. Confidence is held above 65% because: (a) direct adaptation from published methodology; (b) materials-to-AI-infrastructure causal chains are well-documented in primary sources; (c) the symmetric treatment principle is methodologically sound; (d) 11 blocked candidates demonstrate the gap is real and consequential.

Validation commitment: Monitor Option G applications for scope creep. If events qualifying through Option G are systematically tangential to AI infrastructure coordination (general industrial policy presented as minerals strategy), tighten Condition 1 or add a third condition. Review the AI-Critical Materials Register at each methodology version milestone.

3.7 PIET — Post-Implementation Evidence Track

Manual Section: §3.5.1

Category: MIXED — components vary; see sub-sections

Rationale: PIET addresses a bounded documentation gap for Policy restriction events on physical AI-critical materials or physical technology products where official instruments use no AI-specific language. The mechanism is grounded in the WTO de facto discrimination doctrine — the principle that facially neutral measures are classified by their demonstrated effects — and adapts this to the Stage 2 AI connection question. All calibration thresholds are WEO-original.

3.7.1 Gate Architecture and 3-of-4 Pass Logic

Manual Section: §3.5.1 — Pass Logic

Category: CATEGORY CREATOR (MEDIUM-HIGH)

Rationale: No established international framework uses explicit N-of-M partial pass logic — confirmed across ten surveyed frameworks (Basel RCAP, IOSCO, ICD 203, UK PHIA, EMA MCDA, HACCP, Moody's/S&P, Wassenaar, WTO DSU, UN Sanctions). FATF's 4th/5th Round Mutual Evaluation Procedures demonstrate threshold-based multi-criteria assessment as a recognised governance design principle, but FATF's ALL-of-N structure is the structural inverse of PIET's N-of-M. The 3-of-4 threshold is WEO-original: N=1 risks false positives; N≥3 provides independent corroboration whilst tolerating one structurally unavailable gate.

Precedent	Contribution
FATF 4th/5th Round Mutual Evaluation Procedures	Threshold-based multi-criteria assessment as recognised governance design principle; ALL-of-N contrast
Survey of ten frameworks	Confirmed absence of N-of-M pass logic across all surveyed frameworks

Confidence Assessment: MEDIUM-HIGH. **Validation Commitment:** Review against cumulative PIET event data.

3.7.2 Trade Flow and Price Thresholds

Manual Section: §3.5.1 — Gates 1–2

Category: CATEGORY CREATOR (MEDIUM-HIGH) for both

Rationale: No institutional framework defines percentage disruption levels from export restrictions — confirmed across WTO, OECD, BIS/EAR, Wassenaar, OFAC, EU, and IMF. The >50%/>90% trade flow thresholds are calibrated as conservative detection floors. The >20% price threshold derives from composite reasoning: Cashin & McDermott (2002, *IMF Staff Papers*) document annual $\sigma = 12.6\%$ for real industrial commodity prices (placing 20% at $\sim 1.6\sigma$); the LME 15% daily circuit breaker — introduced as emergency interim measures during the March 2022 nickel crisis; 15% daily price limit formally designated permanent in March 2023 following the Oliver Wyman Independent Review — provides institutional reference for disorderly price movement; IMF WP/2011/283 confirms no canonical threshold exists.

Precedent	Contribution
Cashin & McDermott (2002, <i>IMF Staff Papers</i>)	Annual $\sigma = 12.6\%$; $20\% \approx 1.6\sigma$
LME 15% daily circuit breaker (introduced as emergency interim measures during the March 2022 nickel crisis; 15% daily price limit formally designated permanent in March 2023 following the Oliver Wyman Independent Review)	Institutional reference for disorderly price movement
IMF WP/2011/283	Confirmed absence of canonical threshold

Confidence Assessment: MEDIUM-HIGH. **Validation Commitment:** Review against each qualifying PIET event; thin-market sensitivity analysis.

3.7.3 Assessment Windows

Manual Section: §3.5.1 — Pending Designation and Gates 3–4

Category: 12-month: WEO SYNTHESIS (MEDIUM-HIGH); 18-month: ADAPTED (MEDIUM-HIGH)

Rationale: The 12-month floor synthesises: WTO Safeguards Agreement Article 7.4 (one-year liberalisation threshold); IMF Article IV consultation cycle; IMF WP/24/77 (mineral commodity adjustment 18–24 months). The 18-month outer bound adapts the SEC 10-K cycle: maximum ~14–15 month lag from fiscal year start (SEC Rules 33-8128/33-8644). **Known structural limitation:** Enacted standalone statutory legislation typically requires 24–60+ months — documented outer-bound exception.

Precedent	Application
WTO Safeguards Agreement, Article 7.4	1-year progressive liberalisation threshold
IMF Article IV / WP/24/77	12-month review cycle; 18–24 month adjustment for mineral commodities
SEC Rules 33-8128/33-8644	10-K mechanics — maximum ~14–15 month lag
WTO DS431	Government response timing corroboration

Confidence Assessment: MEDIUM-HIGH for both windows.

3.7.4 T2 Impact-Demonstrated Mechanism

Manual Section: §5.5 — T2 Restriction Test: Impact-Demonstrated Pathway

Category: WEO SYNTHESIS (MEDIUM-HIGH)

Rationale: Draws on the WTO de facto discrimination doctrine. *Japan–Alcoholic Beverages II* (WT/DS8/AB/R, 1996) originated the "design, architecture, and structure" framework under GATT Article III:2 — whether a measure affords protection is determined objectively from its design, architecture, and structure, not stated intent. *US–Clove Cigarettes* (WT/DS406, 2012) extended this to TBT Article 2.1. *Canada–Autos* (WT/DS139/AB/R) confirmed de facto discrimination under Article I:1. PIET adapts this from product-level trade discrimination to state-level strategic targeting: Gate 3 government response patterns serve as the effects evidence. The domain gap is acknowledged, correctly limiting confidence to MEDIUM-HIGH.

Precedent	Application
<i>Japan–Alcoholic Beverages II</i> (WT/DS8/AB/R, 1996)	Originated "design, architecture, and structure" framework under GATT III:2
<i>US–Clove Cigarettes</i> (WT/DS406, 2012)	Extended framework to TBT Article 2.1
<i>Canada–Autos</i> (WT/DS139/AB/R)	De facto discrimination — Article I:1; facially neutral measures by demonstrated effects

Confidence Assessment: MEDIUM-HIGH. **Validation Commitment:** Review as T2 Impact-Demonstrated events accumulate.

3.8 OEP — Operational Evidence Protocol

Manual Section: §3.5, Stage 2 Option D

Category: MIXED — components vary; see sub-sections

Rationale: OEP resolves Stage 2 qualification gaps for Infrastructure events where official primary documentation is absent, generic, or application-agnostic. The single-pathway pass logic reflects the binary nature of the AI connection question: one confirmed pathway is sufficient to establish that a facility serves AI workloads.

3.8.1 Pathway A — Technical Output Analysis

Manual Section: §3.5, Option D — Pathway A

Category: EMPIRICALLY GROUNDED (HIGH)

Rationale: Chip decapsulation is direct physical measurement of fabricated silicon — observational methodology, not inference. CSET/ETO's institutional data reliance and US government response to the September 2023 Kirin 9000S analysis confirm TechInsights' de facto primary source status. China's Commerce Ministry Unreliable Entity designation (October 2025) constitutes indirect adversary-government confirmation. **Coverage limitation:** TechInsights is the sole provider at this standard; the designation may impair access to Chinese chips with no identified substitute.

Precedent	Application
CSET/ETO dataset (eto.tech/dataset-docs/chipexplorer)	Direct institutional data reliance — US government-funded research
TechInsights Kirin 9000S analysis (September 2023)	Calibration event; BIS investigation trigger
BIS ECCN 3A090 (87 FR 62,186 and subsequent)	AI chip definition anchor

Confidence Assessment: HIGH.

3.8.2 Pathway B — Regulatory Designation Logic

Manual Section: §3.5, Option D — Pathway B

Category: ADAPTED (HIGH)

Rationale: Adapts BIS AI-enabling product designation logic to Stage 2 AI connection. The foundational instrument is 87 FR 62,186 (October 2022), creating ECCN 3A090 and listing AI capability denial as one of four stated rationales. All subsequent BIS rulemakings cite this instrument. Adaptation is explicit: where BIS uses product-type designation for export control jurisdiction, OEP uses it for AI connection. PROBABLE confidence reflects the inferential step — facility-level AI allocation inferred from product type, consistent with IC analytic practice under ICD 203 (ODNI, 2015). Regulatory currency check required: 90 FR 4,544 was rescinded May 2025.

Precedent	Application
87 FR 62,186 (October 2022)	Foundational instrument — ECCN 3A090; AI capability denial rationale
89 FR 96,790 (December 2024)	HBM addition; BIS AI-enabling language
IC analytic practice under ICD 203 (ODNI, 2015)	PROBABLE confidence basis

Confidence Assessment: HIGH.

3.8.3 Pathway C — Hardware Deployment Evidence

Manual Section: §3.5, Option D — Pathway C

Category: WEO SYNTHESIS (MEDIUM-HIGH) for rack density; EMPIRICALLY GROUNDED (HIGH) for MIIT framework; CATEGORY CREATOR (PROVISIONAL) (MEDIUM-HIGH) for 10% threshold

Rationale: No formal AI rack density classification standard exists. The ≥ 30 kW threshold reflects the air cooling upper limit (ASHRAE TC 9.9 H1); ≥ 60 kW is calibrated to 2022–2025 AI training configurations (NVIDIA DGX H100/H200 ~40.8 kW/rack; GB200 NVL72 ~120 kW/rack). Uptime Institute 2024 confirms both thresholds substantially exceed deployment norms. **These hardware references represent the calibration basis at time of V1.3 publication (April 2026) and are subject to the hardware generation review commitment documented in Manual §3.5, Pathway C. Hardware generation note:** GB200 NVL72's 2:1 GPU-to-CPU ratio generates false negatives for the $\geq 4:1$ indicator.

The MIIT framework is established by 工信部联通信〔2023〕180号 (October 2023, six joint ministries, gov.cn). NDA and MIIT have both cited 智能算力 at 90 EFLOPS (FP32 precision, 换算为 FP32) at end-2024 as a separate national statistical category. **Regulatory currency note:** MIIT's 智算中心 / 智能算力 designation framework is subject to ongoing regulatory evolution; currency of the governing instrument should be verified at the time of OEP assessment.

The 10% threshold has no external precedent. Three structural analogies: EAR de minimis rule (15 CFR §734.4); USMCA Chapter 4 RVC (19 U.S.C. §4531); USDA Organic Certification (7 CFR Part 205). PROVISIONAL status retained.

Precedent	Application
ASHRAE TC 9.9 H1; Uptime Institute 2024	Air cooling upper limit; confirms ≥ 30 kW and ≥ 60 kW exceed deployment norms
NVIDIA DGX H100/H200; GB200 NVL72 specifications	Empirical calibration for rack density and GPU-to-CPU indicators
工信部联通信〔2023〕180号 (October 2023)	Formal MIIT tripartite classification framework
EAR de minimis (15 CFR §734.4); USMCA RVC (19 U.S.C. §4531); USDA Organic (7 CFR Part 205)	Structural analogies for percentage-of-composition classification

Confidence Assessment: MEDIUM-HIGH (rack density); HIGH (MIIT); MEDIUM-HIGH PROVISIONAL (10% threshold). **Validation Commitment:** Hardware generation review at major GPU platform transitions; 10% threshold reviewed at 10-event OEP Pathway C milestone.

3.8.4 Confidence Gradation (STRONG/PROBABLE)

Manual Section: §3.5, Option D — Pass Logic
Category: WEO SYNTHESIS (MEDIUM-HIGH)

Rationale: Established prior art for evidence-to-confidence mapping includes ICD 203 (ODNI, January 2015, Section D.6.e.(2)), which provides the conceptual basis for confidence as a function of evidence quality and quantity — operationalised in IC assessment practice (2007 NIE on Iran; 2017 ICA on Russian Activities). IPCC uses five confidence levels; NATO STANAG 2511 uses a 6×6 reliability-credibility matrix. The specific two-level implementation (STRONG/PROBABLE) is WEO-original, suited to OEP's binary classification context.

Precedent	Application
ICD 203 (ODNI, January 2015), Section D.6.e.(2)	Conceptual basis — confidence as function of evidence quality and quantity
2007 NIE; 2017 ICA	IC assessment practice operationalising ICD 203
IPCC five levels; NATO STANAG 2511	Prior art for multi-level confidence gradation

Combination Rules Derivation: The combination rules follow from ICD 203's evidence quality primacy principle (Section D.6.e.(2)): confidence reflects evidence quality, not evidence quantity. Two PROBABLE pathways do not aggregate to STRONG because cumulative inferential evidence remains inferential — the AI connection question is binary, and convergent inference about a binary fact does not become direct observation. One STRONG pathway raises overall confidence to STRONG regardless of additional pathways because observational evidence is established. This design prevents confidence inflation through accumulation of weaker evidence types, consistent with IC analytic practice under ICD 203.

Confidence Assessment: MEDIUM-HIGH.

3.9 6-Month Verification Lag

Manual Section: 3.9

Category: STANDARD PRACTICE

Rationale: The 6-month lag trades recency for verification quality, aligning with established practice across multiple domains:

Domain	Standard Lag	Rationale
Financial infrastructure	90-180 days	Post-implementation credit ratings
Physical infrastructure	6-12 months	Operational history requirement
Policy/regulatory	6-12 months	Impact assessment period
Academic/think tank	6-12 months	Standard publication lag (OECD, World Bank)

3.10 Currency Windows

Manual Section: 3.4.8

Category: PRECEDENT-BASED (HIGH)

Rationale: Currency windows govern source publication dates (distinct from event implementation dates which are governed by the 6-month verification lag). These follow established financial reporting standards and regulatory requirements:

Data Type	Currency Window	Precedent
Infrastructure operational status	≤24 months	Financial reporting standards
Policy/regulatory status	≤12 months	SEC/Basel reporting requirements
Financial data	≤12 months	SEC/Basel reporting requirements
Technical specifications	≤36 months	Technical documentation practices

Interaction with 6-Month Verification Lag: Both requirements must be satisfied:

1. **Event implementation date:** Within verification window (Jan 2022 to 6 months prior to analysis)
2. **Source publication date:** Within currency window for data type

A source published today reporting on infrastructure completed 2 years ago is valid — the methodology intends current verification of established reality.

3.10.1 Conflict-Era Source Currency Adjustment

Manual Section: §3.4.12

Category: ADAPTED (MEDIUM-HIGH to HIGH — varies by source type; see below)

Rationale: Armed conflict produces a documented bifurcation in institutional documentation output. Emergency instruments accelerate to revision cycles measured in weeks; routine regulatory instruments decelerate or halt; infrastructure verification shifts from near-real-time documentary evidence to satellite imagery with inherent temporal lag; and financial reporting experiences filing delays proportionate to sanctions disruption and audit access constraints. These patterns are empirically documented across multiple conflict episodes and institutional frameworks.

The adjustment adapts the existing §3.4.8 currency windows to account for this bifurcation. It does not create a new evidentiary framework — it modifies temporal freshness parameters within the established source verification architecture.

Trigger framework:

The conflict-era trigger adapts the World Bank Fragile and Conflict-Affected Situations (FCS) dual-source classification methodology for WEO's documentation-impact purpose. The FCS framework — the most formally codified conflict classification in use — requires quantitative thresholds from both ACLED and UCDP before classifying a country as conflict-affected. WEO adapts this dual-gate principle: criterion 1 provides the external conflict classification (UCDP), whilst criterion 2 establishes the documentation-impact connection that justifies adjusting currency windows.

Precedent	Contribution
World Bank FCS (dual-source quantitative classification, FY24 methodology)	Structural precedent — external quantitative threshold + impact verification before classification activates
UCDP Armed Conflict Dataset (Gleditsch et al., 2002; codebook v.25.1)	Criterion 1 — externally maintained, quantified conflict classification with defined intensity levels
IMF World Economic Outlook database (ad hoc data exclusion, Syria from 2011)	Documented institutional response to conflict-era documentation gaps — validates the premise that conflict degrades documentation landscapes

The criterion 2 sub-criteria are not uniformly quantified. Sub-criterion (a) provides a hard threshold ($\geq 10\%$ chokepoint volume); sub-criteria (b) and (c) are qualitative assessments constrained by the evidence requirement in criterion 3. This asymmetry is documented honestly: no systematic quantitative measurement of regulatory publication rate depression during conflict exists in academic literature. The absence of this measurement was confirmed during deep research and represents an unfilled gap suitable for future research. The current calibration — qualitative assessment with mandatory documented evidence — is the methodologically honest response to that gap.

Adjusted currency ceilings — empirical basis by source type:

Source Type	Adjustment	Category	Confidence	Key Evidence
Emergency executive instruments	≤12mo → ≤6mo	EMPIRICALLY GROUNDED	~80%	EU sanctions: 6 packages in 101 days (17-day cycle, 2022); UNSC Gulf War: 12 resolutions in 153 days (12.7-day cycle, 1990); EPAA: major revision every 12–18 months
Routine regulatory instruments	≤12mo → ≤24mo	ADAPTED	~70%	Ukraine Business Ombudsman bill stalled; IMF Syria data exclusion from 2011; aggregate deceleration documented but not systematically measured
Infrastructure operational status	≤24mo → ≤36mo	ADAPTED	~72%	Kuwait Stock Exchange: 18–20 month closure; Iraq Stock Exchange: ~15 month closure; satellite verification lag weeks–months; ICRC "data vacuum" finding
Financial data	≤12mo → ≤18mo	ADAPTED	~70%	SEC Release 34-88465 (45-day COVID extension); Russia audit firm substitution delays; EU ban on accounting services to Russian entities (June 2022)
Technical specifications	Unchanged	N/A	N/A	No evidence of accelerated technical obsolescence during conflict

All adjusted ceilings exceed the 65% methodology component confidence threshold.

Key design decisions:

1. **Ceilings, not floors.** The adjustment modifies the maximum permissible source age, not the minimum. A newer source is always valid. This is consistent with the §3.4.8 design where the windows are upper bounds.
2. **6-month emergency instrument ceiling (not 3 months).** The observed emergency revision cycles are 2–8 weeks, which would support a ceiling as short as 3 months. The 6-month ceiling provides a conservative buffer. The specific calibration between 3 and 6 months is flagged for refinement pending analysis of EU sanctions revision content — whether successive packages substantively modified predecessors or merely extended them.
3. **CC temporal windows excluded.** Coordination Connection temporal windows (§9.4.1) govern the timing relationship between events, not source freshness. Conflict may accelerate coordination activity, but the temporal windows measure when events occurred relative to each other — this relationship is independent of how old the source documentation is.
4. **Territory-scoped application.** Adjusted ceilings apply to sources produced within or about the conflict-affected territory, not globally. This prevents a conflict in one region from loosening currency standards for sources from unaffected jurisdictions.
5. **Six-month review cycle.** Active determinations are reviewed at six-month intervals — long enough to avoid administrative churn, short enough to deactivate determinations when conditions change. No external precedent governs this interval; it is a CATEGORY CREATOR calibration.

Confidence Assessment: MEDIUM-HIGH to HIGH overall (70–80%). Emergency instrument shortening is the strongest empirically grounded component (~80%); routine instrument and infrastructure extensions are ADAPTED with moderate confidence (~70–72%); financial data extension is ADAPTED (~70%). The trigger framework is ADAPTED from the World Bank FCS methodology with MEDIUM-HIGH confidence (70–75%). The six-month review interval is CATEGORY CREATOR.

Validation Commitment: Review adjusted ceilings against the first conflict-era determination. Monitor whether the 6-month emergency ceiling proves too permissive or too restrictive. Review trigger threshold if UCDP revises its intensity classification methodology. Monitor academic literature for systematic regulatory publication rate measurement during conflict — if published, use to recalibrate the routine instrument extension.

3.11 Non-English Source Verification

Manual Section: 3.3

Category: EMPIRICALLY GROUNDED (HIGH)

Rationale: Enhanced verification protocols apply to non-English source events due to:

1. **Language barriers:** WEO's predominantly English-language research base affects efficient navigation and filtering of primary sources in other languages
2. **Translation verification:** Non-English documentation requires additional corroboration to ensure accurate interpretation
3. **Verification standards:** Intelligence community multi-source validation precedent

These protocols apply to any event where primary sources are not in English, regardless of bloc — including Chinese, Russian, French, German, Japanese, Arabic, and other non-English documentation.

Primary Application — Chinese-Language Sources:

Chinese-language sources represent the most common non-English verification scenario for AI infrastructure events.

- Official State Council documents (english.gov.cn) OR Ministry-level announcements (MIIT, CAC, NDRC) AND
- Third-party confirmation from: World Bank/ADB project databases, UN agency reports, peer-reviewed academic papers, or trade statistics cross-check (China Customs + UN Comtrade)

Infrastructure Claims: State announcement + satellite imagery OR budget allocation + construction permits OR vendor contracts + operational evidence

Financial Claims: Official announcement + IMF/World Bank country data OR China Customs + UN Comtrade cross-verification

Other Non-English Sources:

The same enhanced verification principles apply to all non-English sources regardless of bloc:

- Official government sources in original language + English-language confirmation OR
- Academic research with primary source citations + multilateral organisation data OR
- Trade statistics + third-party reporting from internationally recognised outlets

Precedent Citations:

- ICD 203/206 (Intelligence Community Directive — multi-source validation)
- UK PHIA (Professional Head of Intelligence Assessment)
- OSINT Guidelines 2023
- Jane's methodology (defence intelligence)

3.11.1 CJK Language Conventions

Manual Section: §3.3.1

Category: WEO SYNTHESIS (75% confidence)

Rationale: The four-pattern taxonomy addresses inconsistencies identified during event processing where CJK text appeared without a governing standard — truncated operative terms, inverted ordering (CJK-first vs English-first), and commercial brand names with no methodology-functional role. The taxonomy provides a testable rule for each use case rather than relying on editorial judgement.

Pattern A (verbatim citation) preserves evidentiary integrity for CC qualification. Pattern B (methodology-functional terms) retains operative CJK where removal would degrade analytical precision — particularly for Chinese regulatory citation markers (根据, 关于, 印发) that carry specific legal meaning in CC evidence chains. Pattern C (named programmes) follows standard English-language academic convention for foreign-language proper nouns. Pattern D (direct quantities) preserves source fidelity for numeric evidence.

The methodology-functional test provides a binary decision rule that prevents CJK text from accumulating without analytical purpose whilst preserving it where it serves the methodology.

Confidence: 75%. The taxonomy is grounded in established multilingual documentation practice (ISO 704 terminology principles, academic transliteration conventions) adapted for WEO's specific processing context. Applied retrospectively to existing database entries with consistent results.

Validation commitment: Monitor for CJK patterns that do not fit the four categories. If a fifth pattern emerges from processing, extend the taxonomy. Review at methodology version milestones.

3.12 Bloc Classification

Manual Section: §3.10 (Bloc Classification)

Category: WEO SYNTHESIS (80% confidence)

Rationale: V1.3 §3.12 combined two distinct classification processes — macro-level bloc membership and event-level Non-Aligned attribution — in a single section without explicit separation. This created interpretive ambiguity: the three-priority hierarchy (with financial backing at Priority 1) could be mistaken for the method by which macro-level register classification decisions are made, when in fact macro-level classification is determined by technology ecosystem alignment.

The two processes answer different questions at different analytical scales. Macro-level classification asks "which ecosystem does this nation's AI infrastructure structurally depend on?" — technology ecosystem alignment is the correct primary signal because it captures structural lock-in (switching a chip supply chain takes years and billions of dollars; switching a funding source for a single project takes months). Event-level attribution asks "for this specific event, which bloc's ecosystem does it operate within?" — financial backing is the correct Priority 1 because at the individual event level it provides an objective, measurable, reproducible signal that can be calculated as a percentage. Technology determination (Priority 2) requires more interpretive judgement, so it serves as the fallback when money does not resolve the attribution.

The priority ordering is not contradictory — it is appropriate to each analytical scale. The separation makes this explicit.

Confidence: HIGH (80%). The underlying analytical processes are operationally stable. No classification outcome changes.

3.12.1 Universal Membership Categories

Manual Section: §3.10.1 (category definitions)

Category: ADAPTED (75% confidence)

Source methodology: Systematic survey of eight international tiered classification frameworks: NATO partnership tiers, EU accession framework, FATF grey/black list classification, OECD accession framework, Basel Committee RCAP, Shanghai Cooperation Organisation membership tiers, Non-Aligned Movement (historical), and contemporary technology alliance frameworks (Pax Silica, Chip 4, AISI Network, Wassenaar Arrangement).

Adaptation rationale: No existing framework offers a symmetric, universal set of membership categories applied identically to competing technology ecosystems. Every surveyed operational framework is anchored by one hegemon, with rivals as residual or excluded. The five-category universal structure (Core / Integrated / Engaged / Peripheral / Non-Aligned) is WEO's synthesis of transferable design primitives from these frameworks, adapted for technology ecosystem alignment classification.

Key design decisions:

1. **Five-category count.** The modal tier count across the eight surveyed frameworks is five $\pm$ one. NATO functionally operates with three sub-member tiers plus the member tier; OECD maintains five tiers; EU has four substantive tiers. Five categories provides sufficient discrimination to capture the spectrum from structural embedding (US in Western ecosystem, 7/7 SCP dimensions as of May 2026) to single-vector connection (e.g., governance participation without infrastructure dependency) without classification noise from finer subdivision.
2. **Strength over mechanism.** Categories describe alignment strength (how deeply integrated), not alignment mechanism (how the nation entered the ecosystem). This follows Basel RCAP's design — the only surveyed framework that cleanly separates membership status from compliance assessment. The mechanism (formal agreement, voluntary technology choice, infrastructure lock-in, political alliance, governance participation) is recorded as a descriptive attribute within each nation's register entry. This resolves the V1.3 asymmetry where Western categories described mechanisms (Conditional, Functional, Governance) and Eastern categories described dependency types (Aligned, Dependent, Leaning).
3. **Symmetry.** Identical definitions, indicators, and distinguishing criteria apply to the Western Bloc, Eastern Bloc, and any future bloc. No category definition references bloc-specific institutions, values, or political systems. This is a genuine methodological contribution — no surveyed operational framework achieves this level of symmetry across competing blocs. The FATF methodology is symmetric by design but asymmetric in application (no G7 grey-listings despite documented deficiencies); the WEO structure is symmetric by both design and definition.
4. **Non-Aligned as a dynamic status.** Non-Aligned is defined by the absence of predominant bloc attribution, verified against observable indicators, not by political declaration. This addresses the Non-Aligned Movement's historically demonstrated failure mode, where formal non-alignment masked operational alignment (Cuba, India-USSR 1971, Vietnam). The dual assessment (declared policy + observable dependency) follows the FATF dual-axis precedent (technical compliance + effectiveness).

Precedent sources:

Precedent	Institution	Design primitive adopted
RCAP four-grade compliance scale	Basel Committee (BIS)	Separation of mechanism from strength; weakest-link aggregation
EU accession framework (2020 revised methodology)	European Union	Phased, proportionate, reversible transitions; cluster organisation
FATF mutual evaluation methodology	FATF	Dual-axis assessment (declared + de facto); action-plan escalation
OECD accession framework	OECD	Graduation pipeline; values filter; Key Partner vs Candidate distinction
NATO partnership tiers	NATO	Three-tier sub-member structure (MAP, EOP, PfP)
Non-Aligned Movement (1961–present)	NAM	Cautionary precedent: declared vs de facto alignment

Confidence: 75%. The five-category structure is grounded in systematic precedent survey across eight frameworks, with convergent evidence supporting the tier count, strength-over-mechanism design, and symmetry principle. The specific indicators for each category involve analytical judgement that will be refined through continued application. The 65% minimum confidence threshold (§2.3) is met.

Validation commitment: Reassess category structure at methodology version milestones. Monitor whether the five-category discrimination remains adequate as the bloc landscape evolves (particularly if a third bloc forms). Review indicator sets after three register snapshots to identify any systematic classification difficulties.

3.12.1a Bloc Assignment Gate

Manual Section: §3.10.1 (Bloc Assignment Gate subsection)

Category: ADAPTED (75% confidence)

Rationale: The two-track bloc assignment gate fills the architectural gap between the general classification principle ("every nation is classified by which ecosystem its AI infrastructure primarily operates within") and the within-bloc category assignment. In V1.3, Non-Aligned actors had a formalised identification mechanism (the primary partner test: financial backing → technology transfer → strategic initiative leadership), but actors assigned to the Western or Eastern blocs had no equivalently formalised assignment process. The bloc assignment was qualitatively sound but procedurally implicit.

Track 1 formalises the observable infrastructure alignment assessment that was already being applied in practice. The three indicators (hardware origin, compute/model platforms, regulatory alignment) are the same signals that informed every existing Bloc Membership Register entry. Track 2 extends the assessment to the full seven-dimension dependency direction analysis for contested cases where the observable indicators are ambiguous or contradictory. The seven-dimension framework draws directly from the Sovereign Capability Profiles (§5.3.2), applied here as a dependency direction test rather than a capability assessment — asking "whose ecosystem does this actor draw on?" rather than "what does this actor independently control?"

The dead heat rule (exact dependency split = Non-Aligned) follows from the operational definition of Non-Aligned status: the absence of predominant alignment to either ecosystem. If a structured dependency analysis across all seven dimensions produces an exact split, the actor has demonstrated through its infrastructure choices that it is not structurally embedded in either ecosystem — which is precisely what Non-Aligned status describes. Event-level attribution for such actors uses the existing three-priority hierarchy (§3.12.3).

The near-dead-heat rule (narrow majority constrains category to Peripheral or Engaged) ensures that the category assignment reflects the strength of structural embedding. An actor with a 4–3 dependency split has a measurably weaker ecosystem attachment than an actor with a 7–0 or 6–1 split. Assigning such an actor to Core or Integrated would overstate the depth of its embedding.

Closest Precedents:

Framework	Relevance	Divergence from WEO
EU SWD(2021) 352 Strategic Dependencies	Single-source dependency identification across ~5,000 products using HHI concentration metrics	Product-level, not actor-level; measures import concentration, not ecosystem alignment
IISS Cyber Capabilities (2021/2023)	Multi-dimensional capability assessment producing tiered classifications	Does not assess ecosystem dependency direction; classifies capability, not alignment
NATO Partnership Tiers	Graduated alliance membership based on integration depth	Security alliance, not technology ecosystem; membership is treaty-based, not infrastructure-based
FATF Grey/Black List	Classification based on observable compliance indicators with dual assessment (declared vs actual)	Financial compliance, not technology alignment; binary pass/fail rather than graduated categories

Key Design Decisions:

1. **Two-track sequential system rather than universal seven-dimension analysis.** Track 1 resolves the vast majority of registered nations through observable indicators that do not require a full SCP-style dependency analysis. Track 2 is reserved for a small minority of genuinely contested cases. This keeps the assessment burden proportionate: the full seven-dimension analysis is analytically powerful but resource-intensive, and applying it universally to nations whose alignment is unambiguous would be methodologically unnecessary.
2. **Dead heat = Non-Aligned, not forced assignment.** The alternative — a forced tiebreaker that assigns a perfectly split actor to one bloc — would impose a false precision that the data does not support. An actor whose infrastructure dependencies are exactly evenly distributed between ecosystems is telling the assessor something real about its structural position. Forcing a bloc assignment would obscure this signal. Non-Aligned status with event-level attribution preserves the analytical information.
3. **Dimension durability tiebreaker as conditional empirical heuristic.** The default hierarchy (hardware > platform > governance) is grounded in documented switching costs: semiconductor fabrication and equipment dependencies exhibit switching timelines measured in decades, whilst governance alignment can shift within a single political cycle. However, the foundational lock-in literature (Shapiro & Varian 1999, Table 5.1) demonstrates that hardware lock-in declines with time as assets depreciate whilst information/platform lock-in rises — the hierarchy is empirically observed for frontier AI infrastructure in 2026, not a universal law. The four override conditions ensure the tiebreaker adapts to cases where the default ordering breaks down.

Evidence Basis:

Source	Type	Contribution
Shapiro, C. & Varian, H. R. (1999). <i>Information Rules</i> , Ch. 5, Table 5.1	Direct	Canonical seven-category lock-in taxonomy; documents time-dynamics showing hardware lock-in declining and information lock-in rising — grounds the conditional framing
Allen, G. C., et al. (2024). "The True Impact of Allied Export Controls." CSIS	Direct	Documents ~10-year minimum Chinese EUV substitution gap despite US\$150B+ investment — direct evidence that frontier hardware switching costs exceed platform/governance alternatives
Farrell, J. & Klemperer, P. (2007). "Coordination and Lock-In." <i>Handbook of Industrial Organization</i> , Vol. 3, Ch. 31	Direct	Canonical survey distinguishing switching costs from network effects; does not propose a strict hierarchy by magnitude — reinforces the need for conditional framing
UK OBR Brexit Analysis (2020–2025)	Empirical	Institutional gravity: 5+ years, 4% GDP impact, still unable to meaningfully diverge — grounds the institutional gravity override
GAIA-X / EU cloud sovereignty (2019–2026)	Empirical	Failed platform displacement: 7 years, 43 billion euros adjacent spending, US hyperscaler share held at ~70% — grounds the network-effect platform override
Apple x86→ARM transition (2020–2023)	Empirical	Vertically-integrated hardware switching in 35 months — grounds the vertical integration override
Korea-Japan photoresist dispute (2019–2020)	Empirical	Commodity-input substitution in ~6 months — grounds the commodity-input override

Confidence Assessment: 75%. The two-track system formalises existing practice (Track 1) and extends it with established WEO analytical tools (Track 2) for contested cases. The dead heat and near-dead-heat rules follow logically from the Non-Aligned definition and the graduated category structure. The dimension durability tiebreaker is the least certain element — it is grounded in documented switching costs and three primary academic sources but is explicitly contested by the foundational lock-in literature at long time horizons. Confidence is bounded below 80% because the tiebreaker's four override conditions require assessor judgement about when to apply them. Confidence is above 70% because Track 1 and Track 2 are methodologically straightforward, the dead heat/near-dead-heat rules are deterministic, and the tiebreaker is needed only in edge cases.

Validation Commitment: Monitor the dimension durability hierarchy against observed infrastructure transitions. If frontier hardware switching costs decline below platform switching costs (e.g., through successful domestic EUV programmes or alternative lithography development), revise the default ordering. Review the four override conditions after each Bloc Membership Register update to identify any cases where the conditions were invoked and whether they produced analytically sound results. Reassess the tiebreaker calibration at methodology version milestones.

3.12.1.1 Evidence Standard for Register Classification Changes

Manual Section: §3.10.1.1

Category: ADAPTED (80% confidence)

Source methodology: §5.3.4 Sovereign Capability Profiles evidence standard (1 Primary Grade 1/2 + 1 Corroborating Grade 2/3), adapted for bloc membership classification context.

Adaptation rationale: The SCP evidence standard was designed for dimension-level assessment decisions — determining whether an actor meets a specific capability dimension. Bloc membership classification decisions are analogous: they determine whether a nation meets the indicators for a specific alignment category. The same evidence architecture (primary + corroborating, with defined grade requirements per §3.1) applies. The minimum standard (1+1) is appropriate because bloc classification changes are lower-stakes than SCP dimension assessments — a bloc classification does not carry the same analytical weight as a PAA or AIK designation.

Gap addressed: V1.3 did not specify an evidence standard for register classification changes. The register's own documentation stated it was "evidence-based" and required "source citation," but no section of the Manual defined what quality or quantity of sources was required. This created a circular reference: the Manual delegated to the register, but the register had no formal standard to comply with. §3.12.1.1 closes this gap.

Retrospective audit provision: Snapshot 2025.12.14-001 was produced prior to the formalisation of this standard. The retrospective audit provision acknowledges this honestly without backdating the standard. Source documentation for that snapshot has been supplemented to meet the §3.12.1.1 standard; classifications were not altered.

Confidence: HIGH (80%). Directly adapted from the evidence standard defined in §5.3.4 with minimal modification.

3.12.1.2 Register Operations

Manual Section: §3.10.1.2 (versioning, archival, update triggers)

Category: ADAPTED (85% confidence)

Rationale: The register's operational framework addresses three design decisions: version identification, source preservation, and update cadence.

The version identifier format (YYYY.MM.DD-NNN) provides unambiguous temporal ordering and supports multiple snapshots per day during periods of rapid geopolitical change. This follows standard document management convention adapted for a living register that updates at irregular intervals.

The dual archival strategy (Internet Archive + OpenTimestamps) serves complementary purposes: Internet Archive provides public accessibility and content retrieval, whilst OpenTimestamps provides cryptographic proof of document state at a specific point in time. The combination ensures that any register snapshot can be independently verified as unaltered after publication — a requirement for institutional-grade data integrity.

The event-triggered update model (snapshots taken when geopolitical developments warrant reclassification, not on a fixed calendar) reflects the nature of the underlying data: AI infrastructure ecosystem alignment shifts in response to discrete geopolitical events (export controls, bilateral agreements, infrastructure deployments), not on quarterly or annual cycles. Calendar-triggered snapshots would either capture periods of no change or miss significant developments between scheduled dates.

Confidence: 85%. All three design choices adapt established operational practices (document versioning, digital archival, event-driven data management) to the register context. No novel methodology is introduced.

Validation commitment: N/A — operational format, not analytical methodology.

3.12.1.3 Third Bloc Formation — Transitional Classification

Manual Section: §3.10.1.3

Category: WEO SYNTHESIS (65% confidence)

Rationale: The two-bloc structure (Western, Eastern) reflects the empirical reality at V1.4 publication — only two actors satisfy the PAA qualification threshold (§5.3). However, the methodology must accommodate the possibility of a third bloc forming without requiring architectural revision. The most likely formation scenario involves an actor currently classified within an existing bloc approaching PAA qualification through independent sovereign capability accumulation across multiple SCP dimensions, causing a structural split between the new and original blocs.

The structural problem: If a third bloc forms from within an existing bloc, nations previously classified as Core in the original bloc would not cleanly transfer to the new bloc. Their ecosystem dependencies would be split — some dimensions aligning with the new bloc (e.g. regulation, investment, equipment), others remaining with the original (e.g. chip design, compute, models). This is structurally different from Non-Aligned status, which reflects strategic choice or neutrality. Split-alignment during bloc formation reflects unavoidable structural transition where the infrastructure dependencies themselves are divided.

Transitional mechanism rationale: Reclassifying structurally split nations as Non-Aligned during the transition period — and applying the three-priority hierarchy (§3.12.3) per-event — repurposes an existing, tested mechanism for a new scenario. This avoids the need for a new classification framework whilst providing event-level accuracy. The three-priority test is well-suited because it assesses each event individually against measurable criteria (financial backing, critical technology, strategic leadership), which correctly handles the situation where a nation's events may genuinely belong to different blocs depending on which ecosystem capabilities are utilised.

Stabilisation criterion rationale: "Predominance across the seven SCP dimensions" rather than a simple dimensional count (e.g. 4/7) was chosen because dimensions are not equally weighted in practice. Hardware dependencies (D1–D3) create deeper structural lock-in than regulatory alignment (D6), which can shift with a single legislative act. The stabilisation assessment must consider the structural weight of dependencies, not merely count them. This judgement is left to the analyst rather than algorithmically specified, consistent with the indicator-based assessment approach used throughout §3.10.1.

Category accommodation rationale: During the transition period, the nation is classified as Non-Aligned and events are attributed per the three-priority hierarchy. The analyst tracks the dependency pattern across both blocs to assess when stabilisation occurs. Upon reclassification to a specific bloc, the category level reflects the strength of alignment at that point — a nation emerging from transition is unlikely to qualify as Core immediately, as residual dependencies on the former bloc would typically place it at Engaged or Peripheral. The expectation that alignment will stabilise over time is grounded in the economic logic of supply chain consolidation — maintaining parallel dependencies across two ecosystems is costly and strategically unstable.

Scope limitation rationale: Full multi-bloc methodology (formal dimensional dependency assessment, weighted scoring) is deferred rather than attempted. The transitional mechanism is sufficient for initial classification and avoids over-engineering a solution for a scenario that has not yet occurred. If a third bloc forms, the specific dimensional profile of the new PAA and the specific dependencies of affected nations will inform the full methodology — designing it in the abstract risks producing a framework mismatched to the actual scenario.

Confidence: 65%. This is the minimum confidence threshold (§2.3). The transitional mechanism is logically sound and repurposes tested components (Non-Aligned classification + three-priority test), but it has not been empirically tested against a real third-bloc formation scenario. Confidence reflects this untested status. Reassess when a third PAA candidate approaches qualification.

3.12.2 Non-Aligned Nation Identification

Manual Section: §3.10.2 (five-criteria test)

Category: PRECEDENT-BASED (HIGH, 80% confidence)

Rationale: The five-criteria test provides an auditable, explicit identification mechanism for Non-Aligned status, establishing the relationship between macro-level NA classification and event-level attribution.

Dual assessment provision: The requirement that Non-Aligned status be assessed against both declared policy and observable infrastructure dependency addresses a documented failure mode. The Non-Aligned Movement historically demonstrated that declared non-alignment does not reliably predict actual ecosystem dependency (Cuba maintained non-aligned status whilst operating entirely within the Soviet ecosystem; India signed the Indo-Soviet Treaty of Friendship in 1971 whilst remaining NAM chair). WEO's dual assessment follows the FATF dual-axis methodology (technical compliance + effectiveness) as the closest operational analogue for distinguishing what nations declare from what they do.

Confidence: HIGH (80%). The five-criteria test and the dual assessment provision are operationally stable — no classification outcome changes.

3.12.3 Event-Level Attribution Scope Limitation

Manual Section: §3.10.3 (scope paragraph)

Category: WEO SYNTHESIS (75% confidence)

Rationale: The explicit scope limitation — stating that the three-priority hierarchy applies only to event-level attribution for Non-Aligned nations, not to macro-level register classification — resolves the interpretive ambiguity identified in the V1.3 structure. The rationale for the different priority ordering at each scale is documented in §3.12 and in this rationale entry (§3.12 above).

The three-priority hierarchy operates as documented in the Methodology Rationale (§3.12) and the AI model events clarification (§3.12.4), which remains applicable in full.

Confidence: 75%. The scope limitation formalises what was implicitly true in V1.3 — the three-priority test was always designed for event-level classification, not macro-level register decisions. No operational outcome changes. The risk is that explicit scope limitation could be misread as narrowing the test's applicability; the Manual text addresses this by clearly stating the test's purpose within its defined scope.

3.12.4 AI Model Events — Critical Technology Determination

Manual Section: §3.10.3 (Priority 2 — Critical Technology)

Category: ADAPTED

Source methodology: US Bureau of Industry and Security (BIS), Export Administration Regulations (EAR), 15 CFR Parts 730-774.

Adaptation rationale: BIS EAR establishes two principles relevant to AI model bloc attribution:

1. Development location determines technology origin (15 CFR § 734.3)
2. Cloud computing access does not change underlying technology origin (15 CFR § 734.18)

WEO adapts these principles for bloc attribution: the headquarters jurisdiction of the entity that developed the base model weights determines which bloc provides "critical technology" under Priority 2. Deployment infrastructure (cloud provider, hosting location) is not determinative.

Derivative threshold: The >50% new parameters threshold is WEO-original calibration for determining when derivative work constitutes genuinely new technology rather than adaptation. No direct external precedent — this is a category creator sub-component within an otherwise adapted framework.

Confidence: HIGH (85%) — core attribution principle directly adapted from BIS EAR. Derivative threshold is category creator calibration subject to refinement.

3.13 Event Evolution & Succession Protocol

Manual Section: 3.14

Category: PRECEDENT-BASED (HIGH)

Rationale: The transformation-scaling distinction determines whether evolving events warrant new record creation or version history updates. This institutional-continuity approach aligns with established practice across infrastructure databases, treaty registries, and financial platforms.

Core Principle — Transformation vs Scaling:

Transformation (New Event)	Scaling (Rolling Update)
Legal instrument changes	Same legal instrument, expanded scope
Party composition changes	Same parties, increased commitment
Classification changes (Tier, Bloc, Domain, Type)	Classification stable
Entity splits or merges	Single entity persists
Fresh authorisation required	Existing authorisation covers expansion

Precedent — Industry Practice:

Database	Record Creation Trigger	Scaling Handling
World Bank PPI	New contract	Update existing record
Global Energy Monitor	New construction phase	Update wiki/phase record
IJGlobal	New financing event	Same asset record
UN Treaty Collection	New legal regime requiring fresh consent	Amendment attaches to parent
Bloomberg/Refinitiv	New deal (distinct transaction)	Same deal record through all revisions

Numerical Thresholds as Review Triggers:

The 2× budget and 50%+ capacity thresholds function as mandatory review triggers rather than automatic new-event creation:

Threshold	Function	Outcome Determined By
2× budget (100% increase)	Triggers human review	Transformation-scaling criteria
50%+ capacity increase	Triggers human review	Transformation-scaling criteria

Rationale: Quantitative scaling within the same institutional framework represents continuous evolution, not transformation. A coordination mechanism (e.g., a federal award) may remain unchanged whilst associated private investment components scale significantly. New records for such scaling would fragment analytical continuity without corresponding analytical benefit.

Succession Modelling:

When genuine transformation warrants new event creation, explicit bidirectional succession relationships preserve analytical continuity:

- `superseded_by` / `succeeded_from`: Standard succession
- `split_into` / `merged_from`: Entity division or consolidation

Confidence: HIGH (85%)

Precedent Sources:

- World Bank PPI Database methodology
- Vienna Convention on the Law of Treaties (amendment vs new treaty distinction)
- Bloomberg/Refinitiv deal tracking methodology
- Temporal database literature (Snodgrass, Kimball SCD patterns)

3.13.1 Corporate Product Release Diagnostic

Manual Section: 3.14.1 (Corporate Product Release Supplement)

Category: ADAPTED (MEDIUM-HIGH)

Rationale: The transformation-scaling criteria in *Manual Section 3.14.1* were designed for events with identifiable legal instruments, sovereign parties, and regulatory authorisations — characteristics inherent to Coordination, Policy, and Infrastructure event types. Corporate product releases lack these institutional structures. A model release has no treaty, no sovereign parties, and no regulatory authorisation in the sense the criteria intend. Under strict application of the unmodified institutional criteria, successive product releases from the same developer would always resolve as rolling updates (all answerable questions indicate scaling; inapplicable questions provide no transformation signal), regardless of the technical and commercial magnitude of change between generations.

The corporate product diagnostic adapts the same principled boundary — change in kind versus change in degree — to characteristics meaningful for product releases: product identity (is this a separately identifiable product?), technical independence (does this represent independent development or incremental refinement?), and market relationship (does this product coexist with or replace its predecessor?). The boundary remains institutional in the Bloomberg/Refinitiv sense: a "new deal" versus "same deal through all revisions" distinction applied to technology product context.

Adaptation Source: The Bloomberg/Refinitiv deal-tracking methodology cited in §3.13 distinguishes new issuances from amendments to existing issuances. The corporate product diagnostic applies the same logic: a ground-up technical rebuild with independent documentation constitutes a new record, whilst an optimisation variant within the same architecture constitutes an update to the existing record. The underlying principle — institutional continuity determines record architecture — is preserved; only the indicators of continuity differ by event type.

Empirical Validation: Corpus analysis of 17 corporate model events across five sequential processing cycles during the initial database population phase (January–April 2026) confirmed the replacement criteria produce outcomes consistent with all existing database determinations. The framework correctly identifies genuine product transformations (architecture class changes, independent training runs, parallel product lines) whilst routing version updates (weight refreshes, inference optimisations, checkpoint updates) to rolling update status. Two validation scenarios involving closed-source products with limited public technical documentation correctly produced "flag for human review" outcomes — confirming the framework does not auto-resolve genuinely ambiguous cases.

Closed-Source Interaction: Developers who do not publish architecture details may systematically produce Dimension B = Ambiguous outcomes for successor releases. This is an honest interaction with Structural Visibility Asymmetry (*Manual Section 4.11*), not a framework deficiency. The asymmetry in public documentation between open-source and closed-source releases means that open-source succession cases are more readily resolved, whilst closed-source cases more frequently require human review. This mirrors the broader SVA pattern documented in the Manual.

Confidence: MEDIUM-HIGH (70%) — adapted from established precedent (Bloomberg/Refinitiv deal-tracking distinction, already validated in §3.13) with empirical validation against the full corporate event corpus. The specific indicators for Dimension B (what constitutes "independent" versus "incremental" technical development) involve analytical judgement that will be refined through continued application. The 65% minimum confidence threshold (*Section 2.3*) is met.

Precedent Sources:

- Bloomberg/Refinitiv deal tracking methodology (via §3.13)
- WEO processing corpus: 17 corporate model events, 5 processing cycles (January–April 2026)
- CC Type 7 corporate product release exclusion (*Manual Section 9.4.3*; *Rationale §9.5.1*) — establishes precedent for event-type-specific treatment of corporate product releases within the existing methodology

3.14 Post-Qualification Verification

Manual Section: 3.15

Category: PRECEDENT-BASED (HIGH)

Rationale: Events evolve after initial database entry. Infrastructure projects expand, policy frameworks are amended, corporate initiatives scale. The Rolling Updates system addresses this through three mechanisms, each grounded in established practice.

6-Month Lag for Updates

The same 6-month verification standard governing initial qualification applies to post-entry developments. Research on major technology infrastructure announcements documents that 40-60% of large capital projects experience significant delays, modifications, or cancellations. The 6-month window captures immediate failures whilst validating actual implementation.

This aligns with institutional verification practice:

Domain	Standard Verification Lag	Source
Credit ratings	90-180 days post-implementation	Rating agency methodology
Infrastructure bonds	6-12 months operational history	Bond prospectus requirements
Policy effectiveness	6-12 months post-enactment	Legislative impact assessment
Academic infrastructure data	12-24 months	OECD, World Bank methodology

Genesis Data Preservation

Genesis data — the original qualified values at database entry — remains immutable. This provides permanent reference for qualification basis, Coordination Connection verification provenance, historical analysis, and audit trail integrity.

This principle derives from financial data versioning standards where baseline values must be preserved alongside current values for compliance and auditability.

Two-Tier Architecture

The Genesis/Current data separation enables systematic evolution tracking without compromising historical integrity. Current data reflects the latest verified state; Genesis data documents what qualified the event for inclusion. This architecture mirrors version control patterns in compliance-grade financial systems.

Confidence: HIGH (85%)

Integration with Event Evolution

The rolling updates system operates alongside the Event Evolution Protocol (*Manual Section 3.14*). When pending updates are verified, they undergo transformation-scaling assessment:

- **Scaling** (same institutional framework): Incorporate via rolling update
- **Transformation** (classification change, party change, legal regime change): Route to Section 3.14 for succession handling

This integration ensures that quantitative growth within stable institutional frameworks is captured through version history, whilst genuine transformations receive appropriate new-event treatment with explicit succession relationships.

3.15 Source Accommodation and Event Verification Alignment

Manual Sections: §3.1 (Source Accommodations), §3.4.11 (Verification Assessment for Accommodated Sources), §3.5 Option D (OEP dual-purpose evidence), §3.5.1 (PIET dual-purpose evidence), §4.11 (Relationship to Event Verification)

Category: ADAPTED (HIGH)

Rationale: Documentation-based methodologies produce structural gaps where standard primary source categories do not cover the evidence class required for verification. The Manual addresses three such gaps through source accommodations — expanded primary source definitions for specific event classes. Each accommodation specifies qualifying conditions at its point of definition (§4.11, §3.5, §3.5.1). The source accommodation alignment formalises the relationship between these accommodations and §3.1 event verification requirements, and establishes confidence assessment guidance for events verified through accommodated sources.

OEP and PIET dual-purpose evidence. OEP pathway evidence (independent technical laboratory analysis) and PIET gate evidence (trade flow data, commodity prices, government responses, corporate disclosures) each establish AI connection at Stage 2. The same evidence inherently verifies event existence — the analysis that confirms AI-relevant architecture also confirms the manufacturing milestone; the trade flow disruption that confirms AI supply chain impact also confirms the restriction's implementation. The dual-purpose role follows from the evidence structure and requires no additional qualification gate.

§4.11 state media accommodation. Regulatory process architectures that produce structurally less public documentation than jurisdictions with extensive regulatory transparency frameworks — no published regulatory impact assessments, no published consultation responses, shorter operative texts — create a verification gap where standard §3.1 primary source categories (government official documents via .gov domains, regulatory filings, multilateral publications) may not be available for Tier A claims. State media with official documentary function in the issuing jurisdiction — outlets operating with documented state editorial oversight — bridges this gap when reporting named official statements from identified government authorities. The qualifying conditions (official documentary function, named official attribution, verifiable institutional role) adapt established journalism verification practice for source attribution in state-controlled media environments.

Closest Precedents:

Framework	Institution	Relevance	Divergence from WEO
ICD 203 Analytic Standards	US Intelligence Community	Requires explicit disclosure of single-source dependence; distinguishes assessments based on converging vs. single-source intelligence	Intelligence assessment context, not geopolitical event verification
FATF Mutual Evaluation Methodology (2013/2022)	FATF	Graduated effectiveness assessment — single-channel evidence receives lower rating than multi-channel convergent evidence	National regulatory system assessment, not individual event verification; four-point scale, not binary ceiling
Reuters Handbook of Journalism	Reuters	Source attribution requirements scale with claim significance; accommodation of non-standard sources requires explicit flagging and editorial justification	Journalism, not methodology documentation; no formal ceiling mechanism

Key Design Decision — Confidence Term Ceiling:

Tier A claims verified exclusively through a single accommodated source type carry a confidence term ceiling of "Officially reported" (Manual §3.4.5) rather than "Confirmed / Verified." This applies the existing §3.4.5 confidence language framework to a specific evidence configuration. "Officially reported" is defined as evidence "stated by authoritative primary source with independent corroboration" (1P + 2C) — the structural match for sole-reliance on an accommodated primary source supplemented by corroborating coverage. Where standard primary sources independently verify the same claims, the ceiling does not apply.

Source type	Confidence ceiling	Rationale
Standard §3.1 primaries	No ceiling	Converging independent documentation streams
§4.11-accommodated (sole basis)	"Officially reported"	Authoritative primary source with corroboration, not independent multi-source confirmation
OEP (sole basis)	"Officially reported"	Single underlying physical evidence base from one source entity
PIET	No ceiling	Multiple independent evidence streams across 3–4 empirical domains

The PIET exemption follows from multi-stream convergence: a 3-of-4 gate pass across distinct empirical domains is structurally equivalent to standard §3.1 multi-source confirmation.

Confidence Assessment: HIGH. The dual-purpose evidence statements follow from evidence structure and require no novel calibration. The §4.11 state media accommodation adapts established journalism source-attribution practice (Reuters Handbook, AFP Twenty Principles) to WEO's verification context — the qualifying conditions (documentary function, named attribution, verifiable institutional role) are standard source-verification requirements applied to a specific media category. The confidence term ceiling applies existing §3.4.5 terminology to a defined evidence configuration with direct precedent in ICD 203 single-source disclosure requirements.

Validation Commitment: Monitor for (1) changes in regulatory process architecture in jurisdictions where the §4.11 accommodation currently applies that increase public documentation availability, reducing the structural gap; (2) emergence of additional regulatory environments exhibiting structurally similar documentation gaps, requiring assessment of whether the accommodation framework extends; and (3) evolution of intelligence community or journalism source-assessment frameworks that may provide refined models for accommodated-source confidence calibration.

Section 4: Significance Assessment Rationale

Manual Reference: Chapter 4

4.1 Basel 5-Factor Assessment

Manual Section: 4.3-4.5

Category: EMPIRICALLY GROUNDED (VERY HIGH)

Rationale: WEO adapts the Basel Committee on Banking Supervision G-SIB Methodology (2018) — the framework G20 central banks have used since 2011 to assess systemic importance in banking infrastructure.

The five factors (Size, Interconnectedness, Substitutability, Complexity, Cross-jurisdictional Activity) provide proven multi-dimensional assessment for infrastructure significance. WEO adapts thresholds and evidence requirements for AI infrastructure whilst preserving the framework's structural integrity.

2+ Factors Requirement: Prevents single-factor outliers from qualification, aligning with Basel's score-based (not binary) methodology.

Citation: Basel Committee on Banking Supervision. (2018). *Global systemically important banks: revised assessment methodology and the higher loss absorbency requirement*.

4.2 \$500M Physical Infrastructure Threshold

Manual Section: 4.4 (Factor 1: Size)

Category: EMPIRICALLY GROUNDED (HIGH)

Rationale: The \$500M threshold derives from U.S. Federal Highway Administration "Major Project" definition under 23 U.S.C. 106(h), which designates projects exceeding \$500M as requiring enhanced federal oversight.

Exact Statutory Language:

"A recipient of Federal financial assistance for a project under this title with an estimated total cost of \$500,000,000 or more... shall submit to the Secretary... a project management plan"

Legal Status: Current federal law. Established by SAFETEA-LU (2005), lowering the previous \$1B "Mega Project" threshold.

Supporting Convergent Sources:

Source	Threshold	Type
U.S. FHWA 23 U.S.C. 106(h)	\$500M	Federal statute
Hart-Scott-Rodino Act (2025)	\$505.8M	Federal statute
EU Foreign Subsidies Regulation	€500M (~\$525M)	EU regulation
AIIB Accountability Framework	\$300M	Multilateral bank

4.3 5,000+ GPU/Accelerator Cluster Threshold

Manual Section: 4.4 (Quantitative AI Thresholds)

Category: EMPIRICALLY GROUNDED (HIGH)

Rationale: The 5,000+ GPU threshold is empirically grounded through distribution analysis of AI infrastructure investments (2020-2025).

Empirical Findings:

Research Finding	Source
Natural breakpoint at ~\$100M	Distribution analysis (multi-source research)
5,000 GPUs ≈ \$150-200M	GPU pricing correlation (H100 ~\$30-40K); domestic AI accelerator pricing may differ but threshold measures card count, not capital equivalent
\$100M distinguishes technical from strategic	Cross-validated research

Research Convergence:

- GPU clusters show breakpoints at \$50M (project vs institutional scale) and \$200M (national significance); the cleaner infrastructure-scale break occurs at ~\$100M across all categories
- 5,000+ H100s correlate to \$150M-\$200M investment; \$100M identified as "significant deployment threshold"

Analysis confirmed a natural breakpoint exists at ~\$100M for distinguishing technical projects from strategic deployments. 5,000 GPUs sits comfortably above this threshold.

Position in Threshold Hierarchy: Between infrastructure entry (\$100M) and geopolitical significance (\$500M Basel Size Factor), the 5,000+ GPU threshold (~\$150-200M) captures infrastructure-scale AI compute deployments.

4.4 Software Routes 1-4

Manual Section: 4.8

Category: EMPIRICALLY GROUNDED (varies by route)

Route 1: Official Designation

Rationale: Relies on authoritative government frameworks:

- CISA Systemically Important Entities (NSM-22, 30th April 2024)
- EU DMA Gatekeepers (Regulation 2022/1925)
- EU NIS2 Essential Entities (Directive 2022/2555)
- EU DORA Critical ICT Third-Party Providers (Regulation 2022/2554)

Eastern Bloc Finding: No defensible equivalent exists. CIIO lists are confidential by Chinese government policy. EB platforms qualify via Routes 2-4 instead.

Route 2: Investment Threshold

Rationale:

- **\$100M threshold:** Institutional investor "significant commitment" convention. Market transactions cluster in the \$75M-\$150M significance zone (POET Technologies \$75M AI hardware financing, Applied Digital \$112.5M AI data centre drawdown).
- **\$50M threshold:** Government programme participation thresholds (NRCan, EU Cohesion Policy, DoD FMR)

Route 3: Multi-Criteria

Rationale:

- **10,000 dependents:** npm/PyPI ecosystem dependency analysis
- **50% market share:** Sherman Act Section 2 (US) and Article 102 TFEU (EU) dominance thresholds. Note: These derive from case law precedent and Commission/Court guidance (US v. Alcoa, 1945; EU Commission Article 102 guidance) rather than statutory text, which does not specify numeric thresholds.

Route 4: Open-Source AI Models

Rationale: $\geq 100\text{M}$ cumulative downloads, derived from EU DMA's 45M monthly end users threshold with $\approx 2\times$ multiplier for AI-specific context where downloads $\neq$ active users.

R4-Level Thresholds:

R4-Level	Score	Multiplier	G-SIB Precedent
Level 1	≥1.0	1.15×	Bucket 1 (1.0% surcharge)
Level 2	≥4.0	1.40×	Bucket 2 (1.5% surcharge)
Level 3	≥7.0	1.75×	Bucket 3 (2.0% surcharge)

4.5 Route 4 Closed-Source Equivalents (4A-CS to 4D-CS)

Manual Section: 4.8 (Route 4 subsection)

Category: ADAPTED (from DMA, with reduced observability accommodation)

Rationale: Closed-source AI models (GPT-4, Claude, Gemini, Ernie) lack the publicly observable metrics available for open-source models (downloads, derivatives). The 4A-CS to 4D-CS framework establishes equivalent qualification criteria using observable proxy metrics.

Qualification Logic: Any ONE of four criteria qualifies.

Criterion 4A-CS: User Base

Threshold:

-  100M monthly active users (MAU) → Automatic qualification
- 50M-100M MAU → Qualifies with 4B-CS OR 4C-CS

Precedent:

- EU Digital Markets Act: 45M monthly end users threshold
- WEO applies ≈2× multiplier (100M) for automatic qualification
- 50M-100M range requires supporting criterion to prevent borderline misclassification

Measurement Sources: Similarweb, data.ai, company earnings disclosures, third-party analyst reports

Criterion 4B-CS: Geographic Reach

Threshold:

- Available in ≥150 countries, OR
- Available in ALL G7 nations + documented Eastern Bloc presence (PRC, Russia)

Rationale: Geographic availability indicates cross-bloc coordination significance. The G7 + Eastern Bloc test captures strategic positioning even for models with restricted geographic access.

Criterion 4C-CS: Enterprise Integration

Threshold (Composite):

1. Fortune 500 enterprise adoption (≥ 50 documented customers)
2. Government/defence contracts (≥ 3 agencies across ≥ 2 countries)
3. API ecosystem ($\geq 1,000$ registered developers OR ≥ 100 published integrations)

Limitation Acknowledged: Closed-source enterprise integration is less observable than open-source metrics. Composite assessment prevents single-dimension gaming whilst maintaining rigour.

Precedent: DMA business user criteria adapted for AI platform context.

Criterion 4D-CS: Capability Threshold

Threshold:

- $\geq 1B$ parameters (documented or credibly estimated), AND
- Frontier performance on recognised benchmarks (top 10 on MMLU, HumanEval, or equivalent)

Rationale: Capability threshold captures models with significant coordination impact regardless of commercial adoption metrics. Parameter count + benchmark performance provides observable proxy for frontier status.

4.6 PISA-5 Assessment

Manual Section: 4.9

Category: MIXED (factors vary)

Rationale: PISA-5 (Policy and Institutional Significance Assessment) was developed to qualify single-jurisdiction AI policy and institutional events that cannot route through Basel 5-Factor (no physical infrastructure) or Software Routes (no software/platform release).

Factor	Status	Source
F1: J-Tier (Jurisdictional Significance)	EMPIRICALLY GROUNDED	G7/G20 classification; Stanford HAI AI Index
F2: Regulatory Scope	EMPIRICALLY GROUNDED	Legislative analysis methodology
F3: AI Centrality (mandatory gate)	CATEGORY CREATOR	AI-specific filter — no precedent
F4: Enforcement/Accountability	EMPIRICALLY GROUNDED	Regulatory effectiveness research
F5: International Relevance	EMPIRICALLY GROUNDED	Extraterritoriality doctrine; Brussels Effect

Qualification Logic

Standard Pass: $\geq 4/5$ factors satisfied (with F3 mandatory)

Alternative Pass: F1 + F2 + F3 all satisfied

Rationale for Alternative Pass: Major jurisdiction (F1) + binding framework (F2) + AI-specific (F3) captures "minimum viable significance" even without full enforcement (F4) or international reach (F5).

Mandatory Gate: F3 (AI Centrality) must ALWAYS pass — if F3 fails, event fails PISA-5 regardless of other factors.

Novelty Assessment

Category: CATEGORY CREATOR (85% confidence)

Research across 40+ sources confirmed:

- No existing framework combines all five PISA-5 design elements (multi-factor qualification + tiered jurisdiction + mandatory gating + alternative pass logic + human review tiebreaker)
- Basel G-SIB is closest structural analogue but serves fundamentally different purpose (financial institution capital requirements vs policy event qualification)
- Five PISA-5 *design elements* assessed as NOVEL:
 - J-Tier classification (no precedent for G7/G20 + GDP + AI capability combination)
 - F3 mandatory gate (multi-factor frameworks typically use weighted averages, not hard gates)
 - $\geq 4/5$ OR F1+F2+F3 alternative logic (no precedent found)
 - AI Hub exception (J-Tier D)
 - T1-T5 structured tiebreaker criteria
- One component ADAPTED: The 5-factor structure concept from Basel G-SIB (but application to policy events is novel)

J-Tier D (AI Hub Exception): Category creator accommodation for AI-significant smaller jurisdictions meeting 2 of 4 criteria. Flagged for annual designation review. See §4.6.2 for detailed derivation.

4.6.1 Tiebreaker Criteria Derivation (TB1-TB5)

Manual Section: 4.9 (Human Review Framework)

Category: MIXED (TB1-TB2 adapted from governance precedent; TB3-TB5 WEO-original)

The PISA-5 Human Review Framework uses five tiebreaker criteria when events pass exactly 3/5 factors:

Criterion	Threshold	Derivation Source
TB1: Resource Commitment	>\$15M annual budget OR >50 FTE	AI governance institution budget analysis
TB2: Institutional Permanence	Statutory basis OR >3 year mandate	Administrative law institutional design precedent
TB3: Stakeholder Engagement	≥ 3 stakeholder categories	Multi-stakeholder governance best practice
TB4: International Influence	Formal coordination mechanism	Treaty and MOU practice
TB5: Novel Governance	First-of-kind approach	Innovation assessment

TB1 \$15M Threshold Derivation:

Analysis of established AI governance institution budgets (2023-2025) identified \$15M annual operating expenditure as the threshold distinguishing substantive institutional capacity from nominal presence. This threshold is validated against established infrastructure classification frameworks.

Critical Distinction: TB1's \$15M is annual operating budget (recurring expenditure), distinct from Part A Basel Factor 1 (\$500M+ capital investment) and Chapter 3 entry threshold (\$100M infrastructure investment).

Validation Commitment: Review tiebreaker thresholds at 100-event milestone based on classification outcome data.

4.6.2 J-Tier D AI Hub Exception — Derivation

Manual Section: 4.9 (AI Hub Exception)

Category: CATEGORY CREATOR (75% confidence)

Problem Addressed: Standard J-Tier criteria (G7/G20 membership, \$500B GDP) exclude jurisdictions that are demonstrably significant to global AI coordination despite smaller economic footprints. Examples include Taiwan (critical semiconductor manufacturing), Israel (AI startup ecosystem), Singapore (AI governance leadership), and UAE (sovereign AI investment).

Criteria Selection Rationale:

The four J-Tier D criteria were selected to capture distinct pathways to AI significance:

Criterion	Rationale	Precedent
Top-15 AI Index	Captures jurisdictions with demonstrated AI capability disproportionate to GDP	Stanford HAI methodology; Tortoise Global AI Index
Critical AI Infrastructure (>5% global share)	Captures jurisdictions controlling chokepoint infrastructure regardless of domestic AI development	Semiconductor supply chain analysis; CSET chokepoint research
Top-10 AI Investment Destination	Captures jurisdictions attracting significant capital flows indicating market confidence	PitchBook/CB Insights methodology; venture capital concentration analysis
Foundation Model Developer	Captures jurisdictions producing frontier AI capabilities regardless of other metrics	Papers with Code benchmarks; Hugging Face leaderboards

2-of-4 Threshold Rationale:

The 2-of-4 requirement balances inclusivity with rigour:

Threshold	Assessment
1 of 4	Too permissive — single-dimension outliers could qualify (e.g., jurisdiction with one well-funded AI lab but no broader ecosystem)
2 of 4	Selected — requires demonstrated significance across multiple dimensions
3 of 4	Too restrictive — would exclude jurisdictions with genuine but concentrated AI significance
4 of 4	Unnecessary — jurisdictions meeting all four likely qualify via standard J-Tier criteria

Precedent Analysis:

No direct precedent exists for AI-specific jurisdictional significance thresholds. Adjacent frameworks consulted:

- **Global Financial Centres Index (GFCI):** Private index using multi-criteria assessment of financial centre competitiveness; no comparable FSB regime exists for financial centres (FSB assesses institutions, not centres)
- **OECD Digital Economy Outlook:** Ranks jurisdictions by digital indicators but does not create qualification thresholds
- **Stanford HAI Global Vibrancy Tool:** Provides rankings but no threshold-based classification

Conclusion: J-Tier D represents genuine category creator calibration. The 2-of-4 threshold and specific criteria selection are WEO-original, informed by but not derived from external frameworks.

Validation Commitment: Annual designation review; recalibration at 100-event milestone based on classification outcomes.

4.6.3 F3 Pathway Exemption

Manual Section: §4.9, Factor 3 (F3) — pathway exemption

Category: WEO SYNTHESIS (75% confidence)

Rationale: F3 filters general technology and digital economy policies where AI is mentioned incidentally (data protection law with an automated decision-making article, copyright directive with a text-mining exception). Its structural indicators — AI in policy title, AI-specific definitions, AI-specific obligations — are calibrated to regulatory and governance instruments where the legislature chooses whether to make AI the primary subject.

Two AI Connection pathways (Option F and Option G) serve event classes where the legislature structurally cannot make AI the primary subject because the instrument addresses the programme or the material — not the downstream application. Defence appropriation acts fund entire departmental budgets; critical minerals acts address mineral supply chains. In both cases, the AI infrastructure relevance is real and documentable, but the instrument's structural format guarantees F3 failure regardless of the event's genuine significance.

The pathway exemption resolves this by recognising that each pathway's own evidence requirement performs the AI centrality function more precisely than F3's generic indicators. Option F requires explicit AI designation in programme titles within the financial instrument. Option G requires documented causal chains from the controlled material to AI infrastructure applications. Each test is more demanding and more specific than F3's title-or-definitions check.

Evidence basis: Two gap reports — GAP-2026-04-14-001 (Option F, April 2026) and GAP-2026-05-13-001 (supply-chain response events, May 2026) — separately documented the same structural barrier. The §4.9 F3 coverage note acknowledges the pattern: "Application-agnostic Policy instruments controlling physical materials or technology products... will typically fail F3 because the instrument's primary subject is the controlled material or technology, not AI." The pathway exemption implements what the coverage note identified: each qualifying pathway's own evidence standard performs the AI centrality function more precisely than F3's generic indicators.

Pass logic adjustment: For exempt events, PISA-5 is assessed on four factors (F1, F2, F4, F5). Auto-qualification requires $\geq 3/4$ factors. The F1+F2 double-fail exclusion rule is preserved — neither significant jurisdiction nor regulatory scope remains grounds for automatic disqualification regardless of F3 status. The standard PISA-5 has a 60% auto-qualify route (F1+F2+F3) that depends on F3's mandatory-gate status; with F3 removed, the lowest auto-qualify ratio rises from 60% to 75% ($\geq 3/4$), an appropriate tightening given the narrower factor set. Marginal cases (exactly 2/4 with F1+F2 not both failing) route to Human Review under the existing TB1–TB5 tiebreaker framework.

Confidence: 75%. The structural mismatch is documented in two separate gap reports. The resolution is logically sound (pathway evidence subsumes F3's function). The 65% minimum threshold (§2.3) is met. Confidence is bounded below 80% because the exemption has not yet been applied to live events — first application will occur following V1.4 publication.

Validation commitment: Monitor F3-exempt events for quality. If events qualifying through pathway exemption are systematically lower quality than standard PISA-5 events, reassess whether the pathway evidence requirements are sufficiently demanding.

4.7 T2 Buildout Qualification Criteria

Manual Section: 5.5 (T2 Buildout Test)

Category: EMPIRICALLY GROUNDED (HIGH)

Rationale: The four-criterion framework for T2 Buildout classification derives from analysis of explicit cross-bloc targeting language in infrastructure legislation.

Four-Criterion Framework (ONE OR MORE required for T2):

Criterion	Requirement	Evidence Standard
1. Dependency Reduction	Explicit statement reducing dependency on NAMED other-bloc	"Reduce dependence on [Country X]" in official documentation
2. National Security + Co-Mention	"National security" in same paragraph/section with NAMED other-bloc	Security language co-located with named entity
3. Operating Restrictions	Funded entities prohibited from operating in NAMED other-bloc	Binding restrictions on cross-bloc activity
4. Supply Chain Resilience + Geographic Specificity	"Supply chain resilience" co-located with NAMED other-bloc	Resilience language with geographic targeting

Criterion Strength Ranking:

- Criterion 3 (Operating Restrictions) = **STRONGEST** — creates direct legal obligation
- Criterion 4 (Supply Chain Resilience) = **WEAKEST** — often uses euphemistic language

Key Precedent: US CHIPS Act §9902 guardrails (explicit China naming); Banque de France geopolitical trade fragmentation analysis; EU Chips Act implementation criteria.

Section 5: Tier Classification Rationale

Manual Reference: Chapter 5

5.1 Three-Tier Architecture

Manual Section: 5.2

Category: PRECEDENT-BASED (HIGH)

Rationale: The T1/T2/T3 architecture captures three distinct scales of coordination dynamics:

Tier	Scope	Mirror Framework
T1	Cross-bloc (verified adoption or implementation by the principal AI actor from at least two distinct ecosystems)	Multilateral trade/security analysis
T2	Between-bloc (structural separation through restriction, buildout, or regulatory barriers)	Bilateral/adversarial analysis
T3	Intra-bloc (activity within a single bloc)	Alliance theory; bloc coordination

T1 Cross-Bloc Criteria

Category: CATEGORY CREATOR (MEDIUM-HIGH)

Requires ALL THREE:

1. Cross-bloc participation (the principal AI actor from at least two distinct ecosystems)
2. Binding/implemented status
3. Material integration

Operational Definitions:

Criterion	Definition	Observable Marker
Cross-bloc participation	Verified cross-bloc adoption or implementation by the principal AI actor from at least two distinct ecosystems	National standards adoption, operational certification, or equivalent implementation evidence
Binding/implemented	Actual commitments with adoption mechanisms, OR operational systems with implementation evidence, OR standards with formal certification schemes	Published standards/regulations, adoption documentation, certification bodies, implementation records
Material integration	Shared infrastructure, mutual vulnerability, or enforceable agreements creating cross-bloc interdependence	Cross-border implementation, mutual recognition documentation, shared infrastructure usage, certification schemes operating across blocs

Material Integration Examples:

- Shared AI safety testing protocols
- Common model evaluation benchmarks
- Interoperable data governance standards

For full tier classification criteria: See *Manual, Chapter 5*.

5.2 Mandatory Standard Test

Manual Section: 5.5 (T2 Mandatory Standard Test)

Category: EMPIRICALLY GROUNDED (HIGH)

Rationale: The Mandatory Standard Test addresses standard-setting events that impose mandatory compliance requirements with enforcement mechanisms. The three T2 tests — Restriction, Buildout, and Mandatory Standard — capture distinct fragmentation mechanisms: explicit barriers (Restriction), parallel capacity development (Buildout), and mandatory compliance regimes (Mandatory Standard).

Two-Condition Test (both must be met for T2 classification):

Condition	Requirement	Evidence
1. Mandatory Language	Contains "shall," "must," "required"	NOT "recommended," "encouraged," "voluntary"
2. Enforcement Mechanisms	Specific consequences documented	Financial penalties, market access restrictions, licensing consequences, legal liability

Chinese Standard Rule: GB (without /T) = mandatory → test for T2; GB/T (with /T) = voluntary → automatic T3.

Routing Decision: Standard-setting events meeting both conditions route to **Restriction classification**, not to a separate category.

Rationale for Restriction Routing: Mandatory standards with enforcement are structural fragmentation tools — they impose compliance costs as condition of market access, creating non-tariff barriers. This is functionally similar to explicit restrictions.

Precedent:

- WTO TBT Agreement (mandatory vs voluntary standards distinction)
- Brussels Effect (regulatory extraterritoriality)
- Regulatory extraterritoriality doctrine

5.3 Bloc Definitions

Manual Section: 3.10 (Bloc Membership Reference)

Category: PRECEDENT-BASED (HIGH)

Rationale: Bloc definitions follow alliance theory and strategic alignment analysis precedent, adapted for AI/semiconductor technology partnerships.

Western Bloc (WB) Criteria

Core Membership Indicators: NATO membership, Five Eyes intelligence alliance, AUKUS partnership, US bilateral defence treaties (Japan, South Korea); Taiwan Relations Act (non-treaty security framework), EU membership, QUAD participation.

Observable Characteristics: US-led security architecture, integrated technology supply chains (semiconductors, AI, cloud), coordinated export controls, G7 coordination frameworks.

Eastern Bloc (EB) Criteria

Core Membership Indicators: Shanghai Cooperation Organisation (SCO), CSTO (Collective Security Treaty Organisation), China-Russia strategic partnership, Belt and Road Initiative participation (when EB-led).

Observable Characteristics: China-centric or Russia-centric security relationships, alternative ecosystems, SCO coordination frameworks.

Non-Aligned (NA) Determination

All 5 criteria must be met for NA classification:

1. Active security/economic partnerships with BOTH blocs
2. Explicit strategic autonomy or non-alignment doctrine
3. Refusal to join exclusive WB or EB alliances
4. Sovereign infrastructure development (not dependent on either bloc)
5. Voting independence in multilateral forums

Primary NA Nations (see **WEO Bloc Membership Register for current authoritative classifications**): India, Indonesia, Vietnam, Thailand, Malaysia, Philippines, South Africa, Nigeria.

Implementation: The WEO Bloc Membership Register (living document) tracks current classifications with evidence-based changes, categorised as Core, Conditional, Functional, or Leaning.

Relationship Between 5-Criteria Test and Register: The 5-criteria test (above) determines NA eligibility for methodology documentation. The Bloc Membership Register may additionally list nations undergoing alignment transitions that no longer meet all 5 criteria but require per-event assessment during the transition period. Such transitional cases are documented in the Register with "Shifting" or equivalent status indicators, enabling appropriate handling until classification stabilises.

V1.3 Clarification (April 2026): Manual §3.10 clarified that AI ecosystem alignment is the primary determinant for bloc classification, with alliance frameworks and geographic membership providing supporting context. This formalises the hierarchy implicit in the V1.2 framework, where AI ecosystem lists were presented alongside but not subordinate to alliance frameworks. The clarification was prompted by empirical findings during event processing that cross-bloc development participation in standards bodies (SC 42) did not correlate with cross-bloc implementation, highlighting the importance of AI ecosystem alignment over institutional membership as a classification signal.

5.4 Principal AI Actors

Manual Section: §5.3 (Principal AI Actors)

Category: CATEGORY CREATOR (75% confidence)

Rationale: No existing geopolitical, international relations, or technology governance framework determines "principal actor" status based specifically on AI infrastructure capability dimensions. This finding was confirmed through review of CSIS, Brookings, RAND, IISS, Chatham House, OECD, WEF, Stanford HAI, Belfer Center, Carnegie Endowment, and academic international relations literature. Within the established lexicon of international relations theory, "principal actor" remains reserved for the sovereign nation-state operating within the Westphalian state system. However, the substantive theoretical architecture underlying WEO's infrastructure-based determination is present across multiple leading institutions, which have constructed analogous full-stack technological typologies using identical supply-chain chokepoints to define sovereign AI power.

Closest Precedents:

Framework	Institution	Relevance	Divergence from WEO
IISS Cyber Capabilities Assessment (2021/2023)	International Institute for Strategic Studies	Closest methodological precedent: three-tier nation classification using qualitative multi-dimensional assessment	Uses cyber power, not AI infrastructure; AI is a sub-indicator within broader cyber assessment
"Broad Stack Leaders" typology (2026)	Brookings Institution	Closest conceptual analogue: categorises US, China, EU as full-stack AI leaders with subordinate tiers (Coordinated Stack Builders, Upstream AI Shapers, AI System Shapers)	Includes non-infrastructure variables (talent, data, education); uses descriptive categories, not threshold-based discrete classification
Cyber/Technology Power Index (2025)	Harvard Belfer Center for Science and International Affairs	Most comprehensive quantitative framework: composite index covering 4 of WEO's 7 dimensions across five technology sectors	Spans five technology sectors, not AI-specific; produces continuous numerical rankings, not discrete principal/non-principal determination
AI Building Blocks framework (2026)	Chatham House	Identifies "AI Superpowers" (US, China) using primary building blocks (compute, frontier models, data) and secondary building blocks (infrastructure, talent, energy, trust)	Bundles hardware layers under "compute"; includes talent, energy, and trust as co-equal primary/secondary dimensions
G-SIB Methodology	Basel Committee on Banking Supervision	Multi-factor threshold-based classification with discrete designation tiers — methodological precedent already used in WEO significance assessment (Basel 5-Factor)	Financial sector, not technology; the threshold-based qualification pattern is directly analogous to WEO's ≥ 3 of 7 dimensional threshold

Key Design Decisions:

1. **Hardware stack disaggregation.** No existing framework treats accelerator design, semiconductor fabrication, semiconductor equipment, and cloud compute as four separate, co-equal dimensions. Brookings bundles hardware under "compute"; Chatham House uses "compute" as a single primary building block. WEO disaggregates because supply-chain chokepoint analysis requires dimensional granularity — a nation can dominate fabrication (Taiwan) without designing accelerators, or dominate equipment (Netherlands) without fabricating chips. Bundling obscures the distribution of supply-chain power that WEO's bloc classification methodology is designed to track.
2. **Dimensional scope: sovereign control over AI infrastructure outputs.** WEO's seven SCP dimensions — AI accelerator design, advanced semiconductor fabrication, semiconductor manufacturing equipment, cloud AI compute services, frontier AI model development, AI-specific regulatory authority with extraterritorial reach, and sovereign AI infrastructure investment at scale — measure assets, instruments, and capabilities that a sovereign actor can independently control and that determine which AI infrastructure ecosystem the actor operates within. This scope is more focused than comprehensive AI power frameworks (Brookings, Chatham House, IISS, Belfer), which additionally assess variables such as talent, data, and energy. WEO's dimensional focus reflects the specific analytical question the SCP framework is designed to answer: not how much AI capability does this actor possess in aggregate, but what does this actor independently control in the AI infrastructure supply chain, and which ecosystem does that place it within? Variables that contribute to AI capability broadly but do not independently determine ecosystem position — including workforce, training data, and energy capacity — are not assessed as SCP dimensions. These variables do appear in WEO's event-level analysis where they constitute characteristics of AI infrastructure coordination activity: energy through the Energy/Infrastructure domain and Environmental Nexus Tags framework, data governance through policy events, and workforce provisions through event characteristics such as CHIPS Act award conditions. The dimensional exclusion is a scope decision at the actor-assessment level, not an analytical judgement about the importance of these variables to AI development. This focused scope, combined with the three-gate designation structure (§5.4.2), produces materially different actor classifications from broader frameworks: most notably, the EU scores 4/7 on WEO's dimensional assessment (including regulatory authority and sovereign investment) but does not qualify as a principal AI actor because its hardware dimensions cannot be sustained independently under the severance test — a structural dependency that broader typologies such as Brookings' "Broad Stack Leaders" do not assess.
3. **Discrete actor classification with binary threshold.** Most indices (Stanford HAI AI Index, Belfer CET Index, OECD.AI) produce continuous numerical rankings. WEO uses a binary threshold (≥ 3 of 7 dimensions) producing a discrete "principal actor / not principal actor" determination. This follows the methodological pattern established by the Basel G-SIB methodology (already adapted for WEO's Basel 5-Factor significance assessment): multi-factor assessment producing a discrete designation that triggers specific regulatory consequences. The ≥ 3 threshold was selected because: $N=1$ would include critical supply-chain nodes whose single-dimension dominance does not independently shape coordination dynamics; $N=2$ produces no empirically different result from $N=3$ under current assessment; $N \geq 4$ produces no empirically different result from $N=3$ but would create a threshold that structurally excludes nations with emerging but incomplete AI ecosystems from future qualification.

Evidence Basis: The two-ecosystem structure and principal actor identification are grounded in cross-referenced research from three independent AI research inputs with dedicated synthesis (March 2026), examining market share data, fabrication capability, equipment monopolies, cloud deployment, model development, regulatory frameworks, and investment scale across all candidate nations. Findings were unanimous on the bipolar hardware ecosystem structure and the US/China principal actor identification. The category creator research (principal actor as novel framework) was reviewed across two independent AI research inputs surveying CSIS, Brookings, RAND, IISS, Chatham House, OECD, WEF, Stanford HAI, Belfer Center, Carnegie Endowment, and academic international relations literature. The framework comprises a three-gate designation structure (§5.4.2), per-dimension qualification principles (§5.4.1), and is applied across 14 actors in the Batch 1 assessment cycle, producing a distribution (2 PAA, 3 AIK, 2 ACS, 7 Participant) that discriminates meaningfully across all four designations. The ≥ 3 threshold is tested at the boundary — actors scoring 3/7 (Netherlands, Taiwan) and 4/7 (Japan, South Korea, EU) span the AIK/ACS divide, supporting the threshold's discriminating power.

Confidence Assessment: 75%. The seven-dimension framework and discrete threshold classification are WEO-original with no direct precedent. Closest analogues (Brookings "Broad Stack Leaders", IISS Cyber Capabilities, Belfer CET Index) support the approach conceptually but diverge on dimensional scope and classification method. The ≥ 3 threshold is boundary-tested through Batch 1 application across 14 actors, producing meaningful discrimination at the 3/7 and 4/7 boundary.

Validation Commitment: Reassess Principal AI Actor designations at methodology version milestones. Monitor threshold sensitivity as additional nations develop AI infrastructure capability (EU hardware independence trajectory, India sovereign compute expansion). Review dimensional framework if new infrastructure categories emerge (e.g., quantum computing as a separate AI-relevant dimension). Continue applying to Batch 2 actors.

T2 Scope: Principal Actor Requirement Not Extended

Manual Section: §5.3 (T2 scope paragraph)

Category: WEO SYNTHESIS (70% confidence)

Rationale: The principal actor requirement is structurally specific to T1. T1 measures ecosystem-level integration — whether distinct AI ecosystems are operationally aligned through verified cross-bloc adoption or implementation by their respective principal AI actors. Without ecosystem controllers at the table, bilateral contacts between non-principal members from different blocs do not bridge the two ecosystems. T2 measures a different dynamic: structural separation between blocs. The significance of between-bloc fragmentation depends on the structural effect — what is disrupted, displaced, or barred — rather than the identity of the actor creating the separation.

This asymmetry is grounded in two independently established theoretical frameworks. Hegemonic Stability Theory (Kindleberger, 1973; Keohane, 1984; Gilpin, 1981) establishes that systemic cooperation requires hegemonic leadership to overcome collective action problems — Keohane's explicit disconfirmation condition makes actor identity a necessary condition for systemically significant cooperation. Weaponized Interdependence (Farrell & Newman, 2019) establishes that network fragmentation depends on structural position (control of hubs and chokepoints), not on state power or hegemonic status — the chokepoint effect is inherently actor-agnostic. Hirshleifer's aggregation technology framework (1983) provides the formal bridge: best-shot public goods (cooperation outcomes) require at least one highly capable actor, whilst weakest-link public goods (disruption vulnerability) are determined by the least capable or most disruptive node.

The asymmetry holds across all three T2 fragmentation types:

- **Restriction:** Supply-chain cuts can originate from any node with sufficient leverage. Critical supply-chain nodes command chokepoint positions (Netherlands: ~100% EUV lithography; Japan: critical deposition, etching, metrology equipment; Taiwan: ~90% advanced logic fabrication) where unilateral action by a non-principal actor produces fragmentation effects equivalent to or exceeding direct principal actor action. The T2 Restriction tests (Explicit-Targeting and PIET Impact-Demonstrated) measure the effect through named other-bloc targeting, trade flow disruption (>50%/>90%), price impacts (>20%), and government policy responses (≥ 2).
- **Buildout:** Parallel infrastructure development creates structural separation through the existence of alternative capacity, not through the identity of the builder. The Basel 5-Factor framework (\$100M floor, cross-jurisdictional significance, substitutability assessment) directly measures whether the buildout creates structurally significant parallel capacity.
- **Mandatory Standard:** Regulatory barriers create fragmentation through enforcement scope and compliance burden, not through the identity of the legislating entity. The EU currently exercises regulatory authority with extraterritorial reach without qualifying as a principal AI actor under the §5.3 dimensional assessment. The Mandatory Standard Test (mandatory language + documented enforcement) measures the structural barrier directly.

Clayton, Maggiori & Schreger (2024) provide the closest single-framework articulation: their model of economic coercion and fragmentation mathematically demonstrates that global integration is driven by hegemonic coordination of externalities (supporting T1's actor requirement), whilst fragmentation is driven by uncoordinated, structurally-determined pursuit of economic security by actors at any scale (supporting T2's effect-based approach).

Evidence Chain:

Precedent	Type	Contribution
Keohane (1984) <i>After Hegemony</i>	Direct	Cooperation requires great-power leadership; disconfirmation condition makes actor identity necessary for systemically significant cooperation
Kindleberger (1973) <i>The World in Depression</i>	Direct	Systemic integration requires a hegemon willing to absorb asymmetric costs; absence of leadership produces fragmentation
Farrell & Newman (2019) "Weaponized Interdependence," <i>International Security</i> 44(1)	Direct	Coercive/disruptive power derives from structural position in network topology, not state size; chokepoint effect is actor-agnostic
Clayton, Maggiori & Schreger (2024) "A Theory of Economic Coercion and Fragmentation," NBER WP 33309 / BIS WP 1224	Structural	Only identified work modelling both hegemonic integration and structurally-driven fragmentation within a single framework
Hirshleifer (1983) "From weakest-link to best-shot"; Sandler (2004) <i>Global Collective Action</i>	Structural	Best-shot/weakest-link distinction provides formal basis for why cooperation is actor-dependent and disruption is actor-agnostic
Olson (1965) <i>The Logic of Collective Action</i>	Structural	"Privileged groups" provide foundational logic beneath HST; defection as rational default grounds the asymmetry
Strange (1988) <i>States and Markets</i>	Structural	Relational power (actor-centred) versus structural power (position-based) directly maps to T1/T2 actor-identity logic
Basel Committee G-SIB Assessment Methodology	Empirical	Actor-agnostic systemic significance assessment via structural factors; directly adapted by WEO for T2

Empirical observation: All 17 T2 events in the current database (as of May 2026) involve at least one principal AI actor as either the initiating entity or the targeted party. This reflects the current geopolitical landscape where US-China strategic competition drives the majority of between-bloc fragmentation activity. It is an empirical feature of the current database population, not a structural requirement of the methodology.

Confidence Assessment: WEO SYNTHESIS, 70%. Both T1 and T2 components are independently grounded at very high confidence (90%+). The connecting logic has formal precedent in Hirshleifer's aggregation technology and single-framework articulation in Clayton-Maggiori-Schreger. Confidence is bounded below 75% because no framework explicitly articulates the differential actor-identity requirement as a classification principle, and the Middle Power Theory challenge is genuine — if minilateral coalitions achieve significant cross-bloc AI coordination without US/China involvement, the T1 requirement faces empirical pressure. Confidence is above 65% because both components are independently strong, the connecting logic has formal precedent, and no framework was found arguing for uniform actor-identity requirements across both cooperation and fragmentation.

Validation Commitment: Monitor for edge cases where a T2 event qualifies without any principal actor involvement on either side. If such cases arise, assess whether the qualification frameworks correctly identified structural significance without the principal actor signal. Monitor for T1 empirical pressure from minilateral cross-bloc coordination. Review at each database doubling milestone.

5.4.1 Dimension Qualification Principles

Manual Section: §5.3.2 (Dimension Qualification Principles)

Category: CATEGORY CREATOR (75% confidence)

Rationale: The parent PAA framework (§5.3, 7-dimension, ≥ 3 threshold) is documented as CATEGORY CREATOR at 75% confidence in Rationale §5.4. The per-dimension qualification principles operationalise the existing "dominant or independently significant capability" language with specific definitions that resolve edge cases at the boundary of each dimension. No existing geopolitical or technology governance framework publishes explicit per-dimension qualification thresholds for AI infrastructure capability assessment. The frameworks surveyed for §5.4 (IISS Cyber Capabilities, Brookings "Broad Stack Leaders," Belfer CET Index, Chatham House AI Building Blocks, Basel G-SIB) all assess analogous dimensions but none provide the definitional specificity required for reproducible binary classification.

Key Design Decisions:

1. **Qualification principles as sentences, not quantitative thresholds.** Each dimension's qualification principle is expressed as a definitional sentence rather than a numerical threshold (e.g., market share percentage, revenue floor, benchmark percentile). This choice trades precision for durability: the AI infrastructure landscape evolves rapidly, and quantitative thresholds calibrated to 2026 market conditions would require constant recalibration. Definitional sentences permit consistent application across assessment cycles whilst accommodating landscape evolution.

2. **Subsidiary clause harmonisation.** A uniform subsidiary/acquisition clause is applied across Dimensions 1, 3, 4, and 5 — the four dimensions where corporate entities may be controlled from a different territory than their physical operations. This addresses edge cases (Habana Labs/Israel, Graphcore/UK, DeepMind/UK) where corporate control and physical operations diverge. The clause follows the same logic as Dimension 5's existing subsidiary treatment, extended for cross-dimensional consistency.

Sensitivity note: The subsidiary and acquisition clause affects the distribution of PAA scores across actors. Attribution of subsidiary lab outputs to the parent entity's territory (e.g., DeepMind → United States via Alphabet) concentrates Dimension 5 credits in actors hosting parent companies rather than those hosting research operations. As of V1.4 publication (May 2026), no actor would cross the $\geq 3/\geq 2$ -categories threshold under an alternative attribution rule granting strategic autonomy to subsidiary labs — this finding should be reassessed at each methodology version milestone, particularly as new actors develop frontier AI capabilities through subsidiary or acquisition structures.

3. **Subsystem inclusion in Dimension 3.** The equipment dimension extends to "sole-sourced subsystems and components essential to the functioning" of critical equipment, not only complete equipment systems. This covers the Zeiss SMT case (sole-source EUV lithography optics, without which no advanced fab can operate) that four independent AI research inputs unanimously identified as structurally necessary. The "sole-sourced" and "essential to functioning" qualifiers prevent the inclusion of commodity components.
4. **Sovereign control principle generalisation (Dimension 2 → all physical infrastructure dimensions).** The sovereign control principle applies territorial presence as the primary test for all physical infrastructure dimensions (D2 fabrication, D4 compute), reflecting the physical reality that these capabilities are geographically bound — a fab cannot be relocated, and its output serves the territory where it operates regardless of corporate ownership. D4 additionally requires both domestic HQ and territorial compute presence (see key design decision 8 below). Corporate capability dimensions (D1 accelerator design, D5 model development) are assessed by ownership chain — the corporate parent's territory has effective control through strategic direction and IP ownership. D3 (equipment) applies both tests. See key design decisions 7–8 below for the sovereign control principle's full specification and the D4 territorial compute requirement.
5. **Dimension 2/3 process-node threshold recalibration.** Dimension 2 uses a quantitative process-node threshold ($\leq 7\text{nm}$) as the boundary for "advanced" fabrication — an exception to the general design preference for definitional sentences over quantitative thresholds (see point 1 above). This threshold reflects the assessment-date definition of advanced semiconductor manufacturing and is subject to periodic recalibration as the global fabrication frontier advances. The threshold should be reviewed at each methodology version milestone against the current state of leading-edge production. The recalibration trigger is when the specified threshold ceases to discriminate between actors with frontier or near-frontier fabrication capability and those operating at mature commodity nodes — operationally, when ≥ 4 actors can independently manufacture at the threshold node without structural equipment dependencies. Recalibration adjusts the threshold downward (e.g., from $\leq 7\text{nm}$ to $\leq 5\text{nm}$ or $\leq 3\text{nm}$) to maintain analytical discrimination. This recalibration applies equally to Dimension 3, which references $\leq 7\text{nm}$ in its equipment substitutability test — when the Dimension 2 threshold is recalibrated, Dimension 3 must follow to maintain cross-dimensional consistency.

6. **Dimension 5 benchmark saturation monitoring.** The Dimension 5 qualification principle relies on "broadly recognised evaluation benchmarks" as external instruments to discriminate between frontier and non-frontier models. These benchmarks saturate over time — a benchmark on which the top 20 models score within 1–2% of each other no longer discriminates meaningfully. Unlike the $\leq 7\text{nm}$ threshold, this is not a quantitative anchor requiring recalibration but a dependency on external measurement instruments whose discriminating power degrades. The methodology does not prescribe specific benchmarks (this is deliberate — benchmark relevance evolves faster than methodology versions), but assessors should document which benchmarks informed each assessment cycle's D5 verdicts. When the benchmarks available at an assessment date can no longer clearly separate "top tier" from the broader field, this should be flagged as a reduced-confidence assessment rather than treated as an expansion of the qualifying set.

Additional Qualification Principles

7. **Sovereign control principle (governing rule).** Physical infrastructure dimensions (D2 fabrication, D4 compute) are assessed by territorial presence — the sovereign territory where infrastructure physically operates has effective control. Corporate capability dimensions (D1 accelerator design, D5 model development) are assessed by ownership chain — the corporate parent's territory has effective control through strategic direction and IP ownership. D3 (equipment) applies both tests: the entity must be headquartered in the actor's territory AND control the qualifying technology. D6–D7 (governance) are inherently sovereign. *Precedent basis:* International investment law (territorial jurisdiction vs nationality of investment in BIT framework), OECD permanent establishment doctrine, WTO rules of origin, CFIUS screening rationale. *Establishing cases:* Netherlands D4 (Nebius — Dutch corporate domicile, zero Dutch-territory compute → NOT MET); Taiwan D1 (MediaTek — Taiwanese firm, client retains architectural IP under design-service model → NOT MET under ownership chain test); US D2 (TSMC Arizona — Taiwanese-owned fab on US soil → MET under territorial test).
8. **D4 territorial compute requirement.** "Domestically controlled" in D4 requires operational compute on the actor's territory or under the actor's direct sovereign jurisdiction, not merely corporate domicile of the parent entity. Corporate ownership of compute infrastructure entirely abroad does not satisfy D4. Both D2 and D4 assess physical infrastructure; the sovereign control principle applies the same territorial presence test to both. *Establishing case:* Netherlands / Nebius Group N.V. — Dutch-domiciled company with zero qualifying compute on Dutch soil; all operational compute located in Finland (Mäntsälä). D4 NOT MET for the Netherlands.
9. **D6 dual-pathway clarification.** The Dimension 6 qualification principle covers both AI-specific legislation and general-purpose regulatory or trade control frameworks whose primary documented application targets AI systems, AI compute hardware, AI models, or AI-relevant manufacturing equipment. Export control regimes with documented extraterritorial enforcement qualify alongside dedicated AI-specific regulatory frameworks. This clarification ensures that actors exercising AI-relevant extraterritorial authority through trade controls (e.g., US export controls on advanced semiconductors and AI chips) are assessed equivalently to actors exercising authority through dedicated AI legislation (e.g., EU AI Act).

10. **D7 national proportionality standard.** "Nationally transformative" in Dimension 7 is assessed relative to the actor's own GDP, fiscal capacity, and industrial base — not against a global spending threshold. A modest absolute spend that represents a structural reorientation of a smaller actor's industrial policy may satisfy the dimension, whilst a large absolute spend by a major economy that represents marginal budgetary commitment may not. *Establishing case:* Netherlands — modest absolute spend relative to US or China, but representing fundamental reorientation of Dutch industrial and technology policy towards sovereign AI infrastructure capability.
11. **D4 assessment-date-relative threshold.** The D4 hyperscale threshold ("sufficient to train a current-generation frontier foundation model") is assessment-date-relative. "Current-generation frontier" shifts as frontier training scales up; no fixed GPU count is prescribed. At each assessment cycle, the threshold is calibrated against the most recent verified frontier training runs. Clusters significantly below the prevailing frontier training scale are clearly NOT MET; clusters at or above are clearly MET; actors between the two require case-specific evidence (empirical frontier-class training run, or demonstrable technical configuration showing frontier training feasibility). *Establishing cases:* South Korea NOT MET at ~4,000 B200 GPUs, Taiwan NOT MET at ~4,608 GB200, Germany NOT MET at ~10,000 Blackwell (~0.5 ExaFLOPS vs ~5+ ExaFLOPS required), India NOT MET at 16,384 H100 (empirical training run produced sub-frontier model).
12. **D7 general-infrastructure-fund exclusion.** General infrastructure funds not earmarked for AI or technology do not satisfy Dimension 7. The dimension assesses sovereign AI infrastructure investment, not aggregate government capital expenditure. Attribution of a country's entire infrastructure or defence budget to D7 inflates the figure and misrepresents the sovereign AI commitment. *Establishing case:* Germany — €500B Special Fund for Infrastructure and Climate Neutrality is a general infrastructure instrument, not creditable to D7.
13. **D7 announced-vs-executed investment.** Government adoption of committee recommendations or announcement of national project budgets does not satisfy D7 if execution data shows systematic under-delivery. The dimension requires evidence of actually appropriated and disbursed capital, not aspirational targets. Announced envelopes that have been systematically under-executed do not meet the "at scale" threshold. *Establishing case:* Russia — announced budgets exceeded executed spending by 5–20× across multiple budget cycles; Data Economy national project 11.2% executed by March 2025.
14. **D5 single-domain exclusion.** Frontier-competitive performance in a single capability domain (e.g., image generation, speech recognition, code) does not satisfy D5's "≥3 benchmarks across capability domains" requirement. The dimension tests general-purpose frontier capability, not domain-specific excellence. *Establishing case:* Germany — Black Forest Labs FLUX.2 is frontier in image generation but single-domain only.
15. **D2 marketing-node exclusion.** Semiconductor process nodes are assessed by their lithographic and transistor-density characteristics, not marketing labels. Intel's rebranded node names (Intel 7, Intel 4, Intel 3) are mapped to their actual technology class for assessment purposes. Intel 7 is assessed as 10nm-class (DUV, no EUV) and does not qualify for the ≤7nm threshold. *Establishing case:* Israel — Intel Fab 28 at Kiryat Gat operates Intel 7 / 10nm Enhanced SuperFin.

16. **Synthesis majority override rule.** A nominal model majority in the synthesis process can be overridden when the majority's interpretive framework contradicts its own cited evidence. The factual base must be agreed across models; the override applies on interpretive grounds only, where the majority draws a conclusion inconsistent with the evidence those same models cite. *Establishing cases:* France D5 (2-1 MET overridden — 44% GPQA Diamond vs 91-94% frontier is not "top tier" by any defensible reading); Russia D7 (2-1 MET overridden — announced budgets under-executed by 5-20×).
17. **D3 subsidiary clause.** The D3 dual test (HQ in territory + technology control) means foreign-parented subsidiaries with local manufacturing are assessed under the parent's territory, consistent with the subsidiary clause. This is distinct from D2, where the territorial presence exception credits any fab on an actor's soil regardless of ownership. D3 requires the qualifying entity to be headquartered in the actor's territory; a US-parented subsidiary manufacturing in Canada credits to the US, not Canada. *Establishing case:* Canada — Teledyne DALSA (Teledyne Technologies subsidiary, US parent) manufactures EUV pellicle membranes at Bromont, QC; credits to US D3, not Canadian D3. Consistent with exclusion of Applied Materials Israel and KLA Israel from Israel's D3.
18. **Actor Designation Framework temporal disclaimer.** Designation scores are assessment-date-stamped snapshots of current capabilities. They are not retrospective assessments of capabilities at the time each event in the database occurred. Bloc classifications for individual events were assigned under the methodology version current at the time of entry. The Actor Designation Framework does not retroactively alter event-level classifications made under prior methodology versions. This distinction is documented in the platform interface alongside the designation display.

Evidence Basis: Dimension qualification principles were calibrated through multi-agent independent AI research synthesis — a methodology necessitated by the absence of historical empirical data or established expert consensus for sovereign AI infrastructure capability assessment at dimension-level granularity. Multiple independent AI research inputs assessed each actor against each dimension independently; a dedicated synthesis process consolidated convergences, resolved divergences, and applied the majority override rule (addition 16 above) where interpretive disagreement conflicted with cited evidence.

Edge-case stress testing was conducted across four independent AI research inputs with dedicated synthesis (May 2026), producing 84 edge cases (3 per dimension × 7 dimensions × 4 inputs). Three dimensions required structural revision (D1, D3, D6); four required targeted additions (D2, D4, D5, D7). The cross-input convergence on the Zeiss/D3 subsystem gap (4/4 inputs, unanimous) and the subsidiary clause inconsistency (4/4 inputs in summary sections) provided the highest-confidence findings informing the initial design. Full Batch 1 actor assessments (14 actors across 3-4 independent AI research inputs per actor, with dedicated synthesis instances) produced 34 calibration precedents, 18 methodology additions, and 2 instances where nominal input majorities were overridden on evidence-quality grounds (France D5, Russia D7).

Confidence Assessment: 75%. Inherits from the parent PAA framework (75%) with calibration refinements from Batch 1 actor assessments. The edge-case stress test across four independent AI research inputs provided initial validation, and the qualification principles are applied across fourteen actors in the Batch 1 assessment cycle — producing a distribution (2 PAA, 3 AIK, 2 ACS, 7 Participant) that discriminates meaningfully across all four designations. The sovereign control principle, D4 territorial compute requirement, and D7 national proportionality standard are grounded in establishing cases across multiple actors. Thirty-four calibration precedents are established across all seven dimensions, covering the principal boundary conditions (inference-only exclusion, subsystem scope, frontier threshold calibration, operational-at-assessment-date, AI-specificity, foreign-base fine-tune exclusion, national proportionality, general-fund exclusion, announced-vs-executed investment, marketing-node exclusion, subsidiary clause in D3). Residual uncertainty remains at the boundary — the India/D5 (Sarvam-105B self-reported benchmarks) and France/D5 (Mistral ~35% frontier deficit) edge cases set the current D5 threshold but may shift with model releases.

Validation Commitment: Batch 1 assessment complete (14 actors). Continue reviewing qualification principles against Batch 2 actor assessment data. Particular attention to edge cases at the D5 frontier threshold (~35% deficit precedent from France/Mistral) and the D4 assessment-date-relative threshold (which will shift as frontier training compute increases). Threshold sensitivity review when any actor's score changes at the ≥ 3 boundary. Review the Dimension 2/3 process-node threshold ($\leq 7\text{nm}$) against the current fabrication frontier at each methodology version milestone; recalibrate downward when the threshold ceases to discriminate (see key design decision 5). Monitor Dimension 5 benchmark saturation and document which benchmarks informed each assessment cycle's D5 verdicts (see key design decision 6). Reassess subsidiary clause sensitivity at each methodology version milestone (see sensitivity note under key design decision 2). Review sovereign control principle test assignments if new dimensions are introduced (see key design decision 7). Monitor D7 national proportionality assessments for cross-actor consistency (see key design decision 10). Verify disbursement data for D7 where execution rates are flagged (see addition 13).

5.4.2 Three-Gate PAA Designation, Severance Test, and AIK Classification

Manual Section: §5.3.1 (Principal AI Actor Designation: Three-Gate Structure)

Category: CATEGORY CREATOR (70–75% confidence)

Rationale: A capability threshold alone (≥ 3 of 7 dimensions) would conflate capability breadth with ecosystem independence. As additional actors accumulate capability breadth whilst remaining structurally dependent on an existing PAA's ecosystem for critical inputs, the threshold must be supplemented by a resilience assessment. Without the severance test, the designation would potentially classify Japan (deeply embedded in the US-centred ecosystem despite significant sovereign capabilities) as an ecosystem anchor equivalent to the United States. The three-gate structure separates capability (Gate 1), independence (Gate 2), and ecosystem provision (Gate 3) — three analytically distinct properties that a single threshold cannot discriminate.

Closest Precedents:

Framework	Institution	Relevance	Divergence from WEO
Basel III Stress Testing	Basel Committee on Banking Supervision (BIS, 2010/2017)	Closest methodological analogue: institutions must maintain capital ratios (quantitative threshold) AND survive hypothetical adverse scenarios (stress test). Dual-gate structure: quantitative floor + resilience assessment	Financial sector, not technology infrastructure; stress scenarios are macroeconomic, not supply-chain severance; assessment is continuous (capital ratio), not binary
Weaponised Interdependence: chokepoint vulnerability	Farrell & Newman (2019)	Establishes that network fragmentation depends on structural position and that chokepoint control creates asymmetric dependencies — the theoretical basis for why capability breadth alone does not imply ecosystem independence	Descriptive theory, not an assessment methodology; does not produce actor classifications
Strategic Autonomy discourse	European Commission (2021 Strategic Compass; 2023 Economic Security Strategy)	Policy framework asserting that capability without independence is insufficient for sovereign action — conceptual validation of the dual-gate principle	Political discourse, not a codified assessment methodology; "strategic autonomy" is not operationally defined with testable criteria
IISS Cyber Capabilities Assessment	IISS (2021/2023)	Tier 1 designation requires both capability breadth AND the ability to exercise cyber power independently; lower tiers explicitly note dependency limitations	Qualitative tiers without explicit severance criteria; cyber, not AI infrastructure

Key Design Decisions:

1. **Hypothetical withdrawal, not observed disruption.** The severance test uses a hypothetical scenario rather than waiting for actual supply-chain disruption. This follows Basel III stress testing methodology: the test assesses structural resilience, not historical experience. Waiting for actual ecosystem severance before classifying would provide accurate data only after catastrophic events.
2. **24–36 month sustainability window.** The test asks whether met dimensions would remain operationally sustainable within 24–36 months, not indefinitely. This window reflects the approximate timeline for emergency substitution of critical AI infrastructure inputs (alternative chip supply, equipment qualification, model retraining on different compute). A shorter window would over-credit actors with large existing inventories; a longer window would under-credit actors with genuine but time-limited domestic capability.
3. **Three capability categories with ≥ 2 spanning requirement.** The hardware/platform/governance trichotomy prevents an actor from qualifying as a PAA based solely on regulatory authority and investment (governance dimensions) without any hardware or platform independence. An actor exercising extraterritorial regulation and directing sovereign investment but possessing no domestic accelerator design, fabrication, equipment, cloud compute, or frontier models is not an ecosystem anchor — it is a regulatory power operating within another actor's technological ecosystem.
4. **Advanced Capability State as a distinct classification.** Rather than treating the severance test as a simple pass/fail that leaves non-qualifying actors undifferentiated, the methodology introduces a distinct classification tier. This captures the analytically significant difference between an actor scoring 4/7 with ecosystem dependence (e.g., the EU, which fails the severance test due to Intel ownership and Cymer dependency) and an actor scoring 0/7 (e.g., UAE). The "advanced capability state" classification is informative for users assessing supply-chain permission risk.
5. **AI Infrastructure Keystone (AIK) as a distinct classification.** The three-gate structure produces a classification tier between PAA and ACS: actors that score ≥ 3 , pass the severance test (demonstrating genuine ecosystem independence), but hold no sustainable platform dimension (D4 or D5). These actors exercise structural chokepoint leverage within an ecosystem — through equipment monopolies, fabrication capacity, or governance authority — without providing the ecosystem substrate (compute or model development platforms) upon which other actors operate. The AIK concept may be inherently asymmetric: it characterises the distributed Western supply chain (where equipment, fabrication, and governance are dispersed across allied actors) rather than vertically integrated ecosystems (where a single actor controls the full stack by necessity). *Establishing cases:* Japan (4/7, D3/D4/D6/D7, severance PASS; D4 met at assessment date but degraded-but-sustained under severance due to US-designed accelerator dependency), South Korea (4/7, D2/D3/D6/D7, severance PASS), Netherlands (3/7, D3/D6/D7, severance PASS).

Evidence Basis: The gate structure addresses the case where actors accumulate capability breadth whilst remaining ecosystem-dependent — Japan at 4/7 is the defining example. Analysis of Japan's dependency chain demonstrates that its met dimensions (equipment, compute, regulation, investment) include structural dependencies on US-ecosystem inputs (accelerator design, frontier models) — a clear case where capability breadth does not automatically imply ecosystem equivalence with the two PAAs. Basel III stress testing provides the strongest structural precedent for combining a quantitative threshold with a resilience assessment. The third gate (sustainable platform dimension) addresses actors that pass the severance test but hold no sustainable platform dimension (Japan, South Korea, Netherlands) — distinguishing chokepoint leverage from ecosystem anchoring. The AIK classification resolves this by distinguishing actors that shape supply chains from actors that provide the ecosystem substrate. The severance test is applied across seven actors with two distinct outcomes: PASS (US 7/7 sustainable, China D1/D5/D6/D7 sustainable with D2 and D4 degraded-but-sustained, Japan D3/D6/D7 sustainable with D4 degraded-but-sustained, South Korea D3/D6/D7 sustainable with D2 degraded-but-sustained, Netherlands D3/D6/D7 with D3 sustained under stress with multi-vector exposure from Cymer jurisdiction plus five additional US leverage vectors identified in FDPR sub-tier research) and FAIL (EU — D2 collapses on Intel ownership, D3 at structural risk from Cymer and broader US leverage; Taiwan — all three MET dimensions structurally coupled to US ecosystem). The severance test's operational components — ecosystem inputs definition, per-PAA testing, deterrence principle, cascade boundary, and dimension sustainability classifications — are documented in §5.4.3–§5.4.7.

Confidence Assessment: 70–75%. The three-gate structure addresses a genuine analytical gap in the parent framework, and the Basel III stress testing analogy provides strong methodological precedent for the approach. The severance test is applied across seven actors (US, China, Japan, South Korea, Netherlands, EU, Taiwan), producing two distinct outcomes (PASS: US, China, Japan, South Korea, Netherlands; FAIL: EU, Taiwan) that are internally consistent and analytically meaningful. The three-gate structure produces a distribution (2 PAA, 3 AIK, 2 ACS) that discriminates meaningfully among actors scoring ≥ 3 . Confidence is bounded below 80% because: (a) the 24–36 month sustainability window remains a calibration choice without direct precedent in technology infrastructure assessment, though its application across seven actors has not produced anomalous results; (b) the AIK classification is a novel construct whose long-term analytical utility will depend on whether the structural pattern it captures (chokepoint leverage without platform provision) remains stable across assessment cycles and Batch 2 actors; and (c) the three-category spanning requirement (hardware/platform/governance) is a novel analytical construct. Confidence is above 70% because: the structural analogy to Basel stress testing is well-grounded; the underlying theoretical basis (Farrell & Newman's weaponised interdependence) is established at very high confidence; the seven-actor application has produced a distribution that discriminates meaningfully; and the alternative (single- or dual-gate structures) produces demonstrably less useful classifications.

Validation Commitment: Continue applying the three-gate structure to Batch 2 actors. Review the 24–36 month window and category spanning requirement against cumulative results. Monitor for actors that transition between designations as their dependency chains evolve (Dimension Watch items tracked in the register). Assess whether the AIK classification produces stable, useful discrimination or converges with ACS over time. Review at each methodology version milestone.

5.4.3 Ecosystem Inputs Definition

Manual Section: §5.3.1 (Severance Test Criteria — "Ecosystem inputs defined")

Category: CATEGORY CREATOR refined (inherits 70–75% from §5.4.2)

Rationale: The ecosystem inputs definition uses a two-part (a)/(b) structure to draw a precise boundary around what the severance test withdraws: (a) inputs directly produced or controlled by entities under the PAA's sovereign jurisdiction, and (b) inputs produced by entities outside the PAA's jurisdiction but subject to denial through the PAA's extraterritorial regulatory instruments (FDPR, direct EAR jurisdiction). This boundary follows the PAA's sovereign denial capability — its ability to prevent an input from reaching the actor under test — rather than its manufacturing footprint or its geopolitical bloc membership. The China severance assessment established this boundary: Korean HBM blocked by FDPR is treated as a US ecosystem input (the US controls access through FDPR), whilst Zeiss and TRUMPF (both German and Western Bloc members) are treated as unaffected by US withdrawal (the US cannot deny their output).

The definition explicitly excludes bloc-wide withdrawal — "The severance test withdraws ecosystem inputs as defined above; it does not withdraw all inputs originating from actors aligned to the PAA's geopolitical bloc" — because ecosystem inputs and bloc membership are distinct concepts. An actor can be in the same geopolitical bloc as the PAA without its inputs being subject to the PAA's denial capability. The boundary is drawn at what the PAA can deny, not at who the PAA is aligned with.

The (b) formulation including direct EAR jurisdiction was supported by the FDPR sub-tier research (May 2026), which found that ASM International's epitaxy product line is designed at ASM America in Phoenix, AZ — making it subject to direct EAR jurisdiction, not merely FDPR-eligible. The definition's reference to "inputs subject to direct Export Administration Regulations jurisdiction from US-based design or manufacturing activity" captures this finding.

Key Design Decisions:

1. **Boundary drawn at denial capability, not production.** A non-PAA actor's inputs blocked by the PAA's extraterritorial instruments (e.g., Korean HBM blocked by FDPR) are PAA ecosystem inputs because the PAA controls access to them. Inputs from the same non-PAA actor that are not blocked remain available. The boundary follows the PAA's sovereign reach, not its manufacturing footprint.
2. **Explicit bloc exclusion.** The final sentence of the definition — "it does not withdraw all inputs originating from actors aligned to the PAA's geopolitical bloc" — is operationally necessary because without it, an assessor could conflate ecosystem withdrawal with bloc withdrawal, producing trivially incorrect FAIL verdicts for every Western actor.

Evidence Basis: China severance assessment (FDPR ruling on Korean HBM and ASML equipment); Netherlands severance assessment (Zeiss/TRUMPF treated as non-US, unaffected — establishing the ecosystem ≠ bloc boundary); FDPR sub-tier multi-agent AI research synthesis (9 entities across Japan, South Korea, Netherlands; May 2026) confirming the ASM International direct-EAR finding.

Establishing Cases:

- China D1/D2: Korean HBM blocked by FDPR → US ecosystem input (withdrawn under severance)
- Netherlands D3: Zeiss (German) and TRUMPF (German) → not US ecosystem inputs (available under US severance, despite being Western Bloc members)
- Netherlands D3: ASM International epitaxy line designed at ASM America, Phoenix → direct EAR jurisdiction (US ecosystem input regardless of de minimis threshold)

Confidence Assessment: Inherits 70–75% from §5.4.2. The (a)/(b) boundary is grounded in the PAA's sovereign denial capability — a precise, testable criterion supported by the FDPR sub-tier research confirming direct EAR jurisdiction findings.

Validation Commitment: Review if new extraterritorial regulatory instruments emerge beyond FDPR/EAR. Monitor for cases where the ecosystem/bloc boundary produces anomalous results. Reassess if a third PAA emerges, as the definition's applicability to non-US extraterritorial instruments would need testing.

5.4.4 Per-PAA Severance Testing (Option B)

Manual Section: §5.3.1 (Severance Test Criteria — per-PAA testing)

Category: CATEGORY CREATOR refined (inherits 70–75% from §5.4.2)

Rationale: The choice of individual-PAA testing (each PAA tested separately, actor must pass each) over simultaneous-PAA testing (all PAAs withdrawn at once) follows established institutional practice for structural dependency assessment. Individual-scenario testing is the institutional default for classification and certification across financial, infrastructure, and supply-chain frameworks. Combined worst-case scenarios are reserved for non-binding strategic planning and resilience exercises. The methodological principle is that single-source dependency is a structural property of the assessed actor, whilst systemic fragility is a property of the global environment. Conflating them in one test produces a result that cannot be attributed to either cause, destroying the diagnostic value that classification is designed to provide. The severance test asks whether an actor's capability stack is structurally contingent on another actor's permission — a question that requires isolating each potential source of contingency.

Closest Precedents:

Framework	Institution	Scenario Application	Relevance to WEO
NERC Reliability Standard TPL-001-4/5.1	North American Electric Reliability Corporation	Planning Events (P1–P7) tested individually, binding; Extreme Events (cascading) evaluative only	Closest formal codification: Planning Events ↔ WEO Option B (binding individual test); Extreme Events ↔ Option C (evaluative cascading study)
Critical Raw Materials Act, Article 5	European Union (Regulation 2024/1252)	65% single-third-country cap as binding law	Legislatively binding single-source dependency threshold — cleanest legislative formalisation of the principle WEO applies
EBA Guidelines on Recovery Plan Scenarios (EBA-GL-2014-06)	European Banking Authority / FSB	Idiosyncratic, systemic, and combined shocks tested as separate parallel exercises	Explicit principled statement that different shock types must be tested separately because they answer different methodological questions
EU Strategic Dependencies (SWD 2021/352)	European Commission	HHI concentration metrics across ~5,000 products; single-supplier shock modelling	Operationalises single-source dependency through product-level concentration analysis
Basel Large Exposures Framework (BCBS d283)	Basel Committee	25% of Tier 1 capital cap on single-name exposure	Regulatory codification of single-counterparty dependency limit

Key Design Decisions:

1. **Per-PAA individual testing for structural classification.** The severance test isolates each PAA's ecosystem inputs and tests withdrawal individually because the framework identifies structural dependency on specific ecosystems — a property of the actor being assessed. Every institutional precedent surveyed applies individual-scenario testing for classification and certification: NERC tests Planning Events individually (P1–P7), EBA mandates separate idiosyncratic and systemic exercises, and the EU Critical Raw Materials Act sets single-country concentration thresholds. Simultaneous multi-source withdrawal addresses a different analytical question (systemic environmental fragility) and is used in planning contexts, not classification.

Evidence Basis: Deep research across two independent AI research inputs (May 2026) surveying Basel III, IISS, NATO, NIST, NIS2, G-SIB, EU Strategic Dependencies, EU CRM Act, and IMF/OECD trade vulnerability frameworks. Finding: individual-scenario testing is the consensus institutional default for classification. The NERC TPL-001 Planning Events / Extreme Events bifurcation provides the single strongest formal precedent.

Confidence Assessment: Inherits 70–75% from §5.4.2. The per-PAA choice is strongly supported by institutional precedent.

Validation Commitment: Monitor for the emergence of a third PAA, at which point secondary and tertiary PAA testing becomes operationally significant. Review the per-PAA vs simultaneous distinction if any actor's designation depends on which approach is used.

5.4.5 Deterrence Principle

Manual Section: §5.3.1 (Severance Test Criteria — "Deterrence principle")

Category: CATEGORY CREATOR refined (inherits 70–75% from §5.4.2)

Rationale: The deterrence principle classifies a dependency as "sustained under stress" rather than "collapsed" when the withdrawn PAA holds jurisdiction over a subsidiary or facility whose denial would be self-destructive for the PAA. This addresses the analytical problem that some dependencies are theoretically severable but practically unseverable because the PAA cannot exercise its denial capability without harming its own infrastructure capability. The three-part documentation requirement ensures assessor consistency by requiring explicit evidence for each deterrence claim.

Key Design Decisions:

1. **Three-part documentation requirement.** Each deterrence assessment must document: (a) the specific facility at risk with jurisdictional basis, (b) the denial mechanism, (c) the specific cost to the PAA. The requirement ensures consistency across assessments — the Cymer dependency affects both the Netherlands and the EU, requiring explicit documentation of why deterrence applies differently for each (the Netherlands holds direct sovereign jurisdiction over ASML; the EU does not). The three-part structure makes each assessment explicit and auditable.
2. **Multi-vector leverage assessment.** The FDPR sub-tier research (May 2026) revealed that the US holds five independent leverage vectors over Netherlands D3 beyond Cymer: ASML Wilton reticle top module (confirmed physical hardware), Brion computational lithography software (US-developed), HMI e-beam R&D, ASM International epitaxy line (direct EAR jurisdiction), and Besi Kinex cluster (Applied Materials co-development). Multiple leverage points weaken the deterrence argument because the PAA can apply graduated pressure through secondary vectors without triggering the self-harming consequences of disrupting the primary entity. The "sustained under stress with multi-vector exposure" classification captures this narrower margin of sustainability.
3. **Supranational entity clause.** Deterrence based on a member state's sovereign leverage over a headquartered entity does not automatically transfer to the supranational level. ASML is headquartered in the Netherlands (an EU member state), but the EU as a supranational entity does not hold the direct corporate-law jurisdiction, nationalisation authority, or operational sovereignty over ASML that the Netherlands holds as ASML's home state. The EU can regulate ASML (competition law, Foreign Subsidies Regulation) but cannot ensure ASML's continued operation against PAA pressure in the way the Netherlands can. The clause ensures that each actor's deterrence assessment is based on its own sovereign jurisdiction over the relevant entity, producing the correct result: deterrence applies for the Netherlands (home state) but not for the EU (supranational entity).

Establishing Cases:

- Netherlands D3 / Cymer: deterrence applied → "sustained under stress with multi-vector exposure" (five additional US leverage vectors beyond Cymer)
- EU D3 / Cymer: deterrence NOT applied → "at structural risk" (supranational entity lacks direct sovereign jurisdiction over ASML; EU does not satisfy the supranational entity clause)

Evidence Basis: Netherlands severance assessment (Cymer deterrence precedent); EU severance assessment (supranational entity exclusion precedent); FDPR sub-tier multi-agent AI research synthesis (five US leverage vectors beyond Cymer confirmed: Wilton reticle module, Brion software, HMI R&D, ASM International direct EAR, Besi/Applied Materials partnership).

Confidence Assessment: Inherits 70–75% from §5.4.2. The deterrence principle is grounded in standard deterrence logic (mutual assured destruction applied to supply-chain dependencies). The multi-vector extension and supranational entity clause are grounded in deterrence logic and supported by establishing cases.

Validation Commitment: Monitor for changes in PAA leverage vectors (e.g., if Cymer operations are relocated to Europe, if ASML reduces US-origin content). Review supranational entity clause if EU institutional authority over corporate operational decisions strengthens. Monitor for new deterrence cases in Batch 2 assessments.

5.4.6 Cascade Analysis Boundary

Manual Section: §5.3.1 (Severance Test Criteria — "Cascade analysis boundary")

Category: CATEGORY CREATOR refined (inherits 70–75% from §5.4.2)

Rationale: The single-level cascade boundary limits the depth of dependency analysis in the severance test. The assessor checks whether the actor's primary capability-sustaining entities directly depend on the withdrawn PAA's ecosystem inputs. Sub-tier dependencies (e.g., whether a Japanese equipment company's sub-components contain US-origin FPGAs) are documented as constraints but do not determine the severance verdict. This boundary balances analytical rigour with practical tractability: full sub-tier supply chain mapping for every entity in every MET dimension is intractable from public sources, whilst ignoring dependencies entirely would produce unreliable verdicts.

Key Design Decisions:

1. **Single-level analysis is sufficient.** The FDPR sub-tier research (May 2026) supported the single-level standard by demonstrating that entity-level analysis — examining US-origin design centres, US-acquired IP, US-jurisdiction subsidiaries, and US-dominated subsystem categories — identified the structural exposure patterns that determine severance verdicts. The research identified material findings at the single-level (ASM International direct EAR jurisdiction, ASML Wilton reticle module, Hanmi HIGH FDPR exposure) without requiring intractable component-level BOM analysis.
2. **Sub-tier findings documented as constraints, not verdicts.** Sub-tier FDPR/EAR exposure is real but unverifiable from public sources at the component level. Incorporating unverifiable findings into the primary severance verdict would degrade the methodology's evidence standard (§5.3.4). The constraint register captures the finding without degrading the methodology — creating a structured record for future investigation without changing current designations.
3. **Single-level boundary extends to non-PAA supply chain assessment.** When criterion 1 requires assessing whether non-PAA actors' supply chains would remain functional under severance, the same single-level boundary applies: the assessor checks whether the non-PAA actor's primary entities depend on the withdrawn PAA's ecosystem inputs, without tracing sub-tier dependencies.

Evidence Basis: FDPR sub-tier multi-agent AI research synthesis (9 entities across Japan, South Korea, Netherlands). Entity-level (single-level) analysis identified: TEL MODERATE FDPR exposure, Lasertec LOW-MODERATE, SCREEN MODERATE-LOW, HPSP MODERATE, Hanmi HIGH, ASML HIGH (non-Cymer), ASM International HIGH (direct EAR), Besi HIGH (Kinex) / MODERATE (standalone). These findings were consequential for constraint documentation (South Korea and Netherlands flagged for review) without requiring sub-tier BOM analysis.

Confidence Assessment: Inherits 70–75% from §5.4.2. The single-level boundary is supported by the FDPR research demonstrating that material findings emerge at entity level.

Validation Commitment: Monitor whether sub-tier findings (documented as constraints) produce designation-changing evidence in future assessment cycles. If sub-tier FDPR exposure is confirmed to change any entity's sustainability classification, reassess whether the single-level boundary should be tightened for specific tool categories.

5.4.7 Dimension Sustainability Taxonomy

Manual Section: §5.3.1 (Severance Test Criteria — "Dimension sustainability classifications")

Category: CATEGORY CREATOR refined (inherits 70–75% from §5.4.2)

Rationale: The five-level dimension sustainability taxonomy provides a closed set of classifications for assessing each met dimension under each per-PAA severance scenario. The five levels — sustainable, sustained under stress, sustained under stress with multi-vector exposure, degraded-but-sustained, and at structural risk — capture the analytically significant gradations between full independence and structural collapse. The taxonomy includes an explicit counting rule: only the first four levels count towards the ≥ 3 threshold; "at structural risk" does not.

Key Design Decisions:

1. **Five levels rather than binary (pass/fail).** A binary dimension-level classification (sustainable / not sustainable) would lose analytically significant information. The difference between "sustainable" (Japan D3 — no US dependency in equipment itself) and "sustained under stress with multi-vector exposure" (Netherlands D3 — five US leverage vectors but deterrence holds) is material for risk assessment, even though both count towards the ≥ 3 threshold. The taxonomy preserves this gradation.
2. **"Degraded-but-sustained" as a counting level.** A dimension that degrades under severance but remains above the §5.3.2 qualification threshold counts towards the ≥ 3 criterion. This is analytically correct because the severance test asks whether the actor can sustain its capability stack, not whether it can sustain it without any degradation. China's D2 (SMIC on installed DUV equipment, no new EUV access) degrades significantly but retains $\leq 7\text{nm}$ production within the assessment window — this is a meaningful capability, not an analytical fiction.

Establishing Cases:

Classification	Actor/Dimension	Basis
Sustainable	Japan D3 (TEL, Lasertec, SCREEN)	No material dependency on US ecosystem inputs for core product lines
Sustained under stress	(Pre-FDPR) Netherlands D3	Cymer dependency with deterrence applied
Sustained under stress with multi-vector exposure	Netherlands D3	Five additional US leverage vectors beyond Cymer weaken deterrence calculus
Degraded-but-sustained	China D2 (SMIC)	Existing DUV production continues but cannot advance to new nodes
Degraded-but-sustained	China D4 (Ascend + stockpiled NVIDIA)	Existing compute degrades but retains frontier training capacity within window
Degraded-but-sustained	Japan D4 (SoftBank cluster)	Existing NVIDIA GPUs operational, no replenishment under US severance
Degraded-but-sustained	South Korea D2 (Samsung Foundry)	Existing ASML equipment operational, no resupply under US severance
At structural risk	EU D2 (Intel Fab 34)	US-owned facility; Intel could withdraw
At structural risk	EU D3 (ASML)	Cymer + multi-vector exposure; supranational entity clause blocks deterrence
At structural risk	Taiwan D2 (TSMC)	Requires US equipment and EDA tools
At structural risk	Taiwan D3 (Gudeng, GMM, GPTC)	Subsystem value contingent on fab ecosystem; D2 collapse renders subsystem capability moot
At structural risk	Taiwan D7 (NCHC/TSMC programme)	Programme procures NVIDIA GPUs; compute delivery dependent on US ecosystem

Confidence Assessment: Inherits 70–75% from §5.4.2. The five classifications are grounded in the analytical distinctions required by the severance test, with establishing cases across all five levels.

Validation Commitment: Monitor whether the five levels remain sufficient or whether Batch 2 assessments reveal edge cases requiring additional classifications. Review the counting rule (first four count, "at structural risk" does not) if any actor's designation depends on the boundary between "degraded-but-sustained" and "at structural risk."

5.4.8 AI Infrastructure Keystone (AIK) Designation

Manual Section: §5.3.1 (Actor Designation Framework — Gate 3)

Category: ADAPTED (82% confidence, with synthesis elements)

Rationale: The AIK designation classifies nation-state actors scoring ≥ 3 dimensions, passing the per-PAA severance test, but holding no sustainable platform dimension (D4 Cloud AI Compute or D5 Frontier AI Model Development). These actors hold structural chokepoint leverage within an AI ecosystem without providing the ecosystem itself. The structural distinction between ecosystem-providing power and chokepoint-leveraging power is not novel to WEO — it has been independently theorised across international relations, ecology, industrial organisation, network science, and documented in historical cases. WEO's contribution is the bounded application of this well-established structural distinction to sovereign AI infrastructure capability assessment, combined with operational calibration ($\geq 3/7$ dimensional threshold, 24–36 month severance test, Gate 3 platform-dimension test).

The designation was supported by cross-disciplinary deep research across four independent AI research inputs with dedicated synthesis (May 2026), surveying frameworks across five domains. Eight independent intellectual traditions converge on essentially the same bifurcation between ecosystem-providing/dominant/hub/platform power and chokepoint-leveraging/keystone/bridge/bottleneck power, suggesting the distinction tracks a real structural feature of complex networked systems.

Closest Precedents:

Framework	Author/Institution	Year	Relevance	Divergence from WEO
Vertical vs Horizontal Foreclosure	Rey & Tirole	2007	Primary theoretical anchor. Formally separates bottleneck-input power (AIK) from end-market dominance (PAA). Near-isomorphic. Embedded in peer-reviewed IO theory and active antitrust case law (US Section 2; EU Art. 102).	IO/competition framework, not geopolitical; treats firms, not nation-states
Structural vs Relational Power	Strange	1988	Cleanest IR-native binary. PAA = structural power (shapes the game); AIK = relational power (leverages position within the game).	Strange's four structures (security, production, finance, knowledge) do not map precisely to WEO's seven dimensions
Weakest-Link vs Best-Shot Aggregation	Hirshleifer	1983	Most elegant formal encoding. AIK = weakest-link gatekeeper whose removal degrades the system; PAA = best-shot provider whose contribution defines the frontier.	Public goods model; requires adaptation to supply-chain context
Keystone vs Dominant Species	Paine (1966); Power et al. (1996)	1966/1996	Strongest ecological analogue. AIK = keystone species (disproportionate systemic impact relative to abundance); PAA = dominant species (proportionate impact, high abundance). Power et al.'s canonical abundance-impact framework formalises the keystone concept.	Ecological metaphor requires disambiguation from Iansiti & Levien (2004) who use "keystone" for the PAA/hub position
Degree vs Betweenness Centrality	Freeman	1977/1979	Cleanest mathematical formalisation. PAA = high-degree hub (many connections); AIK = high-betweenness bridge (controls flow between clusters).	Pure scale-free networks correlate hub and bridge positions; distinction requires modular network structure (empirically satisfied in semiconductor GVCs)
DMA Gatekeeper vs Art. 102 Essential Facility	EU regulatory architecture	2022	Institutional embodiment: DMA governs PAA-type platform gatekeepers; Art. 102/Bronner governs AIK-type essential facility holders. Distinction embedded in positive law.	Regulates firms within single market, not nation-states across blocs
Weaponised Interdependence: Chokepoint vs Panopticon	Farrell & Newman	2019	Most cited contemporary geopolitical vocabulary. Chokepoint effect (denial = AIK) vs panopticon effect (surveillance/ecosystem provision = PAA).	Under-theorises non-hegemonic chokepoint case: F&N locate both effects in the same state (US), whereas WEO separates ecosystem providers from small-state chokepoint holders
Geopolitical Pivots vs Geostrategic Players	Brzezinski	1997	AIK = geopolitical pivot (importance derives from position, not intrinsic power); PAA = geostrategic player (importance derives from power and strategic vision).	Cold War framing; geographic rather than supply-chain positioning

Key Design Decisions:

1. **ADAPTED classification, not CATEGORY CREATOR.** The PAA/AIK structural distinction is pre-existing across five disciplines. Someone familiar with weaponised interdependence, keystone ecology, or vertical foreclosure theory could independently arrive at a similar classification. WEO's contribution is the bounded application to sovereign AI infrastructure capability and the operational calibration ($\geq 3/7$ threshold, severance test, Gate 3), not the underlying typology.
2. **Rey & Tirole (2007) as primary theoretical anchor.** Rey & Tirole's vertical/horizontal foreclosure bifurcation provides the strongest single formal precedent for three reasons: (a) formal isomorphism — bottleneck-input power vs end-market dominance IS the AIK/PAA distinction in IO theory; (b) operational foundation — embedded in peer-reviewed theory, active case law, and positive-law regulatory architecture; (c) predictive content — generates testable hypotheses about different strategic behaviours for vertical and horizontal power holders.
3. **PAA/AIK as ideal-type endpoints, not mutually exclusive categories.** All five researched domains warn that ecosystem-provision and chokepoint-leverage co-occur in many actors. The US is both PAA (frontier models, cloud compute) and AIK-adjacent (CUDA software stack, EDA tools). The WEO $\geq 3/7$ dimensional threshold and severance test handle hybrid cases procedurally through the Gate 3 platform-dimension test, but the designations capture the dominant mode of structural leverage, not the exclusive mode.
4. **Iansiti-Levien disambiguation required.** Iansiti & Levien's *The Keystone Advantage* (2004) uses "keystone" for platform hub orchestrators (Microsoft, Walmart) — aligning with PAA, not AIK. WEO's "AI Infrastructure Keystone" aligns with Paine's (1966) ecological meaning — disproportionate systemic impact relative to abundance — not the Iansiti-Levien business-strategy meaning. Manual §5.3.1 includes the disambiguation parenthetical.
5. **AIK fragility acknowledged as structural feature, not classification defect.** Historical precedent demonstrates non-hegemonic chokepoint power erodes through substitution (OPEC post-1980), hegemonic coercion (Switzerland post-2008), or technological bypass (canal states). The 24–36 month severance test incorporates temporal bounds. AIK status is inherently provisional and subject to reassessment at each methodology version milestone, distinguishing it from PAA status which is more structurally durable. AIK leverage in the semiconductor supply chain is structurally deeper than commodity leverage — protected by cumulative process knowledge, IP portfolios, and capital barriers measured in decades — but it is not permanent.
6. **Österblom operational criteria adopted for AIK identification.** To inoculate against the keystone-concept vagueness critique (Mills, Soulé & Doak 1993; Cottee-Jones & Whittaker 2012), the designation adopts Österblom et al.'s (2015) four-criteria approach adapted for AI infrastructure: (i) production concentration, (ii) segment control, (iii) network reach, (iv) governance influence.
7. **Formalising the non-hegemonic chokepoint gap in weaponised-interdependence theory.** Farrell & Newman (2019) identified chokepoint and panopticon effects but located both in the same hegemon (US), under-theorising the case of small states hosting non-substitutable chokepoints within a hegemon's broader ecosystem. WEO separates hegemon-hosted chokepoints from small-state-hosted chokepoints that F&N gesture at but do not formalise.

Evidence Chain:

Source	Type	Contribution
Rey & Tirole (2007) "A Primer on Foreclosure," <i>Handbook of Industrial Organization</i> Vol. 3	Direct	Formally separates vertical foreclosure (AIK) from horizontal foreclosure (PAA). Near-isomorphic.
Strange (1988) <i>States and Markets</i>	Direct	Structural power (PAA) vs relational power (AIK). IR-native binary.
Hirshleifer (1983) "From Weakest-Link to Best-Shot," <i>Public Choice</i> 41(3)	Direct	Best-shot (PAA as frontier provider) vs weakest-link (AIK as supply-chain bottleneck).
Paine (1966) "Food Web Complexity and Species Diversity," <i>American Naturalist</i> 100(910)	Direct	Keystone species concept — disproportionate impact relative to abundance = AIK.
Power et al. (1996) "Challenges in the Quest for Keystones," <i>BioScience</i> 46(8)	Direct	Formalised keystone-vs-dominant distinction.
Freeman (1977/1979) centrality measures, <i>Sociometry</i> 40(1); <i>Social Networks</i> 1(3)	Direct	Degree centrality (hub = PAA) vs betweenness centrality (bridge = AIK).
Farrell & Newman (2019) "Weaponized Interdependence," <i>International Security</i> 44(1)	Structural	Chokepoint (AIK) vs panopticon (PAA). Under-formalises non-hegemonic case.
EU DMA Reg. 2022/1925; Case C-7/97 <i>Bronner</i> (1998)	Structural	DMA governs PAA-type gatekeepers; Art. 102/Bronner governs AIK-type essential facilities.
Österblom et al. (2015) "Transnational Corporations as Keystone Actors," <i>PLoS ONE</i> 10(5)	Structural	Peer-reviewed biology-to-political-economy transfer. Four operational criteria.
Netherlands/ASML; CSET (2022) "Preserving the Chokepoints"	Empirical	Contemporary single-state chokepoint precedent.

Establishing Cases (Batch 1, May 2026):

Actor	Score	AIK Profile	Chokepoint Nature
Japan	4/7 (D3, D4, D6, D7)	Front-end equipment monopolies + sovereign compute (D4 degraded-but-sustained under severance)	TEL coater/developer ~90–100% EUV share; Lasertec 100% sole-source actinic mask inspection; SCREEN single-wafer cleans
South Korea	4/7 (D2, D3, D6, D7)	Memory packaging + fabrication + regulation	HPSP sole global source for high-pressure annealing; Hanmi ~71% TC bonder share; Samsung Foundry HVM at 2nm GAA
Netherlands	3/7 (D3, D6, D7)	Core lithography monopoly + export control leverage	ASML 100% EUV monopoly; ASM International ALD/epi; Besi hybrid bonding; Strategic Goods Decree enforcement

Confidence Assessment: 82% (bounded range: 72–90%). The structural distinction is well-grounded across five independent disciplines, with at least one framework (Rey & Tirole 2007) providing near-isomorphic formal correspondence. Institutional embodiment in positive law (EU DMA vs Art. 102/Bronner) provides additional validation. Four independent AI research inputs converge on the distinction's validity. Confidence exceeds the parent three-gate framework (§5.4.2, 70–75%) because the AIK/PAA structural distinction is independently grounded across five disciplines — it does not depend on the three-gate calibration choices (24–36 month window, ≥ 2 category spanning) that bound the parent framework's confidence. The cross-disciplinary convergence provides validation from outside the WEO methodology. Confidence is capped below 90% because: (a) PAA and AIK traits empirically co-occur in some actors; (b) historical chokepoint power erodes through substitution, diversification, or hegemonic coercion; (c) the Iansiti-Levien terminological collision requires active disambiguation. Confidence is held above 72% because: (a) cross-disciplinary convergence across five independent domains is unusually strong; (b) Rey & Tirole provides formal near-isomorphism; (c) the distinction is institutionally embodied in positive law; (d) three establishing cases from Batch 1 application produce a coherent structural pattern.

Validation Commitment: Monitor AIK designation stability across Batch 2 actor assessments. Review whether additional actors qualify and whether the structural pattern (hardware chokepoints without platform provision) remains stable. Monitor for technological developments that could erode AIK positions. Reassess at each methodology version milestone.

5.4.9 Advanced Capability State (ACS) Designation

Manual Section: §5.3.1 (Actor Designation Framework — Gate 2 failure)

Category: ADAPTED (78% confidence)

Rationale: The ACS designation classifies nation-state actors scoring ≥ 3 dimensions but failing the per-PAA severance test — actors with significant sovereign capabilities that nonetheless cannot sustain those capabilities autonomously under withdrawal from the dominant ecosystem. ACS is the structural residual of the severance test: actors that pass the capability gate (Gate 1) but fail the resilience gate (Gate 2). The concept of "capable but dependent" actors is well-established in international relations and technology governance literature.

Closest Precedents:

Framework	Author/Institution	Year	Relevance	Divergence from WEO
IISS Cyber Capabilities Tier 2	IISS	2021/2023	Tier 2 states possess "significant" cyber capability but lack the ability to exercise it independently across all domains	Cyber, not AI infrastructure; qualitative assessment without binary severance test
"Coordinated Stack Builders"	Brookings Institution	2026	Nations with multi-layer AI capability that coordinate within (rather than define) an ecosystem	Includes non-infrastructure variables; descriptive, not threshold-based
Basel III Stress Test Failures	Basel Committee	2010+	Banks meeting capital adequacy thresholds but failing stress scenarios	Financial sector; institutional stress, not geopolitical severance
NATO Interoperability Tiers	NATO	Various	Alliance members with capable military forces that cannot operate independently of the US logistics, intelligence, and command infrastructure	Military, not technology; alliance framework

Key Design Decisions:

1. **ACS is a severance-test outcome, not an independent structural concept.** Unlike AIK (which captures a distinct mode of structural power), ACS captures a dependency condition. The designation exists because the severance test produces a meaningful binary: actors whose capabilities survive ecosystem withdrawal (PAA/AIK) vs actors whose capabilities collapse (ACS).
2. **ACS does not imply weakness.** Both current ACS actors (EU 4/7, Taiwan 3/7) hold globally significant capabilities. The EU enforces the AI Act with extraterritorial reach; Taiwan hosts TSMC. ACS flags structural dependency, not capability deficit.
3. **ACS actors are attributed to a specific PAA ecosystem.** The severance test identifies the primary PAA dependency (for both current ACS actors: the United States). This attribution is documented in the register.

Establishing Cases (Batch 1, May 2026):

Actor	Score	Severance Failure Mechanism
European Union	4/7 (D2, D3, D6, D7)	D2 collapses (Intel Fab 34 — US-owned); D3 at structural risk (Cymer + multi-vector US leverage; supranational entity clause blocks deterrence)
Taiwan	3/7 (D2, D3, D7)	All three MET dimensions structurally coupled to US ecosystem: D2 requires US equipment and EDA tools; D3 subsystem value contingent on fab ecosystem; D7 programme procures NVIDIA GPUs

Confidence Assessment: 78%. The "capable but dependent" concept is well-established. The application via the severance test is a bounded domain transfer. The two establishing cases produce intuitively correct results.

Validation Commitment: Monitor whether ACS actors transition to AIK (if severance resilience improves) or to Participant (if capabilities degrade). The EU's severance profile may shift if Cymer operations are relocated or if European EDA alternatives emerge. Taiwan's profile is structurally stable absent fundamental changes to TSMC's equipment dependency. Review at each methodology version milestone.

5.4.10 Participant Designation

Manual Section: §5.3.1 (Actor Designation Framework — below Gate 1)

Category: ADAPTED (85% confidence)

Rationale: The Participant designation classifies all actors scoring fewer than 3 dimensions. These actors operate within ecosystem(s) shaped by PAAs and AIKs. No further subdivision is applied below the ≥ 3 threshold because the structural distinctions at this level are better captured by per-dimension data (which specific dimensions are met) than by additional designation categories.

Key Design Decisions:

1. **Single designation below ≥ 3 .** A single "Participant" designation covers all actors scoring fewer than 3 dimensions because: (a) the structural distinction between a 1/7 actor with a chokepoint and one without is already captured by which dimension is met; (b) subdividing below ≥ 3 introduces "designation competition" consuming attention disproportionate to the structural significance; (c) the G-SIB methodology provides precedent: banks below the systemic threshold are simply "non-G-SIB."
2. **Dimension profiles carry the analytical detail.** A Participant scoring D3 only (Germany — Zeiss/TRUMPF) has a fundamentally different structural profile from a Participant scoring D7 only (India — IndiaAI Mission). This difference is visible in the dimension data without requiring separate designations.

Establishing Cases (Batch 1, May 2026):

Actor	Score	Profile
United Kingdom	1/7 (D7)	Sovereign investment only
France	2/7 (D6, D7)	Governance dimensions only
Germany	2/7 (D3, D6)	Equipment chokepoints (Zeiss, TRUMPF) + export controls
India	1/7 (D7)	Sovereign investment only
Israel	1/7 (D3)	Inspection/metrology equipment only (Nova, Camtek)
Russia	0/7	No qualifying dimensions under sanctions
Canada	0/7	No qualifying dimensions; Tenstorrent redomiciled, Teledyne DALSA US-parented

Confidence Assessment: 85%. The concept of a residual category below a structural threshold is universal across comparable frameworks. The decision to use a single designation is conservative and well-supported.

Validation Commitment: Monitor whether per-dimension data alone provides sufficient analytical detail. If demand for subdivision emerges from user feedback, reassess. Review at each methodology version milestone.

5.4.11 Sovereign Capability Profiles Register

Manual Section: §5.3.3 (Sovereign Capability Profiles Register)

Category: ADAPTED (80% confidence)

Rationale: The register format directly adapts the operational pattern established by the WEO Bloc Membership Register (blocs.html), extending it to actor-level capability assessment. The versioning, dating, living-document status, and periodic reassessment mechanisms are carried over without modification. The addition of per-dimension source citations, constraint documentation, and the severance test result extends the register to accommodate the richer analytical content of capability assessment versus bloc membership classification.

Key Design Decisions:

1. **Dual-pathway register inclusion prerequisite.** The register inclusion prerequisite uses a dual pathway: event presence in the WEO Events Database, or ≥ 1 met dimension as defined in §5.3.2. The dual pathway ensures that actors with demonstrated AI infrastructure capability are eligible for inclusion regardless of database processing sequence. An actor that meets one or more AI infrastructure dimensions but has not yet had a verified event processed into the Events Database should not be excluded from the register solely due to event-processing timing — the capability assessment is independently grounded in the §5.3.2 qualification principles and §5.3.4 evidence standard. The dual pathway removes the sequencing dependency between event verification and register inclusion, allowing both analytical processes to operate independently.

Confidence Assessment: 80%. Direct adaptation of an established WEO operational pattern. The dual-pathway prerequisite removes the sequencing dependency between event verification and register inclusion.

Validation Commitment: N/A — operational format, not analytical methodology.

5.4.12 Evidence Standard for Dimension Qualification

Manual Section: §5.3.4 (Evidence Standard for Dimension Qualification)

Category: ADAPTED (75% confidence)

Rationale: WEO's event verification and coordination connection standards (1 primary + 2 corroborating) are grounded in journalism and research best practice (Reuters triangulation norm, AP verification standards). The dimension qualification evidence standard required separate research because structural capability claims are epistemically distinct from event claims — a difference recognised across the frameworks surveyed but formalised only by ICD 203 (US Intelligence Community Directive on Analytic Standards), which doctrinally separates confidence levels (for structural judgments) from likelihood language (for events).

Closest Precedents:

Framework	Institution	Evidence standard	Relevance
ICD 203 Analytic Standards	US ODNI	Quality-over-quantity; confidence tied to source quality for structural judgments; likelihood for events	Only framework with formal structural/event doctrinal separation
Transparency International CPI	TI	≥3 of 13 vetted sources	Strongest published mathematical rationale ($\sigma/\bar{n}$); applies to perception composites, not binary structural claims
TechInsights market research	TechInsights	≥3 independent data vectors	Triangulation for commercial data where single-vendor self-reporting is unreliable
Reuters Handbook of Journalism	Reuters	≥2 sources	Century-old convergent journalism norm; strongest for event claims
ITU Global Cybersecurity Index	ITU	1 government submission + expert verification	Closest operational model to 1+1 for structural state-level capability
Sovereign credit rating agencies (S&P, Moody's, Fitch)	S&P/Moody's/Fitch	No fixed per-claim minimum; standardised datasets + committee judgement	Structural assessment anchored to authoritative primary data

Key Design Decisions:

1. **1+1 base standard.** The cross-framework finding is that structural claims are evidenced by fewer, higher-authority sources than event claims. A MET dimension anchored to a Grade 1 canonical instrument (statute, benchmark output, audited filing) carries its own verification — the primary instrument is durable, unambiguous, and publicly accessible. One independent corroboration confirms the claim is not misinterpreted. This standard is consistent with the operational practice of the most analogous frameworks (ITU Global Cybersecurity Index, sovereign credit rating agencies).
2. **Rule-based escalation rather than analyst discretion.** The four escalation conditions (opaque-regime self-reporting, contested claim, unaudited vendor disclosure, no canonical instrument) are defined upfront. This preserves predictability whilst capturing the cases where additional corroboration is genuinely needed.
3. **Mandatory source descriptors (from ICD 203).** Per-source quality annotation — source type, publisher, date, credibility caveats — ensures that each cited source's authority and limitations are transparent to reviewers. The combination of quantity (1+1), hierarchy (tiered authority), and descriptor (per-source quality annotation) provides a layered evidence architecture where source quality is assessed alongside source count.
4. **Structural/event asymmetry is deliberate.** The dimension standard (1+1 base) is lower than the event standard (1+2) because the evidence types are structurally different. WEO applies the same principle as CPI, S&P, and ICD 203: structural capability claims and discrete event claims require different evidence architectures.

Evidence Basis: Deep research conducted across three independent AI research inputs, surveying 23 frameworks across four categories (sovereign/institutional assessments, technology/infrastructure assessments, regulatory/legal assessments, journalism/intelligence best practice). The cross-framework consensus supported a quality-tiered base standard with rule-based escalation.

Confidence Assessment: 75%. The standard adapts well-established precedents to a novel application. Confidence is bounded below 80% because: (a) no civilian assessment framework publishes per-source quality descriptors at the level WEO now requires; (b) the escalation triggers are untested. Confidence is above 70% because: the structural/event distinction is well-grounded; the source hierarchy is consensus; the base standard matches analogous frameworks.

Validation Commitment: Monitor escalation trigger frequency across Batch 2 assessments. If >50% of MET dimensions trigger escalation, review whether the base standard is set too low. Track whether any dimension qualification is subsequently challenged on evidence grounds. Review source descriptor implementation for consistency across assessors.

Section 6: Domain Classification Rationale

Manual Reference: Chapter 6

6.1 Four-Domain Framework

Manual Section: 6.1

Category: WEO SYNTHESIS — Domain categories follow established sectoral frameworks; durability ranges derive from empirical sources (IRS depreciation, NRC licensing, insurance tenors, WTO principles); durability-based tiebreaker hierarchy is WEO-original, operationalising implicit market judgements and path dependence theory into explicit classification methodology.

Rationale: The four domains capture distinct coordination dynamics with materially different durability profiles:

Domain	Durability	Empirical Basis
Energy/Infrastructure	20-60 years	Infrastructure asset lifespan data (IRS depreciation schedules, NRC licensing, TSMC operational evidence and industry analysis)
Security	15+ years effective horizon	Institutionalised alliance persistence; political risk insurance tenors (MIGA 20-year maximum; DFC political risk insurance up to 20 years); implementing agreement cycles (DCAs, arms control)
Technology	2-5 years	Technology lifecycle analysis; Moore's Law adaptation cycles
Trade	3-10 years	Trade agreement lifecycle data (WTO, bilateral FTAs)

Domain Durability Clarification

Technology Domain uses two overlapping ranges:

Context	Range	Typical Events
General classification	2-5 years	Standards, protocols, algorithms
AI Software events	2-7 years	Model releases, platforms, APIs

Rationale: AI software durability is more variable than general technology standards. Model releases (e.g., GPT-4 → GPT-5) have shorter lifecycles than protocol standards, whilst foundational AI platforms may persist longer.

Edge Cases: Foundational protocols (TCP/IP: 44+ years, IPv4: 40+ years) achieved durability through network effects, not design intent. These are exceptional and do not change typical domain classification. Original designers expected replacement.

Security Domain Durability Clarification

Security durability reflects *effective horizon* rather than formal treaty terms. Framework alliances (NATO, US-Japan, US-ROK) are formally perpetual with one-year withdrawal provisions, but institutionalised alliances create substantial practical lock-in through reputational costs, infrastructure entanglement, and institutional path dependence.

The "15+ years effective horizon" framing derives from convergent evidence:

Source	Finding	Implication
Political risk insurance	MIGA offers 20-year maximum coverage; DFC political risk insurance up to 20 years	Institutional investor consensus on reliable forecasting horizon for security commitments
Historical alliance persistence	Major post-WWII alliances (NATO, US bilateral treaties) have persisted 70+ years	Institutionalised alliances substantially exceed formal term structures
ATOP dataset	Historical bilateral alliance duration data (1815-2018, v5.1)	Historical volatility confirms alliances are not structurally perpetual
Withdrawal precedents	No NATO member has formally exited in 75 years; partial exits (France 1966-2009) and reversal of withdrawal attempts (Philippines VFA 2020-2021) confirm exit remains practically possible	High barriers but not absolute lock-in

Hierarchy Rationale: Security ranks below Energy/Infrastructure because the lock-in mechanisms differ fundamentally. Physical infrastructure creates *irreversible* sunk costs — a data centre, fab, or power plant cannot be "un-built." Security commitments create *substantial but reversible* reputational and legal sunk costs — a determined government can exit an alliance, however costly, whereas it cannot dismantle physical infrastructure retroactively. This irreversibility asymmetry, rather than formal duration, justifies the domain hierarchy.

6.2 Technology vs Energy/Infrastructure Distinction

Manual Section: 6.2

Category: EMPIRICALLY GROUNDED (HIGH)

Rationale: This distinction is material for classification. Physical AI infrastructure events carry stronger coordination signals than software events due to:

Characteristic	Energy/Infrastructure	Technology
Durability	20-60 years	2-5 years
Capital intensity	\$500M-\$50B+	\$10M-\$100M
Reversibility	Low (cannot relocate fabs)	High (software updates)
Coordination mechanism	Irreversible	Malleable

Government Framework Alignment:

- CFIUS reviews semiconductor fabs under national security authority
- CISA classifies data centres under the Information Technology and Communications critical infrastructure sectors
- Export controls target production-enabling capabilities, including physical equipment, technical data, software, and know-how transfers ("deemed exports" under EAR/ITAR). The regulatory focus is on items and knowledge that enable manufacturing and operational capabilities.

6.3 Domain Tiebreaker Hierarchy

Manual Section: 6.7

Category: WEO-ORIGINAL (derivative of durability ranges)

Rationale: The tiebreaker hierarchy follows durability order — domains creating longer-lasting structural commitments take classification precedence in genuinely ambiguous cases:

1. **Energy/Infrastructure** (20-60 years) — Physical buildouts create highest-durability coordination
2. **Security** (15+ years effective horizon) — Treaties and alliances persist across political cycles
3. **Technology** (2-5 years) — Algorithmic/capability events take precedence over economic exchange
4. **Trade** (3-10 years) — Economic agreements

Derivation: This hierarchy is derivative of the empirically-grounded Domain Durability Ranges (*Manual Section 6.6*).

Note: True 50/50 splits are rare. Most events have identifiable primary domains; tiebreaker application is exceptional.

Technology > Trade Ordering Rationale:

Technology ranks above Trade in the tiebreaker hierarchy despite shorter nominal durability (2-5 years vs 3-10 years) based on mechanism analysis. Technology standards and capability events exhibit stronger path dependence and network effects than trade agreements (Arthur, 1989; David, 1985). Technology choices create self-reinforcing lock-in through increasing returns and switching costs. Trade agreements, by contrast, include explicit modification provisions and demonstrate higher practical reversibility.

Citations:

- Arthur, W. B. (1989). "Competing Technologies, Increasing Returns, and Lock-In by Historical Events." *The Economic Journal*, 99(394), 116-131.
 - David, P. A. (1985). "Clio and the Economics of QWERTY." *American Economic Review*, 75(2), 332-337.
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Section 7: Industry Tags Rationale

Manual Reference: Chapter 7

7.1 11-Category Industry Taxonomy with GICS Mapping

Manual Section: 7.3

Category: ADAPTED (HIGH)

Rationale: The 11-category taxonomy adapts the Global Industry Classification Standard (GICS, maintained by S&P/MSCI since 1999) as its primary institutional reference, supplemented by NAICS (2022 revision) and ICB (FTSE Russell) where GICS does not capture Eastern Bloc industries adequately. Eight of eleven WEO categories map to specific GICS sub-industry codes; Financial Services & Investment maps at GICS sector level; Trade & Logistics and Quantum Computing Infrastructure have no direct GICS equivalent. The correspondence provides institutional investor alignment where mappings exist and documents gaps transparently where they do not.

The practice of mapping geopolitical events to industries they address is established. Caldara & Iacoviello (2022, *American Economic Review*) map geopolitical risk events to Fama-French 49-industry portfolios (based on SIC codes). The IMF Global Financial Stability Report (April 2025, Chapter 2) and Shen (2025, Federal Reserve Bank of Boston) conduct sector-differentiated geopolitical analysis using GICS-comparable groupings. Supply chain intelligence platforms (Resilinc EventWatchAI, Everstream Analytics) map events to industries using proprietary taxonomies.

Adaptation Gap: These precedents use industry groupings analytically — mapping events to industries for downstream risk analysis. WEO formalises this as a structured tagging operation within an event database, producing reproducible per-event industry metadata. No formal adaptation of GICS for classifying geopolitical events by addressed industry has been documented. This gap is real but bounded: the measurement concept is established; the operationalisation as structured event metadata is the contribution.

Evidence Chain:

Precedent	Type	Contribution
GICS (S&P/MSCI, 1999)	Direct (source framework)	Primary institutional reference for taxonomy design; 8 of 11 categories map to specific GICS sub-industry codes
Caldara & Iacoviello (2022, <i>AER</i>) — Fama-French 49-industry mapping (based on SIC codes)	Direct	Geopolitical event-to-industry mapping precedent
NAICS (2022 revision) / ICB (FTSE Russell)	Direct (supplementary reference)	Eastern Bloc coverage and cross-referencing
IMF GFSR (April 2025, Ch. 2) + Shen (2025, Fed Boston)	Direct	Sector-differentiated geopolitical analysis using GICS-comparable groupings
GB/T 4754-2017 (China) — 20-division national classification aligned to ISIC Rev. 4	Analogue	National taxonomy adaptation within international frameworks
SASB SICS — taxonomy analogous to GICS in structure, designed for sustainability risk	Analogue	Formal GICS-analogous taxonomy for non-traditional purposes

Key Design Decisions:

GICS was selected over Fama-French classifications (based on SIC codes) despite the latter having stronger geopolitical-event precedent (Caldara & Iacoviello). GICS provides institutional investor alignment — WEO's primary audience uses GICS as its standard reference. The correspondence is documented: a complete GICS-to-WEO mapping table is maintained in the companion *Industry Tagging Reference*.

The 11-category count, category boundaries, structural decisions (Telecoms as secondary-only; Quantum Computing Infrastructure as a separate category with no GICS equivalent; Trade & Logistics defined outside GICS entirely), and the formal adaptation for event-level tagging represent the novel portion (~25%). Category design rationale for each structural decision is documented in Manual §7.3.

Confidence Assessment: HIGH. ~75% grounded through convergent precedent across financial classification standards, geopolitical risk literature, and supply chain intelligence practice. Novel portion is bounded — the measurement concept is established; only the specific taxonomy design and formal operationalisation are WEO-specific.

Validation Commitment: Review taxonomy at 200 and 400 events. Assess whether any category falls below 3% primary tag usage (indicating potential merger), whether any category consistently strains boundaries (indicating potential split), or whether persistent untaggable events indicate a missing category.

7.2 Primary/Secondary Tag Structure (1 Primary + Max 2 Secondary)

Manual Section: 7.4, 7.7

Category: ADAPTED (MEDIUM-HIGH)

Rationale: Each event receives one primary tag (industry whose operational landscape the event's documented provisions most directly address) and up to two secondary tags (industries whose operational landscapes are also identifiably addressed by documented provisions). This structure adapts well-established primary/secondary classification practice from multiple domains.

Evidence Chain:

Precedent	Type	Contribution
ISIC Rev. 4 (UN, 2008)	Direct	Defines "principal activity" alongside secondary activities per economic unit — the canonical primary/secondary classification standard
ICD-10 coding guidelines (adopting UHDDS definitions)	Analogue	One "principal diagnosis" per encounter with secondary diagnoses that "coexist at the time of admission, develop subsequently, or affect the treatment received"
CPC patent classification	Analogue	One "CPC First" (classification most adequately representing the invention as a whole) with additional classifications
GB/T 4754-2017 (China)	Direct (structural)	Principal activity with recognised secondary activities; operational at national scale

Single-Primary Requirement: Universally supported across all precedent systems. The primary tag represents the industry most directly addressed — paralleling ISIC's "principal activity" determination and ICD-10's "principal diagnosis" rule.

Max-2 Secondary Cap: More restrictive than any identified comparable system. ICD-10, CPC, and ISIC all allow unlimited secondaries; GICS allows zero. The 1+2 structure is a parsimony measure preventing tag inflation. Empirical observation from the initial 87-event launch dataset shows median tag assignment of ~2.4 per event, suggesting the cap accommodates typical cases. The constraint may prove too restrictive at scale.

Confidence Assessment: MEDIUM-HIGH. ~72% grounded. The overall architecture (one primary, multiple secondary) is strongly precedented. The specific max-2 cap has no exact precedent — it occupies a unique middle position between systems allowing unlimited secondaries and systems allowing none.

Validation Commitment: After 6 months of coding, analyse the distribution of events where analysts identified 3+ candidate secondary tags but were constrained to 2. If >15% of events strain the cap, revisit whether the ceiling should be raised.

7.3 Deterministic Rule-Based Assignment from Metadata

Manual Section: 7.4

Category: STANDARD PRACTICE (HIGH)

Rationale: Rule Sets A, B, C, and D map existing WEO metadata (domain + event type) to primary industry tags through deterministic rules, with keyword disambiguation for ambiguous combinations. The approach — deterministic rule-based classification from structured metadata — is one of the most established practices in professional classification.

Evidence Chain:

Precedent	Type	Contribution
CMS DRG Grouper (US Medicare, operational since 1982–83)	Direct	Assigns diagnosis-related groups from ICD codes, procedures, age, sex, and discharge status using purely deterministic rules
Merchant Category Codes (ISO 18245)	Direct	Deterministic assignment by payment processors based on business type metadata
EPO patent auto-classification (over 90% accuracy at team level)	Direct	Automated classification from structured patent text
ICD auto-coding literature (JMIR 2021, PMC 2022)	Direct	Confidence-threshold workflows matching WEO's green/amber/red approach

Auto-Assignment Rate: Fully deterministic assignment (metadata-only, no keyword disambiguation) covers approximately 70–75% of events — comparable to documented performance ranges for rule-based classification systems:

System	Auto-Assignment Rate	Source
Measure (2014, BLS) — automated occupational coding	73.2%	<i>Proceedings of the Joint Statistical Meetings 2014</i>
Comparative Agendas supervised ML classification	Precision/F1 surpassing 80% "remained elusive"	Sebők & Kacsuk (2021, <i>Political Analysis</i>)

Including keyword-assisted routing, the initial 87-event launch dataset shows a ~7% human review rate (~93% resolved without manual intervention). Sample size is small; the rate may shift as the database grows. The 70–75% conservative estimate for fully deterministic assignment (excluding keyword-assisted routing) aligns with external benchmarks.

Separately, the Australian Bureau of Statistics achieved ~70% quality relative to human-created occupation code lists using LLM-assisted classification (UNECE HLG-MOS, March 2025), demonstrating feasibility of AI-assisted coding in official statistics contexts.

Key Distinction: The *technique* (deterministic metadata-to-category mapping) is standard practice. Only the *specific rule content* (domain × event type → primary tag mappings) is WEO-specific. This distinction is why the component is categorised as STANDARD PRACTICE rather than WEO SYNTHESIS — the novelty resides in mapping content, not mapping technique.

Confidence Assessment: HIGH. ~82% grounded through convergent precedent across clinical coding, news classification, patent classification, and statistical coding.

Validation Commitment: Track auto-assignment accuracy against human review outcomes for the first 200 events. If override rate exceeds 20% at the 150-event checkpoint, revise rules.

7.4 Mechanism Statement Requirement

Manual Section: 7.7

Category: ADAPTED (with novel elements) (MEDIUM-HIGH)

Rationale: Each secondary tag requires a one-sentence mechanism statement identifying the specific, document-verifiable pathway from the event's documented provisions to the secondary industry's operations. The statement identifies what the provisions contain and which industry's operations that content addresses — it does not claim what effect the provisions produce.

The *principle* of requiring justification for classification decisions is well-established across multiple domains. The *universal application* — every secondary tag, every time, not only ambiguous cases — and the specific mechanism-statement format go beyond what established systems typically require.

Evidence Chain:

Precedent	Type	Contribution
PRISMA 2020 (Item 5)	Direct	Requires authors to document eligibility criteria with rationale — closest parallel to mandatory justification for every classification decision
ICD coding guidelines / MedDRA	Analogue	Clinical justification linking diagnosis to code; mechanism documentation for adverse event classification
Structured Analytic Techniques (Heuer & Pherson, 2014, 2nd ed., CQ Press/SAGE)	Analogue	Describes techniques as mechanisms by which internal thought processes are externalised in systematic and transparent form (paraphrase)
ISA 230/500 (Audit Standards)	Analogue	Documented reasoning for material judgements
WEO CC evidence standards (Manual Chapter 9)	Internal consistency	Already requires documentary justification for CC classification (CC-V, CC-E, CC-A confidence levels)

Novel Element: PRISMA comes closest to WEO's universal requirement (exclusion reasons for every item). Most classification systems require justification only for borderline cases. The mechanism-specific element — requiring *how* an event's provisions address an industry, not just *that* they do — is the genuine contribution. This ~32% novel portion explains the MEDIUM-HIGH rather than HIGH confidence.

Confidence Assessment: MEDIUM-HIGH. ~70% grounded. The *principle* of classification justification is strongly precedented. The *scope of application* (universal, mechanism-specific) is more demanding than comparable systems.

Validation Commitment: Review mechanism statements at tag review checkpoints (100, 150, 200 events). Assess whether statements add genuine analytical value or become formulaic. Measure inter-analyst agreement on statement quality.

7.5 Cross-Sector Transmission Frequency (CSTF)

Manual Section: 7.9 (quality standards context). Operationally specified in *Towards Coordination Science* §III.3; rationale grounded independently below via E-I Index equivalence

Category: ADAPTED (HIGH)

Rationale: When a Coordination Connection links two events with different primary industry tags, that CC represents a cross-sector transmission. CSTF measures the proportion of such cross-sector CCs:

$$\text{CSTF} = \text{Cross-industry CCs} \div \text{Total CCs}$$

This metric is mathematically near-identical to two established network metrics. It is a linear rescaling of the **E-I Index** (Krackhardt & Stern, 1988, *Social Psychology Quarterly*) and the complement of the diagonal trace of **Newman's mixing matrix** (2003, *Physical Review E*, 67, 026126):

Established Metric	Relationship to CSTF
E-I Index = $(E - I) / (E + I)$	$CSTF = (E - I + 1) / 2$
Newman mixing matrix $Tr(\mathbf{e})$	$CSTF = 1 - Tr(\mathbf{e})$

Both the E-I Index and Newman's mixing matrix are implemented in standard network analysis packages (UCINET, igraph). When a metric is a linear transform of a 1988 publication and the complement of a 2003 formalism, it is adapted — not novel.

Evidence Chain:

Precedent	Type	Contribution
Krackhardt & Stern (1988, <i>Social Psychology Quarterly</i>) — E-I Index	Direct (mathematical equivalence)	CSTF is a linear rescaling; both measure external-to-total ratio for categorical network partitions
Newman (2003, <i>Physical Review E</i> , 67, 026126) — mixing matrix	Direct (mathematical equivalence)	$CSTF = 1 - Tr(\mathbf{e})$; modularity Q provides null-model-corrected version
Diebold & Yilmaz (2012) — spillover index	Analogue	Cross-sector transmission ratios using continuous economic flows
Leontief Input-Output framework	Analogue	Inter-industry flow measurement; technical coefficients for cross-sector transmission
Guimerà & Amaral (2005, <i>Nature</i>) — participation coefficient	Analogue	Same concept at node level rather than network level

Novel Portion: ~25% — primarily naming ("CSTF") and domain application (geopolitical AI infrastructure coordination events with typed CCs and industry tags). No published work applies cross-category network metrics to this domain.

Known Gap: CSTF omits the null-model baseline correction that both Newman (2003) and Krackhardt & Stern (1988) emphasise. Without this correction, a high CSTF could result from unequal group sizes (more events tagged "Semiconductors" than "Quantum Computing Infrastructure") rather than genuine cross-sector coordination. The rescaling choice (CSTF's 0–1 range vs. E-I's –1 to +1) is justified for the target audience — institutional investors and policy analysts are more familiar with proportion-based metrics.

Confidence Assessment: HIGH. ~75% grounded through demonstrated mathematical equivalence to established network metrics. The null-model gap is a genuine weakness relative to established E-I Index implementation, preventing a higher confidence level.

Validation Commitment: (1) Report expected CSTF under random tag assignment alongside observed values, enabling users to distinguish genuine cross-sector coordination from artefacts of unequal tag distribution. (2) Cite the E-I Index explicitly as the established precedent. (3) At operational threshold, assess whether CSTF values are sensitive to reasonable tagging alternatives.

7.6 Cross-Sector Transmission Matrix

Manual Section: 7.9 (quality standards context). Operationally specified in *Towards Coordination Science* §III.3; rationale grounded independently below via Leontief/Newman structural equivalence

Category: ADAPTED (HIGH)

Rationale: An 11×11 matrix where rows and columns are industry categories and cell values are CC counts between each pair. The matrix visualises how coordination dynamics flow between industries through documented Coordination Connections.

The inter-category flow matrix is one of the most thoroughly precedented analytical structures in quantitative social science. It is structurally equivalent to the Leontief Input-Output transactions table (1936/1941; Nobel Prize 1973), where the Bureau of Economic Analysis publishes standard ~70×70 I-O tables for the US economy.

Evidence Chain:

Precedent	Type	Contribution
Leontief Input-Output transactions table (1936/1941)	Direct (structural equivalence)	N×N matrix where rows and columns represent industries, cells contain flow values
Newman mixing matrix (2003, <i>Physical Review E</i>)	Direct (mathematical equivalence)	e_{ij} fraction of edges connecting node types i and j ; WEO matrix uses raw counts rather than proportions
Scherer (1982, <i>Review of Economics and Statistics</i>) — inter-industry technology flows	Direct (structural analogue)	Patent-based inter-industry technology flow classification between sectors — closest structural match
Diebold & Yilmaz (2012, 2015) — volatility spillover tables	Analogue	N×N matrices of cross-variable forecast error variance shares
UN Comtrade bilateral trade matrices; bibliometric citation matrices; bilateral migration flow matrices	Analogue (structural form)	Established analytical form across multiple disciplines

Novel Portion: ~25% — entirely domain application. The analytical structure is canonical. The data substrate — geopolitical AI coordination event relationships flowing between industry categories through typed CCs — has no existing implementation. No identified platform globally offers cross-sector transmission mapping for geopolitical events as a structured analytical product. The novelty resides in what fills the matrix, not in the matrix form.

Confidence Assessment: HIGH. ~80% grounded. The inter-category flow matrix is an established analytical structure applied to new data, not a novel construction.

Validation Commitment: (1) At operational threshold + 150 events, generate first transmission matrix and assess whether patterns are statistically significant vs. noise. (2) Test whether the matrix is dominated by a single industry category, which would limit analytical value. (3) Consider whether row/column normalisation (as in I-O technical coefficients) or raw counts is more informative.

7.7 Adversarial Quality Test

Manual Section: 7.9

Category: STANDARD PRACTICE (HIGH)

Rationale: The adversarial quality test asks: "If a domain professional filtered by this industry tag and saw this event, would they recognise it as correctly tagged — that this event's provisions genuinely address their industry?" This is an established evaluation methodology under at least three disciplinary names.

Evidence Chain:

Precedent	Type	Contribution
Face validity testing (Mosier, 1947; Nevo, 1985)	Direct (structural equivalence)	"The extent to which a test is subjectively viewed as covering the concept it purports to measure" — structural match is exact
TREC relevance assessment (NIST, since 1992)	Direct	Human assessors judge whether retrieved items are relevant to information needs; Voorhees (2000, 2005) demonstrated relative rankings remain stable across assessors
Information retrieval precision from user perspective	Direct	Of items retrieved (events tagged with industry X), what proportion would a user judge as correctly tagged?
Inter-rater reliability testing (Cohen's kappa, Krippendorff's alpha)	Analogue	Standard methodology for assessing classification agreement

Structural Equivalence: Face validity tests whether a measure "looks right" to the intended user. TREC relevance assessment tests whether retrieved items match user expectations. The WEO adversarial test applies the same logic — testing tag quality against domain-professional judgement. The methodology is standard; only the specific domain context (geopolitical AI infrastructure events) is new.

Terminology Note: The "adversarial" label reflects devil's-advocate testing rather than ML adversarial evaluation (red-teaming). The test is closer to classical face validity assessment than to AI safety adversarial testing.

Confidence Assessment: HIGH. ~83% grounded through demonstrated structural equivalence to face validity methodology and TREC relevance assessment practice.

Validation Commitment: (1) Implement with multiple assessors per tag to enable inter-rater reliability measurement (Cohen's kappa). (2) Apply at tag review checkpoints (100, 150, 200 events). (3) Document pass/fail rates per industry tag. If systematic failure patterns emerge for specific industries, this signals taxonomy or tagging rule revision.

7.8 Workflow Integration (Semi-Automated Classification)

Manual Section: 7.4 (rule-based assignment), 7.9 (quality standards)

Category: STANDARD PRACTICE (HIGH)

Rationale: Industry tagging integrates into the existing AI-assisted event verification workflow. Deterministic rules (Rule Sets A, B, C, and D) are applied first; human review handles secondary tags, overrides, and mechanism statements. This LLM-first, human-review-second pattern is the dominant AI-assisted classification architecture as of 2025.

Evidence Chain:

Precedent	Type	Contribution
Reuters AVISTA / AP automated earnings	Direct	News auto-classification with journalist review
CliniCoCo (<i>Nature, npj Digital Medicine</i> , 2024) / MedCodER (2024)	Direct	AI-human collaborative clinical coding; CliniCoCo reduces coding time by 40%
UNECE HLG-MOS Report (March 2025)	Direct	Statistical organisations using LLMs for classification with human verification; ABS achieving ~70% quality relative to human-created lists
Barberá et al. (2021, <i>Political Analysis</i>) — Comparative Agendas Project	Direct (methodology)	"Classify by machine but verify by humans"
Financial regulatory compliance workflows	Direct	Identification, triage, impact analysis, action assignment with human-in-the-loop governance
MIIT gradient cultivation system (China)	Analogue	Integrated evaluation workflows with automated quantitative screening followed by expert committee review

Novel Portion: ~12%. Only the specific domain context and the integration of mechanism statements into the review step are WEO-specific. The workflow architecture is entirely standard.

Confidence Assessment: HIGH. ~88% grounded — the most firmly grounded component in the industry tagging methodology. The LLM-first, human-review-second classification workflow is extensively documented and operationally validated across clinical coding, news classification, patent classification, and statistical coding.

Validation Commitment: Track analyst review time per event, override rates, and inter-analyst agreement. No special validation required — this is process design, not analytical methodology.

Section 8: Environmental Nexus Tags Rationale

Manual Reference: Chapter 8

8.1 Five-Type ENT Taxonomy Structure

Manual Section: §8.3

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%)

Rationale: The five Environmental Nexus Tag types (Resource Demand, Environmental Enablement, Environmental Consequence, Environmental Dependency, Environmental Coordination) preserve directional coherence across three analytical axes: AI → Environment (ENT-1 demand, ENT-3 consequence); Environment → AI (ENT-4 dependency); Governance mediation (ENT-2 enablement/intent, ENT-5 operationalisation).

The directional logic maps to Ranganathan's PMEST faceted classification: ENT-1 (Matter — intrinsic property), ENT-2 (Energy — action/justification), ENT-3 (Energy — measured outcome), ENT-4 (Space — contextual/relational condition), ENT-5 (Personality — core governance purpose). The mix of intrinsic and relational properties within a single tag layer is theoretically grounded — Dublin Core's 15-element metadata set accommodates analogous variation between `Type` (intrinsic), `Subject` (topical), and `Relation` (external linkage).

Evidence Chain:

Precedent	Type	Contribution
Ranganathan PMEST model (1967)	Structural	Faceted classification grounding mixed intrinsic/relational properties within a single layer
Dublin Core 15-element metadata set	Structural	Combining intrinsic description, topical classification, and relational linkage in one framework
EU ESRS double materiality (ESRS 1, ESRS 2, 2023)	Analogue	Impact materiality (inside-out: ENT-1/ENT-3) and financial materiality (outside-in: ENT-4) as separate perspectives within a single framework
MITRE ATT&CK technique taxonomy	Analogue	Flat taxonomy covering heterogeneous vectors within a unified classification layer
Preliminary calibration (109 events, 34 qualifying)	Empirical	Five types capture 31% of events without redundancy; ENT-1-dominant distribution reflects empirical reality

Key Design Decisions:

Why five and not four? Merging ENT-2 and ENT-5 would lose the distinction between aspirational environmental language and binding operational constraints — fundamentally different governance phenomena.

Why five and not six? A sixth type capturing physical geographic enablement would trend towards trivial universality (projected 50–67% at scale) and overlap with ENT-1 and ENT-4.

Why is the skewed distribution acceptable? The ENT-1-dominant distribution (~27 of 34, as of April 2026) reflects rapid expansion where resource demand dominates and governance response is a lagging indicator. Quantifying this gap is itself the primary analytical finding.

Confidence Assessment: WEO SYNTHESIS. ~65% grounded. Faceted classification theory and mixed-property architecture draw on established information science (Ranganathan, Dublin Core) and environmental risk frameworks (ESRS double materiality). The ~35% novel element is the specific five-type taxonomy design.

Validation Commitment: Review distribution at 150-event milestone. Reassess if any type falls below 2% of tagged events, ENT-1 exceeds 60%, or ENT-5 remains at zero beyond 2028.

8.2 ENT as Analytical Enrichment Layer (Not Classification)

Manual Section: §8.2

Category: ADAPTED (HIGH 75–90%)

Rationale: ENTs occupy the same architectural position as industry tags: a post-classification enrichment layer that inherits upstream pipeline assurance and produces metadata consumed by downstream analysis. This classify-first-enrich-second pattern adapts the Basel Committee on Banking Supervision's treatment of climate-related financial risks — not as a standalone risk category but as a cross-cutting driver manifesting through existing core categories (BCBS d532 FAQs, December 2022). BCBS 239 Principles 4 (Completeness) and 6 (Adaptability) formalise the pipeline: classify by core dimensions first, then filter by supplementary dimensions without disrupting aggregation logic.

The intelligence community mirrors this with ICD 203's two-tier architecture: five core Analytic Standards applied universally, then nine supplementary Analytic Tradecraft Standards. MSCI's ESG architecture (GICS industry classification first, environmental Key Issues as overlay) further confirms the pattern.

Evidence Chain:

Precedent	Type	Contribution
Basel III climate risk overlay (BCBS d532; FAQs December 2022)	Direct	Classify-first-enrich-second for environmental overlays on existing risk categories
BCBS 239 Principles 4 and 6	Direct	Core classification first, supplementary filtering second; adaptability without disruption
ICD 203 (ODNI, January 2015)	Structural	Two-tier architecture: core standards first, supplementary tradecraft standards second
MSCI ESG architecture	Analogue	GICS industry classification first, environmental Key Issues as overlay

Confidence Assessment: ADAPTED. ~80% grounded. Directly adapted from Basel III climate risk treatment — the strongest institutional precedent for environmental overlays on non-environmental classification systems. The ~20% novel element is application to a geopolitical event database.

Validation Commitment: Confirm at 200-event milestone that ENT assignment has never modified upstream classification and that the ENT layer remains computationally removable. Monitor for architectural drift — if WEO outputs treat ENT as a de facto domain, the independence principle has been violated.

8.3 Binary Qualification with Evidence-Basis Descriptor

Manual Section: §8.2

Category: ADAPTED (HIGH 75–90%)

Rationale: ENT qualification is binary (present/absent) with an evidence-basis descriptor ("Documented" or "Assessed") adapting the IPCC's confidence gating model. The IPCC establishes a hierarchy where confidence is prerequisite to probability estimation; the ENT design adapts this: binary qualification is the primary gate, with the descriptor providing a simplified confidence indicator.

"Documented" means qualifying evidence appears in the event's source documentation or primary regulatory sources (standard for ENT-1, ENT-2, ENT-5). "Assessed" means the analyst applied authoritative third-party data to documented facility characteristics (standard for ENT-4, and ENT-3 where measurement relies on independent monitoring). The two-value system was chosen over three-level confidence to minimise cognitive overhead for a supplementary metadata layer, consistent with ICD 203 permitting differentiated confidence architectures for supplementary assessments.

Why not three-level confidence? ENTs are supplementary metadata. Replicating the full CC confidence architecture (CC-V/CC-E/CC-A) would imply equivalent analytical weight. The evidence-basis descriptor captures the most analytically significant variation without over-engineering the confidence layer.

Evidence Chain:

Precedent	Type	Contribution
IPCC confidence/likelihood gating model (AR5, AR6)	Direct	Binary gate before graded assessment; validates hierarchical approach
ICD 203 (ODNI, January 2015, Section D.6.e.(2))	Structural	Differentiated confidence architectures permitted for supplementary assessments

Confidence Assessment: ADAPTED. ~75% grounded. The ~25% novel element is the specific two-value evidence-basis descriptor applied to environmental intersection tagging.

Validation Commitment: Review at 200-event milestone whether the Documented/Assessed distinction remains analytically productive. If >90% carry the same evidence basis, the descriptor adds no information.

8.4 Architectural Guardrails

Manual Section: §8.2

Category: ADAPTED (HIGH 75–90%)

Rationale: Four guardrails ensure ENTs function as a supplementary enrichment layer without becoming a competing classification system.

Guardrail 1 — Read-only dependency mirrors the Regnology ESG platform pattern, where additional attributes are included without modifying existing data.

Guardrail 2 — Independent removability adapts the computational removability test from modular overlay architectures (LSEG "Plus" layer, ISO Annex SL).

Guardrail 3 — No aggregation by ENT mirrors MSCI's architecture, where environmental Key Issues are assigned within GICS industry classification rather than as an alternative grouping principle.

Guardrail 4 — Explicit documentation of layer role follows ESRS "cross-cutting" terminology, distinguishing supplementary from categorical layers.

Evidence Chain:

Precedent	Type	Contribution
MSCI ESG architecture	Direct	Environmental Key Issues within GICS industry, not as alternative grouping
Regnology ESG platform	Analogue	Additional attributes included without modifying existing data pool
ESRS cross-cutting standards (ESRS 1, ESRS 2)	Structural	"Cross-cutting" terminology distinguishing supplementary from categorical layers

Confidence Assessment: ADAPTED. ~80% grounded. Each guardrail has direct institutional precedent. The ~20% novel element is the specific four-guardrail formulation.

Validation Commitment: Confirm all four guardrails hold at each database doubling milestone.

8.5 ENT-1 Resource Demand — Definition and Threshold

Manual Section: §8.3.1

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: The $\geq 100\text{MW}$ energy threshold derives from convergent institutional classification of "hyperscale" power demand. The International Energy Agency characterises large hyperscale data centres as those with capacity of 100MW or more (IEA, 2024; IEA, *Energy and AI*, April 2025); the United States Congressional Research Service uses the same characterisation (CRS R48646, August 2025), attributed to industry analysts; industry analysts (Synergy Research Group, Uptime Institute) converge on 100MW as the hyperscale classification boundary.

The threshold captures a qualitative shift: facilities exceeding 100MW typically interconnect directly to high-voltage transmission grid (115–230kV), a structural marker of environmental significance. Two alternative energy qualifiers (documented grid impact assessment, documented power purchase agreement) provide context-sensitive qualification below 100MW.

The $\geq 100\text{MW}$ energy threshold prevents trivial universality — without it, virtually every physical AI infrastructure event would qualify via the energy vector. The water, land, and minerals vectors do not need numerical thresholds because regulatory and planning systems determine what is environmentally significant enough to require dedicated documentation, providing a self-calibrating gate.

Non-Western validation confirms no jurisdiction uses 100MW as a regulatory boundary — regulatory triggers operate at lower levels (South Korea 10MW, Indonesia ~5MW equivalent, Japan energy-consumption-based). The threshold validates through industry convergence, not regulatory adoption, correctly identifying hyperscale environmental significance regardless of the host jurisdiction's classification approach.

Why 100MW rather than 50MW or 200MW? 50MW risks trivial universality at scale. 200MW would exclude lower-range hyperscale and sovereign compute. 100MW has the strongest convergent institutional anchoring.

Evidence Chain:

Precedent	Type	Contribution
IEA "Data centres and energy" (2024); "Energy and AI" (April 2025)	Direct	100MW characterised as "large hyperscale" capacity baseline
US Congressional Research Service R48646 (August 2025)	Direct	>100MW hyperscale characterisation in US government analytical report, attributed to industry analysts
Industry consensus (Synergy Research Group, Uptime Institute, CBRE)	Direct	100MW+ = market-standard hyperscale power category
DOE ANOPR RM26-4-000 (October 2025)	Structural	>20MW = "large load" — confirms 100MW is well above regulatory floor
EU EED Article 12 (Directive 2023/1791)	Structural	≥500kW IT power = environmental reporting floor — confirms 100MW is appropriately higher for geopolitical significance

Confidence Assessment: EMPIRICALLY GROUNDED. ~80% grounded. The ~20% novel element is application to geopolitical significance context and multi-vector architecture.

Validation Commitment: Review threshold at each database doubling. Check for false negatives at 200 events; check for tag inflation if ENT-1 exceeds 50% of infrastructure events.

8.6 ENT-2 Environmental Enablement — Definition and Linguistic Threshold

Manual Section: §8.3.2

Category: WEO-ORIGINAL (CATEGORY CREATOR)

Rationale: ENT-2 captures where environmental intent has been codified in AI infrastructure governance. The qualifying threshold imposes four cumulative criteria: environmental objectives must be (a) explicit, (b) in primary source documentation, (c) in operative provisions (not solely preambles), and (d) directed at the AI infrastructure action. The "operative provisions" test — requiring environmental language in legislative articles, binding conditions, or funding criteria rather than solely in preambles, recitals, or aspirational statements — is the critical discriminator preventing inflation from standardised environmental language. The "documented co-benefit" qualifier ensures that even environmental co-benefits of the AI infrastructure action must be explicitly stated in operative provisions to qualify, preventing unstated environmental side-effects from satisfying the threshold.

The threshold excludes physical substrate enablement (geographic proximity to renewable energy sources) because it trends towards trivial universality at scale (projected 50–67%), overlaps with ENT-1 and ENT-4, and provides analytical value better captured through geographic metadata.

Evidence Chain:

Precedent	Type	Contribution
EU AI Act Annex XI, Section 1, Paragraph 2(e)	Empirical	Qualifying exemplar — binding energy documentation in operative provisions
Canada SCIP (Sovereign Compute Infrastructure Program)	Empirical	Qualifying exemplar — environmental footprint minimisation as documented evaluation criterion in funding
Germany EnEfG (November 2023)	Empirical	Qualifying exemplar — mandatory PUE ≤ 1.2 for new data centres from July 2026, 100% renewable by 2027, waste heat reuse in operative provisions
Singapore DC-CFA2 (December 2025)	Empirical	Qualifying exemplar — PUE ≤ 1.25 , 50% green energy as capacity allocation conditions
Chile PDATA (National Data Centers Plan)	Empirical	Qualifying exemplar — clean production agreements and renewable zone mapping in sovereign planning
Meta Hyperion campus, Louisiana	Empirical	Exclusion exemplar — corporate pledge without sovereign mandate
Hedden (2021) <i>The Accidental Taxonomist</i>	Structural	Metadata-not-tag approach

		for physical geography preserves analytical value without trivially universal tags
GB 40879-2021 (数据中心能效限定值及能效等级), Standardization Administration of China, October 2021. Mandatory national standard (强制性国家标准)	[EB Benchmark]	Binding PUE limits as Chinese mandatory national standard; demonstrates test works for mandatory technical standard form
Inner Mongolia 14th FYP Energy Conservation Plan (内政办发〔2022〕11号), Chapter 4, Section 6: "新建数据中心须达到绿色数据中心建设标准, PUE值不超过1.3"	[EB Benchmark]	Provincial binding implementation using mandatory language (须...不超过); confirms cascade model
Energy Conservation Act (省エネルギー法), Japan, revised April 2023. PUE ≤1.3 binding for new data centres from FY2029	[WB Additional]	Demonstrates test works across Japanese benchmark architecture
AI Basic Act (인공지능 발전과 신뢰 기반 조성 등에 관한 기본법), South Korea, Act No. 20676, effective 22 January 2026. Article 25	[WB Additional]	Correctly-excluded exemplar — sole data centre provision contains zero environmental language; demonstrates discriminating power
Or equivalent binding environmental provisions in operative governance instruments, meeting the same evidentiary verification standard	[Accommodation]	Opens evidence chain to all bloc members following the "or equivalent" pattern (Manual §3.1, §4.6, §4.7)

Confidence Assessment: CATEGORY CREATOR with non-Western validation. ~60–65% grounded. The four-criteria linguistic threshold is WEO-original, but cross-jurisdictional validation across five distinct regulatory architectures (EU statutory, Japanese benchmark, Singapore approval-gating, Chinese cascade, cross-cutting environmental legislation) confirms the operative provisions test produces correct results globally. The ~35–40% novel element remains the specific application to environmental intersection tagging, but the test's discriminating power is empirically demonstrated (correctly qualifying Japan, Singapore, UAE, Indonesia, China; correctly excluding India, South Korea AI Basic Act, Japan Watt-Bit Collaboration).

Validation Commitment: Track ENT-2 population at 200 events. If the EU regulatory stack produces qualifying events at ~5–10% of events at scale, the threshold is functioning.

8.7 ENT-3 Environmental Consequence — Definition and Activation Model

Manual Section: §8.3.3

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%)

Rationale: ENT-3's two-category activation model ensures WEO relies on verifiable empirical data, consistent with the evidence-only observation standard (Manual §1, Core Principles; §3.9). Acute consequences are evidence-gated because qualifying data (enforcement actions, independent monitoring) becomes available rapidly. Cumulative impacts require a 24-month maturation window because single-cycle reporting data is insufficient — the EU EED first assessment (July 2025, Borderstep Institute) identified significant data quality challenges.

The 24-month window aligns two annual reporting cycles with environmental monitoring practice. The governance instrument scope limitation references instruments producing facility-level environmental operational data (EU EED, EnEfG), not model-level energy documentation (EU AI Act Annex XI).

Evidence Chain:

Precedent	Type	Contribution
xAI Memphis / EPA ruling (January 2026)	Direct	Acute enforcement within ~6–8 months; validates evidence-gated approach
EU EED first assessment (July 2025, Borderstep Institute)	Direct	Single reporting cycle yields data with significant quality challenges
US EPA NPDES permit practice	Analogue	12–24 month operational monitoring before definitive discharge limits
EU EED Delegated Regulation EU/2024/1364	Direct	24 KPIs for annual data centre reporting across three annexes (energy/sustainability, ICT capacity, data traffic); defines data accumulating over 2 cycles

Confidence Assessment: WEO SYNTHESIS. ~65% grounded. Acute evidence-gating follows from verification lag and regulatory practice (~80%). The 24-month cumulative window synthesises EU EED experience, monitoring practice, and reporting cycle mechanics (~60%). The two-category architecture is WEO-original with analogue in NEPA's tiered review hierarchy (~50%).

Validation Commitment: Reassess 24-month window after EU EED third reporting cycle (~mid-2026). Track Building Chips in America Act regulatory gap through CHIPS-funded fab HVM commencement.

8.8 ENT-4 Environmental Dependency — Definition and Evidence Indices

Manual Section: §8.3.4

Category: ADAPTED (HIGH 75–90%)

Rationale: ENT-4 deliberately outsources risk determination to authoritative external indices rather than defining WEO-specific risk thresholds. Three authoritative global indices are designated as default sources: WRI Aqueduct Water Risk Atlas (Version 4.0) for facility-level water stress; ND-GAIN Country Index for country-level climate vulnerability; Munich Re NatCatSERVICE for empirical catastrophe loss data. Together they cover hydrological risk, climate vulnerability, and historical catastrophe exposure.

The key exclusion — generalised global climate risk does not qualify — ensures ENT-4 maintains specificity, aligning with TCFD's emphasis on entity-specific physical risk assessment. Where national environmental data sources provide demonstrably superior resolution, they are designated as complementary sources alongside the global defaults. The global baseline is preserved for cross-jurisdictional comparability.

Evidence Chain:

Precedent	Type	Contribution
TCFD Recommendations (2017; updated 2021)	Direct	Physical climate risk assessment using entity-specific exposure analysis
WRI Aqueduct Water Risk Atlas 4.0	Direct	Authoritative hydrological risk classification; facility-level granularity
ND-GAIN Country Index	Direct	Country-level climate vulnerability; academic and development finance recognition
Munich Re NatCatSERVICE	Direct	Empirical catastrophe loss data; insurance industry standard
J-SHIS (Japan Seismic Hazard Information Station), NIED, 250m mesh resolution	[WB Additional]	Seismic hazard mapping orders of magnitude more granular than Munich Re for Japanese facilities
CGWB NAQUIM (National Aquifer Mapping), Central Ground Water Board, India. 1:50,000 scale	[WB Additional (Non-Aligned)]	Block/aquifer-level groundwater data addressing WRI Aqueduct gaps in fossil groundwater
First National Disaster Risk Census (第一次全国自然灾害综合风险普查), Ministry of Emergency Management (应急管理部), China. County-level multi-hazard	[EB Benchmark]	County-level disaster risk mapping superior to global indices for Chinese facilities
Or equivalent authoritative national environmental data sources providing demonstrably superior facility-level resolution, meeting WEO source quality requirements (Manual §3.1)	[Accommodation]	Preserves global baseline whilst accommodating jurisdiction-specific high-resolution data

Confidence Assessment: ADAPTED. ~80% grounded. Designated indices are authoritative, widely recognised, and freely accessible. Cross-jurisdictional validation confirms global architecture works with complementary national sources. The ~20% novel element is application to a geopolitical event database.

Validation Commitment: Review index designations at each database doubling milestone. If evidence of operational disruptions emerges for ENT-4-tagged events through WEO's standard rolling update process, this would strengthen validation of the index-based identification approach.

8.9 ENT-5 Environmental Coordination — Definition and Medium Scope

Manual Section: §8.3.5

Category: WEO-ORIGINAL (CATEGORY CREATOR) for scope selection; ADAPTED (HIGH) for Pathway A evidence architecture

Rationale: ENT-5 captures where environmental governance has been operationalised into binding requirements on AI infrastructure (Pathway A) or where dedicated AI infrastructure has been deployed for environmental purposes at sovereign or multilateral scale (Pathway B). The definition resolves a scope spectrum:

- **Narrow scope yields an empty set** in the current database (projected ~3–8 events at 400–600 scale). Binding environmental regulations for AI infrastructure exist (EU EED, EnEfG) but narrow scope would exclude them because they govern the environmental footprint OF AI infrastructure rather than deploying AI FOR environmental purposes.
- **Broad scope creates unacceptable overlap with ENT-2.** Every broad ENT-5 event would also receive ENT-2 (since all events with environmental governance provisions also contain environmental language).
- **Medium scope** produces the analytically productive ~10:1 ENT-1/ENT-5 ratio quantifying the governance gap whilst acknowledging that governance response has begun.

Evidence Chain:

Precedent	Type	Contribution
EU AI Act Article 53/Annex XI	Empirical	Pathway A — binding energy documentation for GPAI providers
EU EED Article 12 (Directive 2023/1791)	Empirical	Pathway A — mandatory annual reporting for data centres $\geq 500\text{kW}$
Germany EnEfG (November 2023)	Empirical	Pathway A — mandatory PUE ≤ 1.2 , 100% renewable by 2027
Singapore DC-CFA2 (December 2025)	Empirical	Pathway A — capacity-gated green requirements
DestinE Phase 3 (February 2026 confirmation)	Empirical	Pathway B — dedicated EuroHPC infrastructure for climate digital twins
France Jean Zay supercomputer	Empirical	Exclusion exemplar — domestic waste-heat qualifies ENT-2 not ENT-5
Keohane coordination definition (1984)	Structural	ENT-5 Pathway B requires sovereign or multilateral scale — aligning with WEO's foundational coordination definition
EDWC programme (发改高技〔2021〕709号) with provincial binding implementation via GB 40879-2021 and approval-process PUE conditions	[EB Benchmark] (Pathway A)	Binding environmental requirements on compute infrastructure at sovereign scale; first EB exemplar; qualification through administrative approval rather than standalone environmental legislation
Supercomputer Fugaku environmental programmes (RIKEN/JAMSTEC): tsunami prediction, earthquake simulation, extreme weather prediction; J-Alert integration	[WB Additional] (Pathway B)	Strongest Pathway B exemplar globally — governance-scale dedicated environmental AI infrastructure
Or equivalent binding environmental operational requirements or dedicated environmental compute infrastructure at sovereign or multilateral scale, meeting the same evidentiary verification standard	[Accommodation]	Covers all bloc members

Key Design Decision:

Why require binding/enforceable constraints for Pathway A? Non-binding pledges are captured by ENT-2 (intent). ENT-5 specifically tracks operationalisation — the point at which intent becomes obligation. Why sovereign/multilateral scale for Pathway B? Corporate AI-for-environment deployments are product offerings on commercial infrastructure, not governance events.

Confidence Assessment: CATEGORY CREATOR for scope selection (~50%); ADAPTED (~80%) for Pathway A evidence architecture. The ~50% novel element is the specific medium scope boundary definition, grounded in empirical population analysis and Keohane's coordination framework. Non-Western validation confirms Pathway A correctly identifies binding requirements across diverse governance architectures (EU statutory, Chinese administrative approval, Japanese benchmark, Singaporean capacity-gating, cross-cutting environmental legislation).

Validation Commitment: If ENT-5 remains at zero beyond 2028 despite documented regulatory maturation, reassess medium scope. Track regulatory pipeline: EU Data Centre Energy Efficiency Package, Japan Energy Saving Act PUE mandate (post-2029), Singapore Digital Infrastructure Act.

8.10 ENT-2/ENT-5 Boundary (Governance Maturity Spectrum)

Manual Section: §8.3.2, §8.3.5

Category: WEO-ORIGINAL (CATEGORY CREATOR)

Rationale: ENT-2 marks intent; ENT-5 marks operationalisation. An event may carry both tags when environmental language (ENT-2) generates binding constraints (ENT-5). The dual-tag is informative, not redundant — it measures different positions on the governance maturity spectrum.

Scenario	ENT-2	ENT-5	Rationale
EU AI Act Annex XI — binding energy documentation	Yes	Yes	Environmental mandate in operative provisions + legally binding
Canada SCIP — environmental footprint as evaluation criterion	Yes	Borderline	Environmental language in funding; contractual enforceability arguable
France Jean Zay — waste-heat recovery	Yes	No	Environmental provisions in domestic sovereign compute without binding environmental governance targeting AI infrastructure
GAIP \$100B private equity partnership	No	No	Corporate ESG language without sovereign mandate; fails ENT-2's primary source and operative provisions tests

Confidence Assessment: CATEGORY CREATOR. ~45% grounded via governance maturity concepts; ~55% novel calibration. No existing framework applies a comparable intent-versus-operationalisation spectrum to environmental governance in AI infrastructure.

Validation Commitment: At 200 events, assess whether all ENT-5 events also carry ENT-2 (as predicted). If ENT-5 events exist without ENT-2, reassess boundary logic.

8.11 Chinese Cascade Model — Governance Architecture for "Operative Provisions" Test Application

Manual Section: §8.3.2

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: Chinese data centre environmental governance operates through a directive-to-enforcement cascade: central policy documents use directional language (引导 — "guide"; 鼓励 — "encourage") but gain operative force through (a) mandatory national standards (GB 40879-2021, 强制性国家标准) providing legal enforcement baselines, and (b) provincial implementation plans replacing directional language with mandatory language (须 — "must"; 应当 — "shall"; 不低于 — "no less than"; 不得超出 — "must not exceed").

The cascade is confirmed across four provinces spanning two enforcement pathways.

Administrative-hierarchy (Inner Mongolia): DRC links project approval (立项审批) to compliance with energy budget quotas and mandatory PUE standards. **Market-mechanism** (Ningxia): data centres designated as entities subject to renewable energy consumption obligations within provincial electricity trading, with State Grid Ningxia responsible for verification and dispatch. The ENT-2 "operative provisions" test applies correctly to the implementing instrument where binding language appears, consistent with the principle that EU directives require member-state transposition before creating binding obligations.

Evidence Chain:

Precedent	Type	Contribution
Three-Year Action Plan (工信部通信〔2021〕76号, MIIT, 4 July 2021): PUE ≤1.3 for new large data centres (引导)	Empirical	National directional layer — sets targets using guidance language
GB 40879-2021 (Standardization Administration of China, October 2021): mandatory PUE grades (1.5/1.3/1.2)	Empirical	National mandatory standard — legal enforcement baseline
Greater Bay Area hub approval (发改高技〔2022〕66号, NDRC, 7 February 2022): PUE ≤1.25	Empirical	Hub-level binding condition — approval-process enforcement
Inner Mongolia 14th FYP Energy Conservation Plan (内政办发〔2022〕11号): "须...不超过1.3"	Empirical	Provincial mandatory conversion — administrative-hierarchy enforcement
Ningxia 2026 Power Trading Notice (宁发改电力〔2025〕752号): data centre green electricity quota	Empirical	Provincial mandatory conversion — market-mechanism enforcement
EU directive transposition requirement	Structural	Analogous cascade: EU directives require member-state implementation
Chinese 1+N regulatory structure (Methodology Rationale §9.2)	Structural	Precedent for documenting Chinese cascade governance in WEO methodology

Confidence Assessment: EMPIRICALLY GROUNDED. ~80% grounded. The cascade is confirmed across 4 provinces, 2 enforcement pathways, 7 instruments with quoted operative language. EU transposition analogy and established CC Type 6 validation provide structural precedent. The ~20% novel element is application to environmental governance provisions.

Validation Commitment: Confirm cascade in additional EDWC hub provinces as coverage expands. Track whether market-mechanism pathway expands as green electricity trading markets mature.

Section 9: Coordination Connections Rationale

Manual Reference: Chapter 9

9.1 CC Taxonomy Overview

Manual Section: 9.1, 9.2

Category: WEO-ORIGINAL (Verified Novel IP)

Rationale: The 7-type Coordination Connections taxonomy represents WEO's primary novel intellectual property contribution. Deep research across 25+ platforms (Stratfor/RANE, Eurasia Group, GDELT, ICEWS, ACLED, Stanford HAI, Sayari, Bloomberg, etc.) confirmed no existing product implements typed event-to-event relationship categories comparable to CC.

Key Finding: "No existing product implements typed event-to-event relationship categories comparable to CC."

Synthesis Sources:

- Chinese governance frameworks (1+N policy cascades)
- OFAC sanctions methodology
- ISO/IEC Directives structure
- Academic policy diffusion literature (Shipan & Volden, Gilardi & Wasserfallen)

Novelty Verification: Examination of 25+ platforms across four categories (geopolitical intelligence, AI policy trackers, event databases, entity relationship platforms) found:

- No platform uses typed event-to-event relationship categories
- Entity relationship platforms (Sayari, Maltego) have typed taxonomies but only for entity-entity connections
- CAMEO codes 204 event types but does not code relationships between events

9.1.1 Distinct Actor Requirement

Manual Section: §9.1.6

Category: WEO SYNTHESIS (80% confidence)

Rationale: The distinct actor requirement formalises a principle that was implicit in the CC framework's design but not explicitly stated. Coordination, by definition, involves multiple parties — a single actor coordinating with itself is internal decision-making, not inter-actor coordination. The absence of an explicit statement created ambiguity during processing: same-corporation product lineage events (successive model releases) and intra-ministerial policy sequences generated CC candidates that passed the mechanical qualification tests (temporal proximity, thematic similarity) but did not represent coordination between distinct actors.

A corpus-wide investigation across the event database confirmed that same-entity connections, when mechanically qualified, produce analytically misleading results — they inflate the connection count without documenting genuine coordination dynamics. Seven same-entity CC candidates were permanently excluded from the holding pool following this investigation.

The subsidiary clause (consistent with §5.3.2) addresses the boundary case: subsidiaries operating under direct parent-company control are the same actor; subsidiaries with documented strategic autonomy in the relevant domain are distinct actors.

Confidence: 80%. The principle is definitionally grounded (coordination requires multiple parties) and supported by the holding pool investigation. No existing CC in the database is affected — the requirement formalises what was already operationally applied.

Validation commitment: Monitor for edge cases at the subsidiary boundary. If strategic autonomy assessments for CC purposes diverge from SCP subsidiary clause assessments, reconcile.

9.2 Seven Connection Types

Manual Section: 9.2.1

Category: WEO-ORIGINAL (synthesised from established precedents)

Rationale: Each connection type was validated across 3+ independent systems before inclusion.

Type	Name	Definition	Precedent Inspiration
Type 1	Policy Cascade	Event B cites Event A as legal/policy basis	Chinese 1+N Framework, CSET policy analysis
Type 2	Joint Issuance	Single document with multiple issuing authorities	Chinese multi-ministry governance, OECD, EU regulatory practice
Type 3	Funding Flow	Event A provides funding enabling Event B ($\geq 10\%$ threshold)	OECD Due Diligence Guidance, Government grant standards
Type 4	Regulatory Enablement	Event A establishes framework Event B operates within	ISO/IEC Directives Part 2, Administrative law precedent
Type 5	Response	Event B is documented reaction to Event A	OFAC sanctions methodology, GSDB, CSET
Type 6	Framework Family	Events are part of same standards/institutional series	ISO/IEC standards families, UN framework structures
Type 7	Parallel Policy	Substantive similarity without documented linkage ($\geq 3/4$ criteria)	Shipan & Volden (2008), Gilardi & Wasserfallen (2019)

Academic Literature Foundation:

- Shipan, C. R., & Volden, C. (2008). "The mechanisms of policy diffusion." *American Journal of Political Science*, 52(4), 840-857.
- Gilardi, F., & Wasserfallen, F. (2019). "The politics of policy diffusion." *European Journal of Political Research*, 58(4), 1245-1256.
- Berry, F. S., & Berry, W. D. (1990). "State lottery adoptions as policy innovations." *American Political Science Review*, 84(2), 395-415.

Type 1: Policy Cascade — Authority Hierarchy Test

Category: ADAPTED (synthesised from administrative law precedents)

Source Frameworks:

- Western administrative law: Constitutional hierarchy principles
- Chinese governance: 1+N framework analysis (CCICED)
- ISO/IEC Directives Part 2: Normative vs informative reference distinction

Authority Hierarchy (Descending):

AL	Authority Level	Examples
AL-1 (Highest)	International treaties, multilateral binding agreements	UN Security Council resolutions, WTO agreements, binding ISO/IEC standards
AL-2	Constitutional provisions, enabling statutes	US Constitution, Basic Law, foundational organic laws
AL-3	Primary legislation	Acts of Congress/Parliament, Chinese NPC laws (法律)
AL-4	Regulations, Executive Orders, Ministerial Decrees	Federal Register rules, State Council regulations (法规), EU Regulations
AL-5	Agency guidance, implementation rules, departmental orders	Ministry notices (通知), agency guidance documents, rules (规章)
AL-6	Awards, grants, individual determinations	Funding awards under frameworks, permits, licences

Decision Rules:

- 1. **≥2 AL difference** → Policy Cascade
 - Example: Primary legislation (AL-3) → Agency implementation rule (AL-5)
- 2. **1 AL difference** → Examine language
 - Mandatory language ("pursuant to," "implementing," 根据, 落实) → Policy Cascade
 - Permissive language ("authorised under," "enabled by," 依照) → Regulatory Enablement
- 3. **Same Level** → Default to Regulatory Enablement unless explicit hierarchical language present
- 4. **Cross-jurisdictional** → International agreements (AL-1) creating national implementation (AL-3 to AL-5) = Policy Cascade

Chinese Framework Equivalent:

Level	Chinese Term	Authority
Central	中央/国务院	Highest domestic
Ministry	部委	Departmental
Provincial	省级	Regional
Local	地方/市级	Municipal

Document Type Hierarchy:

法律 (Law) > 法规 (Regulation) > 规章 (Rule) > 通知 (Notice) > 意见 (Opinion)

≥2 AL Threshold Rationale:

Clear hierarchy (2+ AL difference) provides unambiguous Policy Cascade classification. Adjacent ALs require linguistic marker analysis because hierarchy alone is insufficient to distinguish mandatory from permissive relationships.

Type 6: Framework Family — Evidence Pathway Grounding

Manual Section: 9.2.1

Category: WEO SYNTHESIS

Rationale: The three Type 6 evidence pathways (common numbering, explicit cross-reference, parent framework document) combine three independently established concepts from standards management and legal instrument classification into a single evidence framework for geopolitical event relationship classification. Each pathway has an identified external basis; the combination and application to WEO's domain is novel.

Research basis: A targeted survey of standards relationship classification systems was conducted across ISO/IEC Directives (Parts 1 and 2), NISO Standards Tag Suite (ANSI/NISO Z39.102), ISO Online Browsing Platform relationship categories, and national standards body concordance systems (BSI, ANSI, DIN, CEN/CENELEC).

Key findings:

1. **ISO/IEC does not formally define "standards family."** The only formally defined structural concept is the multipart standard (ISO/IEC Directives, Part 2, Clause 5.1). ISO's colloquial use of "family" (e.g., "ISO 27000 family") is defined by number range, not committee membership. No surveyed standards body provides a normative definition.
2. **The normative reference is the only relationship type creating binding technical dependency.** ISO/IEC Directives Part 2 defines normative references as documents "referred to in the text in such a way that some or all of their content constitutes requirements of this document." Informative references (bibliography, body text, informative annexes) provide contextual information without imposing requirements.
3. **Committee membership is administrative metadata, not an inter-standard relationship.** The NISO STS schema classifies inter-standard relationships through the `<std-xref>` element — relationship types include "revises," "supersedes," "amends," "adopted-from," and "std-family." Committee attribution and ICS classification appear as metadata but do not constitute formal inter-standard relationships.
4. **Cross-body adoption creates no transitive relationships.** The ISO-IEEE PSDO agreement (2007) defines fast-track adoption and joint development mechanisms. Adoption renumbers a standard into the adopting body's catalogue but creates no normative links to other standards in that catalogue.

Synthesis — pathway mapping to external sources:

WEO Pathway	External Basis	Source	Boundary
Common numbering/naming convention	Informal "family/series" number-range convention used by ISO, BSI, ANSI	ISO Online Browsing Platform; BSI Standards Catalogue	Shared number range required; different numbering under same committee insufficient
Explicit cross-reference	Normative reference as defined by ISO/IEC Directives Part 2, Clause 2	ISO/IEC Directives Part 2 (2021), §6.2.2	Normative Section 2 references only; informative references excluded
Parent framework document	Formal legislative/institutional designation as used in legal instrument classification	Administrative law hierarchy (see §9.2 Type 1 rationale); Vienna Convention on the Law of Treaties	Binding legal instruments and institutional founding documents qualify; administrative committee catalogues do not

Sources:

Institutional/Standards Sources:

- ISO/IEC Directives, Part 2 (2021): Normative vs informative reference distinction (§6.2.2)
- NISO Standards Tag Suite: ANSI/NISO Z39.102-2022, inter-standard relationship taxonomy (element)
- ISO Online Browsing Platform: Standards relationship display and committee attribution metadata
- ISO-IEEE Partner Standards Development Organisation (PSDO) Cooperation Agreement (2007): Fast-track adoption and joint development mechanisms
- BSI Standards Catalogue; ANSI Essential Requirements; DIN 820; CEN/CENELEC Internal Regulations: Surveyed for "standards family" normative definitions (none found)

Confidence: HIGH (85%) — each pathway individually grounded in ISO/IEC Directives Part 2, NISO STS Z39.102, and established legislative instrument classification. Combination into three-pathway structure is WEO synthesis without direct external precedent.

Validation commitment: Review pathway boundaries at 100-event milestone based on classification consistency data.

Non-Western pathway validation: A targeted research investigation was conducted to verify whether the three evidence pathways, derived from Western standards body and legislative practice, adequately capture structural dependencies in non-Western AI regulatory and standards architectures. The Chinese ecosystem was assessed as the primary Eastern Bloc standards and regulatory system relevant to the current event population, with comparative assessment of Japanese (METI/MIC), South Korean (AI Basic Act/MSIT), and Indian (MeitY) regulatory architectures.

Non-Western key findings:

1. **Pathway 2 confirmed applicable to Chinese GB/T standards.** SAC/TC260 standards use a 规范性引用文件 (Normative Reference Documents) section with binding language ("构成...必不可少的条款" — "constitutes an indispensable clause") functionally equivalent to ISO/IEC Directives Part 2 normative reference requirements. No pathway modification required.
2. **TC260 AI standards exhibit a decentralised mesh topology** with domain-specific cluster hubs (data annotation around GB/T 42755, large models around GB/T 45288 series, federated learning around GB/T 41867) rather than the hub-and-spoke pattern observed in ISO/IEC SC 42 (centred on ISO/IEC 22989). The topology differs but the pairwise evidence standard is identical.
3. **Chinese 1+N regulatory structures confirmed compatible with Pathway 3.** Provincial and ministerial implementation documents cite the parent State Council directive by specific document number (e.g., "国发〔2017〕35号") in operative provisions. Where "N" documents cite enabling legislation instead, this constitutes a separate connection chain rather than a direct Framework Family link.
4. **No new pathway required.** Mechanisms proposed as requiring methodology expansion (implicit hierarchical delegation, structural convention, experimental frameworks) represent thematic proximity or administrative relationships rather than formal structural dependencies meeting the CC-V evidence standard.

Comparative pathway compatibility:

Regulatory Architecture	Pathway Compatibility	Notes
Chinese GB/T (SAC/TC260)	Pathway 2 — confirmed	规范性引用文件 equivalent to ISO/IEC Section 2
Chinese 1+N (State Council → ministry/provincial)	Pathway 3 — confirmed	"N" documents cite "1" directive by document number in operative text
Japanese (METI/MIC, AI Promotion Act)	Pathways 2 and 3 — confirmed	Statute-driven model; explicit legislative designation
South Korean (AI Basic Act/MSIT)	Pathways 2 and 3 — confirmed	Comprehensive statutory framework; explicit institutional designation
Indian (MeitY, fragmented model)	Inconclusive	Multi-layered, soft-law-reliant architecture; further assessment required at scale

Non-Western sources:

Standards Architecture:

- SAC/TC260 standards documentation: 规范性引用文件 section structure and binding language conventions
- GB/T 42755-2023 (AI Data Annotation for Machine Learning): Normative reference patterns within TC260 ecosystem
- GB/T 45288 series (AI Large Models): Cluster hub cross-reference patterns
- GB/T 41867-2022 (Federated Machine Learning): Domain-specific normative referencing

Regulatory Architecture:

- State Council Next Generation AI Development Plan ("国发〔2017〕35号"): 1+N directive structure
- Interim Measures for the Management of Generative AI Services (CAC, 2023): Legal basis citation patterns (Cybersecurity Law, Data Security Law, Personal Information Protection Law)
- Provincial implementation plans (Gansu, Liaoning, Jilin): Operative text citation of parent directive by document number

Updated confidence: HIGH (85%) — unchanged. Non-Western validation confirms existing pathways are not Western-specific; no expansion required for current event population. Indian regulatory architecture flagged for reassessment at scale.

Type 7: Parallel Policy — Jurisdiction Clarification

Manual Section: 9.2.1

Category: EMPIRICALLY GROUNDED (HIGH)

Rationale: The "different jurisdictions" requirement was clarified to explicitly exclude same-body standard pairs and same-programme events. This definition is derived directly from the policy diffusion literature that underpins Type 7:

Source citations:

- Shipan, C. R., & Volden, C. (2008). "The mechanisms of policy diffusion." *American Journal of Political Science*, 52(4), 840-857. Study of policy diffusion across US cities — "jurisdiction" means governmental entities with territorial authority.
- Gilardi, F., & Wasserfallen, F. (2019). "The politics of policy diffusion." *European Journal of Political Research*, 58(4), 1245-1256. Define diffusion units as "country, state, city, etc."
- Graham, E. R., Shipan, C. R., & Volden, C. (2013). "The diffusion of policy diffusion research in political science." *British Journal of Political Science*, 43(3), 673-701. Define policy diffusion as "the process by which policymaking in one government affects policymaking in other governments."
- Simmons, B. A., Dobbin, F., & Garrett, G. (2006). "Introduction: The international diffusion of liberalism." *International Organization*, 60(4), 781-810. "Policy decisions in a given jurisdiction are systematically conditioned by prior policy choices made in other jurisdictions" — jurisdictions are always governmental entities.

Key finding: No source in the policy diffusion literature applies the "jurisdiction" concept to individual standards or individual policy outputs within a single decision-making body. Hooghe, L. & Marks, G. (2003) "Unraveling the Central State, but How? Types of Multi-level Governance" (*American Political Science Review*, 97(2), 233-243) — the closest theoretical analogue for treating a standards body as a jurisdiction-like entity — recognises "task-specific jurisdictions" as functionally specific governance entities, but even these are organisational entities, not individual outputs.

Confidence: HIGH (95%) — directly derived from five independent academic sources with zero scholarly precedent for alternative interpretation.

9.3 CC Confidence Levels

Manual Section: 9.3

Category: ADAPTED (from ICD 203/206, ISO/IEC frameworks)

Rationale: The three-tier CC confidence system (CC-V/CC-E/CC-A) provides graduated evidentiary standards for inter-event relationship claims.

Level	Definition	Derivation Source
CC-V (Verified)	Documentary proof of connection	ISO/IEC normative reference requirements; legal citation standards
CC-E (Established)	Authoritative statement documents connection	ISO/IEC informative reference standard; ICD 206 sourcing requirements
CC-A (Analytical)	Analyst assessment based on observable evidence	Policy diffusion research methodology (Shipan & Volden); 4-criteria assessment framework

Design Principle: CC-A is the only confidence level permitting analyst assessment rather than documentary proof. This acknowledges that some coordination relationships (particularly Parallel Policy) are observable but not explicitly documented.

CC-A Evidence Model for Parallel Policy

Category: CATEGORY CREATOR

Type 7 (Parallel Policy) connections capture structural similarity between independently-developed policies. The "connection" is an analytical determination, not a documentary relationship — accordingly, the evidence model differs from CC-V and CC-E.

CC-A Evidence Requirements:

Component	Requirement
Event A documentation	≥1 primary source establishing event exists with documented characteristics
Event B documentation	≥1 primary source establishing event exists with documented characteristics
Criteria assessment	All 4 similarity criteria evaluated (mechanism, target, scope, timing)
Criteria rationale	Explicit evidence cited for each criterion determination
Threshold	≥3/4 criteria met

Rationale: Rigour for Parallel Policy derives from systematic criteria assessment with explicit threshold requirements, not source multiplication. The 4-criteria framework operationalises the policy diffusion literature (Shipan & Volden, 2008; Gilardi & Wasserfallen, 2019) into reproducible analytical determination.

9.4 Evidence Requirements

Manual Section: 9.4

Category: STANDARD PRACTICE

Rationale: Evidence requirements per connection type follow established verification principles from Chapter 3, adapted for inter-event relationship claims.

9.4.1 Temporal Windows by Domain

Category: EMPIRICALLY GROUNDED (MEDIUM-HIGH)

Rationale: Temporal windows reflect domain-specific implementation cycles and response dynamics.

Response CC Derivation Sources:

Domain	Window	Primary Source
Energy/Infrastructure	365 days	Infrastructure project timeline norms; multi-year project timelines
Security	30 days	OFAC enforcement patterns — MidFirst Bank FOV "six hour window" as mitigating factor; transactions "days after designation" treated more severely than "hours after"
Technology	180 days	Product development cycle research; minimum iteration period
Trade	60 days	WTO DSU Article 4.7 consultation period (60 days before panel request)

Parallel Policy Derivation Sources:

Windows set at 2× Response CC windows for same domain, reflecting policy diffusion literature findings that adoption takes "months to years" (Shipan & Volden, 2008).

Domain	Response CC	Parallel Policy CC	Derivation Source
Energy/Infrastructure	365 days	24 months	Multi-year project timelines
Security	30 days	6 months	OFAC enforcement timing patterns
Technology	180 days	12 months	Product development cycles
Trade	60 days	12 months	WTO DSU Article 4.7 consultation period

Academic Literature Foundation:

- Berry, F. S., & Berry, W. D. (1990). "State lottery adoptions as policy innovations." *American Political Science Review*, 84(2), 395-415.
- Shipan, C. R., & Volden, C. (2008). "The mechanisms of policy diffusion." *American Journal of Political Science*, 52(4), 840-857.
- Mallinson, D. (2021). "Determinants of adoption change as the diffusion process unfolds."

Status: Provisional calibration pending empirical validation at 100-event milestone.

Chinese Regulatory Temporal Accommodation

Manual Section: §9.4.1 (Temporal Windows)

Category: ADAPTED (78–85% confidence)

Rationale: The 根据 clause in Article 1 of a Chinese departmental regulation is a mandatory drafting feature established by the *Regulations on Procedures for the Formulation of Rules* (《规章制定程序条例》, State Council Order No. 322, revised 2017). The NPCSC *Legislative Technical Standards (Trial)* (No. 2) (法工委发 [2011] 5号, §13) draws an explicit distinction: 根据 is for citing the Constitution or a parent statute as a legislative basis in Article 1, whilst 依据 (yījù) is for citing specific clauses in operative provisions. This confirms that 根据 establishes an authority dependency — analogous to US CFR authority citations (1 CFR §21.40–.43) or EU "Having regard to" preambles — rather than a content dependency analogous to ISO/IEC Clause 2 normative references.

Chinese regulatory families constructed through 根据 chains exhibit 24–42 month spans between parent and child instruments. The CAC AI regulatory family demonstrates this pattern: Algorithm Recommendation Provisions (March 2022) → Deep Synthesis Provisions (January 2023, ~10 months) → Generative AI Measures (August 2023, ~17 months) → AI Content Labelling Measures (September 2025, ~42 months from parent). Comparative data confirms these timescales are normal: EU Level-1 financial-services acts average 25 months from proposal to publication with a range extending to 51 months (Clifford Chance, February 2025); US rulemaking under enabling statutes commonly takes 3–5 years.

The accommodation resolves a routing asymmetry identified through multi-source research (unanimous convergence across independent inputs). The same functional relationship — a parent authority instrument and a child implementing regulation separated by a multi-year gap — is classified as different CC types depending on jurisdiction. Western authority chains (e.g., an enabling statute and the agency rule issued under it) typically qualify as Type 6 (Framework Family) through Pathway 3, which has no temporal window, or as Type 1 (Policy Cascade), which has longer windows (up to 60 months for Energy/Infrastructure). Chinese 根据 chains are classified as Type 4 (Regulatory Enablement) — the child regulation operates within the framework the parent created — where the Technology domain window is 24 months. The result is that functionally equivalent authority chains pass or fail temporal qualification based on which CC type they are routed through, not on the strength of the documented relationship.

The three-constraint design (explicit full-title citation, CC-V grade, 48-month bound) ensures the accommodation is narrow: it applies only to formally documented authority chains, not to thematic proximity or informal policy influence.

The FATF Mutual Evaluation Methodology provides the strongest precedent for legal-tradition accommodation, requiring assessors to account for "the nature of the legal system (civil law or common law) and institutional framework." GDPR adequacy assessments apply functional equivalence rather than formal identity (*Schrems II*, Case C-311/18, ¶105).

Sources: *Regulations on Procedures for the Formulation of Rules* (State Council Order No. 322, 2017); NPCSC *Legislative Technical Standards (Trial) No. 2* (法工委发 [2011] 5号); Clifford Chance, *How Long is the EU Legislative Process?* (February 2025); FATF *Procedures for Fourth Round Mutual Evaluations*; CJEU, *Schrems II* (Case C-311/18).

Confidence: 78–85%. Authority-dependency characterisation well-sourced through NPCSC drafting standards (2/3 models with stronger sourcing). Temporal normality conclusively established with comparative data. Absence of dedicated peer-reviewed scholarship on the 根据 clause as a legal institution prevents confidence above 85%.

Validation Commitment: Monitor for published Chinese administrative law scholarship analysing the 根据 clause. If empirical evidence emerges that 根据 functions as content dependency rather than authority dependency, reassess whether the accommodation belongs in §9.4.1 or should migrate to Type 6 Pathway 2.

9.4.2 Funding Attribution Threshold (10%)

Category: ADAPTED (multi-source precedent)

Threshold:

- **Primary:** ≥50% = Strong funding dependency
- **Significant:** 10-49% = Meaningful funding connection
- **Minor:** <10% = Not documented as Funding Flow

Primary Precedent Sources:

Authority	Threshold	Application
Basel III Large Exposure	≥10% Tier 1 Capital	Large exposure definition threshold; 25% single-counterparty limit applies
IMF BPM6, paragraph 6.12	≥10% voting power	Direct investment classification — ownership of equity entitling investor to 10% or more of voting power
US GAAP ASC 280-10-50-12	≥10%	Segment reporting materiality
IFRS 8 paragraph 13	≥10%	International accounting materiality
Basel III Significant Investment	>10% ownership	Investment significance threshold
CFIUS Passive Investment	<10% safe harbour	31 CFR §800 — below 10% = passive
WTO SCM Agreement Article 11.9	1% de minimis	WEO's 10% is 10× the international minimum
SEC SAB 99	5% preliminary screen	Makes 10% (2×) clearly material

WEO Application:

The 10% threshold prevents trivial linkages whilst capturing meaningful dependencies. Aligned with international financial reporting standards for "significant" contribution versus "causing" relationships.

Note: OECD Due Diligence Guidance provides conceptual framework ("contributes to" vs "causes" vs "directly linked") but no specific percentage. 10% threshold derived from convergent international standards listed above.

WTO Threshold Alignment: The 10% threshold is independently validated by convergent regulatory practice (SEC ASC 280, Basel III, IMF BPM6). WEO's alignment with this cross-regulatory consensus — at 10× the WTO SCM de minimis (1%) and 2× the SEC SAB 99 preliminary screen (5%) — ensures funding attribution captures relationships that are unambiguously material across financial, trade, and investment frameworks.

Denominator clarification (V1.4): The denominator for funding attribution is the direct funding instrument value, not total project cost. This follows standard grant-accounting practice: when a government award of US\$7.86B enables a US\$100B total project (including private investment and debt), the attribution question is "what percentage of the award traces to the enabling policy?" — not "what percentage of the total project does the award represent?" The former correctly captures the funding dependency; the latter dilutes it with capital the policy did not provide. This clarification resolves the CHIPS Act cluster inconsistency where total-project framing produced sub-threshold attribution (7%) for awards that are 100% traceable to the enabling legislation.

9.4.3 Linguistic Marker Reference

Category: EMPIRICALLY GROUNDED (from policy citation research)

Source: Zhang & Guan (2022); Chinese policy citation literature

High-Confidence Markers (validated across multiple sources):

Chinese	Pinyin	Meaning	CC Type
根据	gēnjù	According to / Based on	Policy Cascade
落实	luòshí	Implement / Put into effect	Policy Cascade
贯彻	guàncè	Carry out / Execute	Policy Cascade
依照	yīzhào	In accordance with	Policy Cascade / Regulatory Enablement
联合	liánhé	Joint / Coordinated	Joint Issuance

Medium-Confidence Markers (require additional verification):

Chinese	Pinyin	Meaning	Notes
配套	pèitào	Supporting / Complementary	May indicate Framework Family
配套政策	pèitào zhèngcè	Supporting policies	Framework Family indicator
针对	zhēnduì	In response to / Targeting	Requires verification of intent
卡脖子	qiǎbózi	Bottleneck / Chokepoint	Security/technology context
拨款	bōkuǎn	Appropriation / Funding	Funding Flow indicator
资助	zīzhù	Subsidise / Fund	Funding Flow indicator
授权	shòuquán	Authorise	Regulatory Enablement indicator
系列	xìliè	Series	Framework Family indicator
回应	huíyìng	Response	Response CC indicator

English Marker Equivalents:

CC Type	English Markers
Policy Cascade	"pursuant to," "in accordance with," "implementing," "as required by"
Regulatory Enablement	"authorised under," "enabled by"
Response	"in response to," "following," "retaliation for"
Framework Family	Same numbering series, explicit family membership

Evidence Quality by Confidence Level:

Level	Label	Evidence Standard
CC-V	Verified	Official document explicitly establishes connection (ISO/IEC normative reference standard)
CC-E	Established	1 Primary + 2 Corroborating sources (ICD 206 corroboration standard)
CC-A	Analytical	Analyst assessment based on observable evidence (Parallel Policy only)

Chinese Policy Citation Research:

Ba et al. (2022) establish that policy citations "perform the legal function of delimiting the policy rights or legislative reference" by identifying and formally linking back to earlier instruments in the Chinese policy hierarchy.

Zhang & Guan (2022) formalise Chinese policy bibliometrics by classifying policy citation types and constructing citation and co-citation networks to trace how focal policies shape subsequent policymaking. Their analysis demonstrates that regional policymakers tend to cite national policies in their local policy documents, with citation patterns reflecting hierarchical governance structure.

Citations:

- Ba, Z., Ma, Y., Cai, J., & Li, G. (2022). "Spatio-temporal dynamics and determinants of new energy vehicle diffusion in China." *Journal of Cleaner Production*.
- Zhang, C., & Guan, J. (2022). "How policies emerge and interact with each other? A bibliometric analysis of policies in China." *Science and Public Policy*, 49(3), 441-459.

9.5 Parallel Policy Similarity Criteria

Manual Section: 9.4.3

Category: CATEGORY CREATOR (derived from academic framework)

Four Similarity Criteria:

Criterion	Assessment
Mechanism Similarity	Same policy tool type (subsidy, restriction, standard, etc.)
Target Similarity	Same industry, technology, or activity scope
Scope Similarity	Comparable geographic/sectoral reach
Timing Overlap	Events within domain-specific temporal window (see §9.4.1)

Threshold: ≥3 of 4 criteria must be met for Parallel Policy classification.

Academic Foundation:

- Shipan & Volden (2008): Established policy diffusion mechanisms (Learning, Competition, Imitation, Coercion)
- Gilardi & Wasserfallen (2019): Refined criteria for identifying parallel policy adoption

WEO Threshold (≥3/4) — Category Creator Calibration:

The 3-of-4 threshold is WEO-original calibration. Academic literature establishes the conceptual framework for similarity assessment but does not specify quantitative thresholds:

Threshold	Classification	Rationale
4/4	Very strong similarity	Rare; may indicate explicit coordination
3/4	Strong enough to document	Parallel Policy threshold
2/4	Insufficient	Could be coincidental
1/4 or 0/4	No connection	No documented relationship

Threshold Selection Rationale: Selected to balance:

1. Sufficient stringency (avoids spurious connections)
2. Practical applicability (4/4 too restrictive)

Threshold Derivation — Empirical Basis

Academic Foundation:

The policy diffusion literature establishes that parallel policy adoption occurs through four mechanisms (Shipan & Volden, 2008):

Mechanism	Description	Observable Indicators
Learning	Jurisdictions observe and adopt successful policies from others	Similar policy design, temporal sequence
Competition	Jurisdictions adopt policies to maintain competitive position	Similar targets, similar timing
Imitation	Jurisdictions copy policies for legitimacy without deep analysis	High mechanism similarity, rapid adoption
Coercion	External pressure drives adoption	Explicit conditionality, power asymmetry

All four mechanisms produce observable similarity across the same dimensions: mechanism, target, scope, and timing. The academic literature does not specify quantitative thresholds for "sufficient similarity" — this is the gap WEO fills.

Threshold Calibration Analysis:

Research on policy diffusion patterns (Berry & Berry, 1990; Gilardi & Wasserfallen, 2019) suggests:

Similarity Level	Observed Pattern	Interpretation
4/4 criteria	Rare (<5% of policy pairs)	Strong indicator of explicit coordination; may warrant Joint Issuance investigation
3/4 criteria	Uncommon (10-20% of policy pairs)	Consistent with diffusion mechanisms; sufficient for Parallel Policy classification
2/4 criteria	Common (30-40% of policy pairs)	Likely coincidental; policies in same domain often share some characteristics
1/4 or 0/4 criteria	Very common (>50% of policy pairs)	No meaningful similarity

WEO Calibration Decision:

The 3-of-4 threshold was selected based on:

1. **Distinctiveness:** 3/4 similarity is uncommon enough (~10-20% of policy pairs) to indicate meaningful connection rather than coincidence
2. **Observability:** All four criteria are objectively assessable from policy documentation
3. **Conservatism:** Setting threshold at 3/4 rather than 2/4 reduces false positives whilst accepting some false negatives (missed connections)
4. **Analytical designation:** Parallel Policy is the only CC type permitting CC-A (Analytical) confidence, explicitly acknowledging the inferential nature of similarity-based connections

Sensitivity Analysis:

If threshold were...	Effect
2/4	~2-3× more Parallel Policy connections; higher false positive rate; diluted analytical signal
4/4	~60-70% fewer Parallel Policy connections; captures only near-identical policies; misses genuine diffusion

Citations:

- Berry, F. S., & Berry, W. D. (1990). "State lottery adoptions as policy innovations." *American Political Science Review*, 84(2), 395-415.
- Gilardi, F., & Wasserfallen, F. (2019). "The politics of policy diffusion." *European Journal of Political Research*, 58(4), 1245-1256.
- Shipan, C. R., & Volden, C. (2008). "The mechanisms of policy diffusion." *American Journal of Political Science*, 52(4), 840-857.
- Mallinson, D. (2021). "Policy innovation adoption across the diffusion life course." *Policy Studies Journal*, 49(2), 335-358.

Validation Commitment: Review at 100-event milestone based on classification consistency data.

9.5.1 Criterion 1 — Corporate Product Release Exclusion

Manual Section: 9.4.3 (Criterion 1: Mechanism Similarity)

Category: STANDARD PRACTICE

Rationale: The five mechanism categories in Criterion 1 (Subsidy/Incentive, Restriction/Control, Standard/Mandate, Framework/Governance, Investment/Buildout) are exclusively policy instruments. Corporate product releases are market actions, not policy tools, and do not fit any enumerated category. Follows from strict application of existing criteria.

9.5.2 Criterion 3 — Scope Comparator Hierarchy

Manual Section: §9.4.3 (Criterion 3: Scope Similarity)

Category: ADAPTED (instrument-specific comparators, 3× threshold, binary discipline) + WEO SYNTHESIS (hierarchical ordering, inapplicable/indeterminate distinction)

Rationale: V1.3 specified scope similarity through geographic matching and a financial scale ratio test ("Within 3× of each other") without codifying metric selection across event types where financial data may be unavailable or inapplicable, and without distinguishing between events where financial comparison is meaningless by nature and events where data would be meaningful but is undisclosed. V1.4 introduces a two-level metric selection hierarchy, the inapplicable/indeterminate distinction, and codification of range-straddling and structural comparability provisions.

Metric Selection Hierarchy (WEO SYNTHESIS)

The practice of matching comparability metrics to instrument type is established across apex institutions: IEA/IRENA use MW/GW for energy programmes (IRENA, *Renewable Capacity Statistics 2026*); IMF uses GDP-share for fiscal policy (*Fiscal Monitor April 2026*); IEG uses segmented methodology by project type (*Project Performance Ratings Codebook*, September 2015). No institution codifies a metric-selection hierarchy — the ordering is a WEO contribution consistent with the OMB Circular A-4 (November 2023) monetise → quantify → qualify cascade, which governs assessment depth rather than cross-programme comparator selection.

The two-level ordering (financial → physical capacity) reflects a precision hierarchy. When neither metric is available, scope is assessed on geographic matching alone. Across 63 Parallel Policy CCs at V1.4 publication, all 22 assessments lacking quantitative data returned Met, consistent with the expectation that events passing mechanism and target similarity at the same jurisdictional level are inherently scope-comparable.

3× Ratio Threshold (ADAPTED)

The geometric ratio of 0.33× to 3.0× is the established institutional standard for defining comparable organisational scale in executive compensation benchmarking, universally applied by tier-one consulting firms: FW Cook ("1/3× to 3× revenue," Lum and Jordan, 14 October 2021), Meridian Compensation Partners (Eaton Corporation plc DEF 14A, 2023/2025), Pearl Meyer (0.33×–3.0× revenue), and Compensia (0.33×–3.0× revenue). ISS applies a tighter 0.4×–2.5× band. The 3× upper bound reflects logarithmic parity — a half order of magnitude ($10^{0.5} \approx 3.16$) — representing the boundary beyond which entities are no longer structurally comparable in scale.

Internal calibration across 35 numerically-assessed Parallel Policy CCs reveals a natural discontinuity:

Ratio Range	Qualifying (Met)	Non-Qualifying (Not Met)
1.0×–2.8×	20	0
2.8×–3.5×	0	2
3.5×–5.0×	0	3
>5.0×	0	10

The 3× threshold falls within the observed gap between qualifying ratios ($\leq 2.77\times$) and non-qualifying ratios ($\geq 3.50\times$). No alternative ratio has stronger grounding in either external precedent or observed data.

Binary Discipline, Range-Straddling, and Structural Comparability

The 3× threshold is applied as strict binary pass/fail, consistent with ISO/IEC 17025 and the ILAC G8 simple acceptance rule. Graduated alternatives exist (Basel III CCB, EU Merger 2023 Notice, FATF) — the binary approach is a deliberate design choice prioritising replicability over case-by-case flexibility.

Range-straddling: where figures are reported as ranges crossing the threshold, the conservative bound governs, consistent with ISO/IEC 17025 conformity assessment practice.

Structural comparability: scope encompasses mechanism-structure comparability beyond numerical proximity. The analyst documents the rationale for any override in the evidence record.

Data-Availability Asymmetry (WEO SYNTHESIS)

Events assessed as indeterminate pass scope on geographic matching alone — the same outcome as events assessed as inapplicable. This means less-documented events face a less stringent assessment than well-documented events. This asymmetry is inherent to evidence-based frameworks: ISS cannot screen companies that do not report revenue; FATF cannot score criteria for which data is absent. Three considerations support the design: penalising regulatory events for lacking budgets would be a category error; penalising events for jurisdictional opacity would create structural visibility asymmetry; and the $\geq 3/4$ threshold ensures scope is one of four gates, not a standalone qualifier. The inapplicable/indeterminate distinction creates an audit trail for milestone review. At V1.4 publication, approximately 35 CCs are quantitative, 20 inapplicable, and 7 indeterminate.

Sources:

Institutional: IRENA (2026); IMF *Fiscal Monitor* (2026); IEG *Codebook* (2015); OMB Circular A-4 (2023); ISO/IEC 17025:2017; ILAC G8:09/2019.

Corporate Governance: FW Cook (Lum/Jordan, 2021); Eaton DEF 14A (2023/2025, Meridian); Pearl Meyer; Compensia; ISS U.S. *Peer Group FAQ* (v.2025.12); CAP S&P 1500 insurer study (0.36×–4.55× observed dispersion).

Academic: Albuquerque, De Franco & Verdi (2013) *JFE*; Bizjak, Lemmon & Nguyen (2011) *JFE*; Faulkender & Yang (2010) *JFE*.

Confidence: MEDIUM-HIGH (72–80%). Instrument-specific practice ~85%. 3× threshold grounded in corporate governance practice, consistent with observed CC data ~78%. Binary discipline ~75%. Hierarchical ordering ~68%. Inapplicable/indeterminate distinction ~70%.

Validation Commitment: Review ratio distribution at database doubling milestones. Monitor for X-fold ratio tests in comparative policy literature. Review structural comparability overrides at 200-event milestone. Review indeterminate-status CCs at 200-event milestone; if >15% of Parallel Policy CCs, assess whether mitigation is required.

9.6 Derivation Documentation

9.6.1 Connection Type Derivation

High-Consensus Types (Multi-Source Validation)

The following connection types appear across 3+ independent systems:

Type	Sources	Consensus
Policy Cascade	Chinese 1+N Framework, CSET, OECD	HIGH
Joint Issuance	Chinese Governance, OECD, OFAC/EU	HIGH
Funding Flow	OECD Due Diligence, Government Grant Standards, Federal Budget Practice	HIGH
Regulatory Enablement	ISO/IEC Directives, OECD, Administrative Law	HIGH
Response	OFAC Sanctions, GSDB, CSET	HIGH
Framework Family	ISO/IEC Standards, UN Framework Families, National Standards Bodies	HIGH
Parallel Policy	Policy Diffusion Literature (Shipan & Volden), Gilardi & Wasserfallen (2019), Berry & Berry (1990)	MEDIUM

Research Sources:

Primary Institutional Sources:

- ISO/IEC Directives, Part 2: Normative vs. informative reference distinction
- OECD Due Diligence Guidance: "Causes," "contributes to," "directly linked" relationships
- OFAC Sanctions Programs: Escalation sequences, response patterns
- Chinese State Council Gazette: 1+N framework documentation practices

Academic Literature:

- Shipan & Volden (2008): Policy diffusion mechanisms (Learning, Competition, Coercion, Emulation)
- Gilardi & Wasserfallen (2019): Politics of policy diffusion
- Berry & Berry (1990): Event history analysis of policy adoption

Policy Analysis:

- CSET AI Policy Research: Sanction cascade, regulatory mimicry, forum shopping response
- CCICED: China 1+N framework analysis
- Carbon Brief: Chinese policy documentation analysis

9.6.2 Threshold Derivation

Temporal Windows:

Domain	Source	Derivation
Energy/Infrastructure (365-day Response)	Policy implementation research	Long-horizon commitment cycles
Security (30-day Response)	OFAC enforcement patterns	Sanctions response typically within 1 month
Technology (180-day Response)	Product development cycles	Minimum iteration period
Trade (60-day Response)	WTO DSU Article 4.7	Consultation period before panel request

Citation Standards:

Standard	Source	Application
Normative reference	ISO/IEC Directives Part 2	Mandatory compliance relationship
Informative reference	ISO/IEC Directives Part 2	Guidance/background relationship
1+N qualification	CCICED analysis	Chinese policy cascade identification

Funding Attribution:

Threshold	Source	Rationale
10% minimum	OECD "contributes to" standard	Meaningful contribution vs. trivial link
50% primary	Government grant reporting	Primary funding source designation

9.6.3 Gap Analysis

Deferred Connection Types:

Type	Gap Reason	Research Needed
Escalation Sequence	Requires serial temporal analysis	Conflict escalation literature
Cross-Agency Referral	Low public documentation visibility	Internal government process research
Genealogical Traceability	Requires supply chain mapping	ISO 9001 traceability implementation

Known Limitations:

Limitation	Impact	Mitigation
GDELT/ICEWS no explicit relationship typology	Cannot benchmark against major event databases	Rely on regulatory/legal framework sources
Chinese informal coordination underdocumented	May miss 部际联席会议 (bùjì liánxí huìyì, inter-ministerial conference) patterns	Enhanced Chinese-language research
Policy diffusion mechanisms difficult to operationalise	"Competition" and "emulation" hard to distinguish from documentation	Conservative classification; defer ambiguous cases

9.7 Novelty Verification

Status: Verified novel via systematic research

Research Scope: 25+ platforms across geopolitical intelligence, AI policy tracking, event databases, and entity relationship platforms

Key Platforms Examined:

- Geopolitical Intelligence: Stratfor/RANE, Eurasia Group, Oxford Analytica, Control Risks
- AI/Tech Policy Trackers: OECD AI Policy Observatory, Stanford HAI, Carnegie
- Event Databases: GDELT, ICEWS, UCDP, ACLED
- Entity Relationship Platforms: Bloomberg, Sayari, Maltego

Conclusion: No comparable typed event-to-event relationship taxonomy found. Coordination Connections (CC) represents genuine novel contribution. The 7-type taxonomy qualifies as copyrightable "creative selection and arrangement."

Section 10: Category Creator Documentation

10.1 Purpose

This section explicitly documents all methodology components where WEO exercises category creator prerogative — novel calibration choices for which no external precedent exists.

10.2 Category Creator Components

Component	Section	Type	Validation Commitment
Level 5 operational evidence (four-pathway gate)	3.5.4	Synthesis	Review as SC 42 database and EU AI Act harmonisation pipeline mature; monitor FATF 5th Round methodology
PIET 3-of-4 gate architecture	3.7.1	Calibration	Review against cumulative PIET event data
PIET trade flow and price thresholds	3.7.2	Calibration	Review against each qualifying PIET event; thin-market sensitivity analysis
OEP 10% hardware deployment threshold	3.8.3	Provisional	Review at 10-event OEP Pathway C milestone
Conflict-era six-month review interval	3.10.1	Calibration	Review interval at first conflict-era determination; assess whether 6 months is operationally appropriate
Third Bloc Formation transitional classification	3.12.1.3	Synthesis	Reassess when third PAA candidate approaches qualification
J-Tier D AI Hub Exception	4.6	Designation	Annual designation review
F3 AI Centrality (mandatory gating factor)	4.6	Threshold	Threshold review at 100-event milestone
Buildout vs Restriction typology	5.2	Classification	Classification consistency monitoring
Principal AI actor framework (7-dimension, ≥ 3 threshold)	5.4	Classification	Reassess at methodology version milestones; threshold sensitivity review as additional nations develop AI infrastructure
Per-dimension qualification principles (7 principles)	5.4.1	Calibration	Review against cumulative actor assessment data; edge-case stress test refinement at each methodology version milestone

Dimension 2/3 process-node threshold ($\leq 7\text{nm}$)	5.4.1	Calibration	Review against fabrication frontier at each methodology version milestone; recalibrate downward when ≥ 4 actors manufacture at threshold node independently
Dimension 5 benchmark saturation dependency	5.4.1	Monitoring	Document benchmarks informing each assessment cycle's D5 verdicts; flag reduced-confidence assessments when benchmark discrimination degrades
Subsidiary clause PAA-score sensitivity	5.4.1	Monitoring	Reassess whether alternative attribution rules would alter the $\geq 3/\geq 2$ -categories result at each methodology version milestone
Sovereign control principle (governing rule)	5.4.1	Classification	Review territorial vs ownership-chain test assignments if new dimension types are introduced or existing dimensions are restructured
D4 territorial compute requirement	5.4.1	Classification	Monitor for edge cases where territorial compute presence and corporate ownership diverge; review against D2 territorial test for consistency
D4 assessment-date-relative threshold	5.4.1	Calibration	Document prevailing frontier training scale at each assessment cycle; review for consistency across actors assessed at different dates
D6 dual-pathway (export controls + AI-specific legislation)	5.4.1	Clarification	Monitor whether the dual-pathway produces inconsistent assessments across actors
D7 national proportionality standard	5.4.1	Calibration	Review proportionality assessments for cross-actor consistency; monitor for threshold drift
D7 general-infrastructure-fund exclusion	5.4.1	Classification	Apply consistently; monitor for borderline cases where general funds have significant AI-adjacent components
D7 announced-vs-executed investment	5.4.1	Calibration	Verify disbursement data where available; flag actors with execution rates below 25%
D5 single-domain exclusion	5.4.1	Classification	Apply consistently; review if frontier model definition expands to include domain-specific excellence
D2 marketing-node exclusion	5.4.1	Classification	Maintain node-name-to-lithography mapping table for Intel and any other manufacturer using rebranded node names
D3 subsidiary clause application	5.4.1	Classification	Monitor for cases where D3 and D2 territorial rules produce divergent results for the same actor; ensure consistency

Synthesis majority override rule	5.4.1	Methodology	Document each override with establishing case; review for overuse
Actor Designation Framework temporal disclaimer	5.4.2	Operational	Maintain platform-facing disclaimer; review if retroactive scoring is implemented
Three-gate PAA designation (capability + severance + platform)	5.4.2	Classification	Continue applying to remaining actors; review gate structure against cumulative results
Severance test 24–36 month sustainability window	5.4.2	Calibration	Review at first severance test application; monitor against actual supply-chain disruption events for calibration validation
Three-category spanning requirement (hardware/platform/governance)	5.4.2	Threshold	Review if any actor scores ≥ 3 from a single capability category
Ecosystem inputs definition ((a)/(b) formulation)	5.4.3	Classification	Review if new extraterritorial instruments emerge beyond FDPR/EAR; monitor ecosystem/bloc boundary
Per-PAA severance testing (Option B)	5.4.4	Classification	Monitor for third PAA emergence; review if any designation depends on per-PAA vs simultaneous approach
Deterrence principle with three-part documentation	5.4.5	Classification	Monitor PAA leverage vectors; review supranational entity clause if EU institutional authority evolves
Multi-vector leverage assessment	5.4.5	Classification	Monitor for changes in US leverage vector count for Netherlands and other actors
Supranational entity deterrence clause	5.4.5	Classification	Review if EU develops direct corporate sovereignty instruments applicable to ASML
Cascade analysis boundary (single-level standard)	5.4.6	Classification	Monitor whether sub-tier constraint findings change any designation; review boundary if FDPR enforcement expands
Dimension sustainability taxonomy (5 classifications)	5.4.7	Classification	Monitor whether five levels remain sufficient; review counting rule at boundary cases
AI Infrastructure Keystone classification	5.4.8	Classification	Monitor analytical utility; assess whether AIK-ACS distinction remains stable and informative across assessment cycles
AIK theoretical grounding (Rey and Tirole primary anchor)	5.4.8	Adapted	Review against emerging IO/IR literature; monitor for frameworks that challenge the vertical/horizontal foreclosure mapping

Advanced Capability State classification	5.4.9	Adapted	Track whether PAA attribution remains stable; monitor for ACS actors developing multi-ecosystem dependencies
Participant classification (single sub- ≥ 3 tier)	5.4.10	Classification	Monitor whether per-dimension data alone provides sufficient analytical detail for sub- ≥ 3 actors, or whether subdivision is warranted
Evidence standard 1+1 base with escalation	5.4.12	Adapted	Monitor escalation trigger frequency; if $>50\%$ of MET dimensions trigger escalation, review base standard
Source authority hierarchy (Grade 1/2/3)	5.4.12	Classification	Review grade assignments against emerging source types
Mandatory source descriptors	5.4.12	Operational	Monitor implementation consistency across assessors
Four escalation triggers (opaque/contested/unaudited/no canonical)	5.4.12	Calibration	Track trigger frequency and outcomes; adjust trigger definitions if producing anomalous results
ENT-2 four-criteria linguistic threshold	8.6	Calibration	Track ENT-2 population at 200 events; test discriminating power against new governance architectures
ENT-5 medium scope boundary	8.9	Classification	If ENT-5 remains at zero beyond 2028, reassess medium scope
ENT-2/ENT-5 governance maturity boundary	8.10	Classification	At 200 events, assess whether all ENT-5 events also carry ENT-2
CC Taxonomy (7 types)	9.2	Novel IP	Protected via Zenodo disclosure
Authority Hierarchy 6-tier structure	9.2	Synthesis	N/A (synthesised from established precedents)
Temporal Windows domain calibration	9.4.1	Provisional	Empirical validation at 100-event milestone
Chinese 根据 temporal accommodation (48-month bound)	9.4.1	Adapted	Monitor for 根据 scholarship; reassess if content-dependency evidence emerges
Parallel Policy 3-of-4 threshold	9.5	Calibration	Review at 100-event milestone
Scope Comparator Hierarchy (two-level metric selection)	9.5.2	Synthesis	Review at database doubling milestones
3 $\times$ scope similarity threshold	9.5.2	Calibration (Adapted)	Review if natural discontinuity shifts

Inapplicable/indeterminate scale assessment distinction	9.5.2	Synthesis	Review indeterminate population at 200-event milestone
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10.3 Validation Framework

All category creator components are subject to:

- 1. **Continuous monitoring** during database growth
 - 2. **Systematic review** at designated milestones
 - 3. **Transparent documentation** of any adjustments
 - 4. **Academic peer review** as methodology matures
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Appendix A: Glossary of Key Terms

Term	Definition
ACS (Advanced Capability State)	Nation-state actor meeting ≥ 3 SCP dimensions but failing the per-PAA severance test (§5.3.1)
AIK (AI Infrastructure Keystone)	Nation-state actor meeting ≥ 3 SCP dimensions and passing the severance test but lacking a sustainable platform dimension (§5.3.1)
Category Creator	Methodology component for which WEO exercises novel calibration prerogative
CC (Coordination Connection)	Typed relationship between two events in the WEO database
CC-A	Coordination Connection - Analytical (analyst assessment with evidence)
CC-E	Coordination Connection - Established (authoritative statement documents connection)
CC-V	Coordination Connection - Verified (documentary proof)
Conflict-Era Source Currency Adjustment	Adjusted currency windows (§3.4.12) that apply to source documentation produced within or about conflict-affected territories during formally defined conflict-era conditions, modifying the peacetime ceilings in §3.4.8
Corporate Strategic Disclosure (Option E)	Stage 2 AI Connection Filter Option E — establishes AI connection for Corporate events through regulated financial disclosure explicitly linking a transaction to AI infrastructure, where Options A, B, and C fail
CSTF (Cross-Sector Transmission Frequency)	Proportion of Coordination Connections linking events with different primary industry tags
Defence Appropriation Evidence (Option F)	Stage 2 AI Connection Filter Option F — establishes AI connection for Policy events through binding defence appropriation documents with explicit AI designation, where Options A, B, and C fail
ENT (Environmental Nexus Tag)	Binary tag identifying documented environmental intersections across five types: Resource Demand (ENT-1), Environmental Enablement (ENT-2), Environmental Consequence (ENT-3), Environmental Dependency (ENT-4), Environmental

	Coordination (ENT-5)
Event	Verified happening (coordination or fragmentation)
Impact-Demonstrated (T2)	T2 Restriction classification basis supported by demonstrated cross-bloc fragmentation impact through ≥ 2 other-bloc government policy responses, rather than explicit other-bloc targeting in primary documentation
Industry Tagging Principle	Governing standard: events are tagged by the industry whose operational landscape their documented provisions most directly address
J-Tier	Jurisdictional significance tier (A/B/C/D/E)
Mechanism Statement	One-sentence statement identifying the document-verifiable pathway from an event's provisions to a secondary industry's operations
OEP (Operational Evidence Protocol)	Stage 2 AI Connection Filter Option D — establishes AI connection for Infrastructure events through operational physical evidence where official documentation is absent, generic, or application-agnostic
Option G (Supply-Chain Response Evidence)	Stage 2 AI Connection Filter Option G — establishes AI connection for Policy events where the instrument controls, secures, or subsidises a documented AI infrastructure input material, and Options A, B, and C fail because the instrument text addresses the material rather than AI (§3.5)
PAA (Principal AI Actor)	Nation-state actor meeting ≥ 3 SCP dimensions, passing the per-PAA severance test, and holding ≥ 1 sustainable platform dimension (§5.3.1)
Participant	Nation-state actor scoring below ≥ 3 on SCP dimensions; the residual designation for actors not qualifying as PAA, AIK, or ACS (§5.3.1)
PIET (Post-Implementation Evidence Track)	Stage 2 qualification pathway for Policy restriction events on physical AI-critical materials or physical technology products where post-implementation empirical evidence establishes the AI connection that documentary sources cannot
PISA-5	Policy and Institutional Significance Assessment (5-factor framework)
Primary Tag	The single industry tag representing the industry most directly addressed by an event's documented provisions
Scale Assessment Status	Three-value taxonomy (Quantitative / Inapplicable / Indeterminate) recording whether Dimension 2 of Parallel Policy scope similarity was assessed with quantitative data (§9.4.3)
SCP (Sovereign Capability Profiles)	Seven-dimension assessment framework for evaluating sovereign AI infrastructure capability, producing scores used for PAA/AIK/ACS/Participant designation (§5.3)

Secondary Tag	Industry tag (maximum 2 per event) representing an industry whose operational landscape is also identifiably addressed by documented provisions
Transmission Matrix	11×11 matrix recording CC counts between industry category pairs
Warmth Engine Ratio (WER)	Ratio of ENT-1 count to ENT-5 count, measuring the balance between documented environmental resource demand and documented environmental governance

Appendix B: Cross-Reference Verification

All Manual cross-references verified against WEO Methodology Manual:

Rationale Section	Manual Reference	Status
§3.1 Source Quality Hierarchy	§3.1	✓ Verified
§3.2 Claim Verification Tiers	§3.4	✓ Verified
§3.3 Four Event Types	§3.6	✓ Verified
§3.4 Event Type × Tier Matrix	§3.6, Table B	✓ Verified
§3.5 Coordination Event Criteria	§3.7	✓ Verified
§3.5.4 Type 2 Level 5 Operational Evidence	§3.7	✓ Verified
§3.6 Two-Stage AI Filter	§3.5	✓ Verified
§3.6.2 Option E — Corporate Strategic Disclosure	§3.5, Option E	✓ Verified
§3.6.3 Option F — Defence Appropriation Evidence	§3.5, Option F	✓ Verified
§3.6.4 Option G — Supply-Chain Response Evidence	§3.5, Option G	✓ Verified
§3.7 PIET — Post-Implementation Evidence Track	§3.5.1	✓ Verified
§3.8 OEP — Operational Evidence Protocol	§3.5, Option D	✓ Verified
§3.9 6-Month Verification Lag	§3.9	✓ Verified
§3.10 Currency Windows	§3.4.8	✓

		Verified
§3.10.1 Conflict-Era Source Currency Adjustment	§3.4.12	✓ Verified
§3.11 Non-English Source Verification	§3.3	✓ Verified
§3.12 Bloc Classification	§3.10	✓ Verified
§3.12.1 Universal Membership Categories	§3.10.1	✓ Verified
§3.12.1a Bloc Assignment Gate	§3.10.1	✓ Verified
§3.12.1.1 Evidence Standard for Register Classification Changes	§3.10.1.1	✓ Verified
§3.12.1.2 Register Operations	§3.10.1.2	✓ Verified
§3.12.1.3 Third Bloc Formation	§3.10.1.3	✓ Verified
§3.12.2 Non-Aligned Nation Identification	§3.10.2	✓ Verified
§3.12.3 Event-Level Attribution Scope Limitation	§3.10.3	✓ Verified
§3.12.4 AI Model Events — Critical Technology Determination	§3.10.3 (Priority 2)	✓ Verified
§3.13 Event Evolution Protocol	§3.14	✓ Verified
§3.13.1 Corporate Product Release Diagnostic	§3.14.1 (Supplement)	✓ Verified
§3.14 Post-Qualification Verification	§3.15	✓ Verified
§3.15 Source Accommodation Alignment	§3.1, §3.4.11, §3.5, §3.5.1, §4.11	✓ Verified
§4.1 Basel 5-Factor	§4.3-4.5	✓ Verified

§4.2 \$500M Threshold	§4.4 (Factor 1)	✓ Verified
§4.3 5,000+ GPU Threshold	§4.4	✓ Verified
§4.4 Software Routes 1-4	§4.8	✓ Verified
§4.5 Route 4-CS	§4.8	✓ Verified
§4.6 PISA-5	§4.9	✓ Verified
§4.6.1 Tiebreaker Criteria (TB1-TB5)	§4.9 (Human Review Framework)	✓ Verified
§4.6.3 F3 Pathway Exemption	§4.9 (Factor 3)	✓ Verified
§4.7 T2 Buildout Criteria	§5.5 (T2 Buildout Test)	✓ Verified
§5.1 Three-Tier Architecture	§5.2	✓ Verified
§5.2 Mandatory Standard Test	§5.5 (T2 Mandatory Standard Test)	✓ Verified
§5.3 Bloc Definitions	§3.10 (Bloc Membership Reference)	✓ Verified
§5.4 Principal AI Actors	§5.3	✓ Verified
§5.4 T2 Scope (Principal Actor Asymmetry)	§5.3	✓ Verified
§5.4.1 Dimension Qualification Principles	§5.3.2	✓ Verified
§5.4.2 Three-Gate Designation, Severance Test, and AIK Classification	§5.3.1	✓ Verified
§5.4.3 Ecosystem Inputs Definition	§5.3.1	✓ Verified

§5.4.4 Per-PAA Severance Testing (Option B)	§5.3.1	✓ Verified
§5.4.5 Deterrence Principle	§5.3.1	✓ Verified
§5.4.6 Cascade Analysis Boundary	§5.3.1	✓ Verified
§5.4.7 Dimension Sustainability Taxonomy	§5.3.1	✓ Verified
§5.4.8 AI Infrastructure Keystone (AIK) Designation	§5.3.1	✓ Verified
§5.4.9 Advanced Capability State (ACS) Designation	§5.3.1	✓ Verified
§5.4.10 Participant Designation	§5.3.1	✓ Verified
§5.4.11 Sovereign Capability Profiles Register	§5.3.3	✓ Verified
§5.4.12 Evidence Standard for Dimension Qualification	§5.3.4	✓ Verified
§6.1 Four-Domain Framework	§6.1	✓ Verified
§6.2 Tech vs E/I Distinction	§6.2	✓ Verified
§6.3 Domain Tiebreaker	§6.7	✓ Verified
§7.1 Industry Taxonomy	§7.3	✓ Verified
§7.2 Primary/Secondary Structure	§7.4, §7.7	✓ Verified
§7.3 Rule-Based Assignment	§7.4	✓ Verified
§7.4 Mechanism Statement	§7.7	✓ Verified
§7.5 CSTF	— (TCS paper; Manual §7.9	✓

	nearest context)	Verified
§7.6 Transmission Matrix	— (TCS paper; Manual §7.9 nearest context)	✓ Verified
§7.7 Adversarial Quality Test	§7.9	✓ Verified
§7.8 Workflow Integration	§7.4, §7.9	✓ Verified
§8.1 Five-Type ENT Taxonomy	§8.3	✓ Verified
§8.2 ENT as Enrichment Layer	§8.2	✓ Verified
§8.3 Binary Qualification	§8.2	✓ Verified
§8.4 Architectural Guardrails	§8.2	✓ Verified
§8.5 ENT-1 Resource Demand	§8.3.1	✓ Verified
§8.6 ENT-2 Environmental Enablement	§8.3.2	✓ Verified
§8.7 ENT-3 Environmental Consequence	§8.3.3	✓ Verified
§8.8 ENT-4 Environmental Dependency	§8.3.4	✓ Verified
§8.9 ENT-5 Environmental Coordination	§8.3.5	✓ Verified
§8.10 ENT-2/ENT-5 Boundary	§8.3.2, §8.3.5	✓ Verified
§8.11 Chinese Cascade Model	§8.3.2	✓ Verified
§9.1 CC Taxonomy Overview	§9.1, §9.2	✓ Verified
§9.2 Seven Connection Types	§9.2.1	✓ Verified

§9.3 CC Confidence Levels	§9.3	✓ Verified
§9.4 Evidence Requirements	§9.4	✓ Verified
§9.4.1 Chinese Regulatory Temporal Accommodation	§9.4.1	✓ Verified
§9.4.3 Linguistic Marker Reference	§9.5.2 (table: Linguistic Markers)	✓ Verified
§9.5 Parallel Policy Criteria	§9.4.3	✓ Verified
§9.5.2 Criterion 3 — Scope Comparator Hierarchy	§9.4.3	✓ Verified

Appendix C: Note on Source Accuracy

This methodology draws on extensive research across academic literature, regulatory frameworks, and institutional documentation. Whilst every effort has been made to ensure accuracy in citations, dates, and quoted material, minor discrepancies may exist due to:

- Translation variations in non-English source material
- Differing publication dates across editions or jurisdictions
- Paraphrasing or summarisation in secondary sources consulted

Where methodology components are classified as **WEO-SYNTHESIS** or **WEO-ORIGINAL**, the contribution represents novel analytical work informed by, but not directly derived from, external sources. These components are explicitly flagged throughout the document.

Identified errors are corrected in subsequent methodology versions. Users identifying potential inaccuracies are encouraged to report them via the contact information provided on the WEO platform.

End of Methodology Rationale

Warmth Engine Observatory | January 2026 (v1.4, May 2026)